

EN



[SIDE CONTROL/NO-OLP]

**OPERATING MANUAL
& SPARE PARTS LIST**

Cargotec Korea Ltd

EN



OPERATING MANUAL

Cargotec Korea Ltd

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1. Foreword

This operator's manual for HIAB 160T-4 teaches you the right way to operate and maintain your crane.

In this Operating manual are :

- Important safety precautions and warnings
- Important directions for use
- An explanation of the arrangement of Operator's manual
- An explanation of the directions in the Operator's manual

▲ DANGER
If you have not studied the Operator's Manual for your crane carefully, this can lead to fatal accidents or serious damage.

Thus you should :

1. Study the entire Operator's Manual carefully
2. Use the crane only after having done so
3. Follow the directions for use, operation and maintenance of the crane exactly

!NOTE

HIAB reserves the right to change specifications, equipment, operating instructions and maintenance instructions without prior notice

2. The Operator's Manual

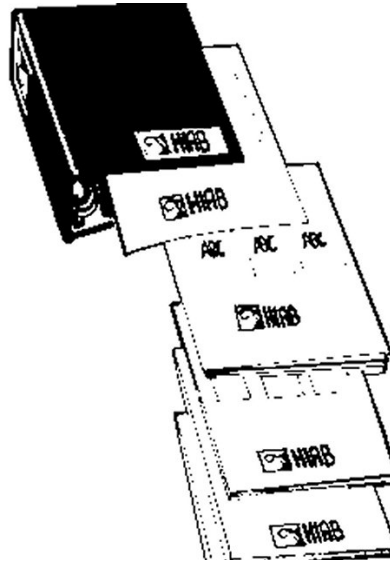
2-1. Structure

The Operator's Manual for your crane consists of :

- The Operator's Manual applies to this HIAB stiff-boom crane.

It contains important safety precautions and warnings. The operator's Manual also contains important general instructions for use.

- Operation and Maintenance of your type of crane
- Technical Data for your crane
- Descriptions of options and extensions for your crane



2-2. Indications in the Operator's Manual

The following indications are used in the Operator's Manual :

! TIP

Tip to make work easy to carry out

! NOTE

Extra information that can prevent problems



CAUTION

Hazard for the crane or crane components.

Follow the instructions carefully



WARNING

Danger of injury to yourself or to bystanders, or
danger of serious damage

to the crane or other objects.

Follow the instruction carefully.



DANGER

Danger to life for yourself or to bystanders.

Follow the instructions carefully.



!NOTE

The sign  above occurs in the text as a black triangle ▲.

The same status is valid for both signs.

2-3. Other indications

Symbol for reference to illustration. ❶

Refers to an illustration.

Symbol for reference to a component in an illustration. ①

Refers to a component in an illustration.

[Option]

Indication for parts that are non-standard for the crane, but an option.

!NOTE

1. Do this

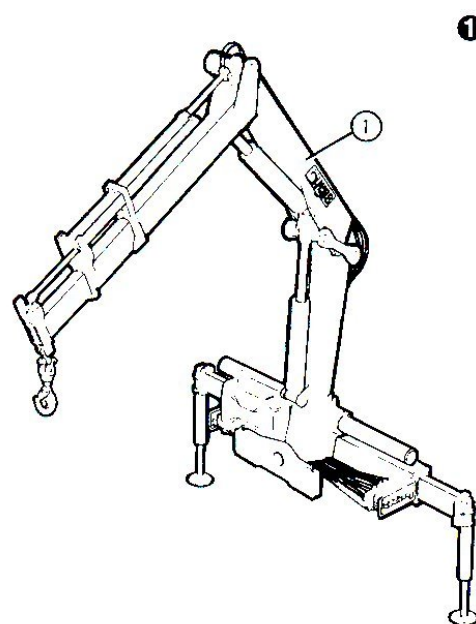
2. Do that

If actions are numbered, you should most certainly carry them out in numerical order!

!TIP

Are you looking for information?

Use the Table of Contents and the Index.



3. Safety precautions and warnings

3-1. Study these instructions carefully!

▲ DANGER

If you have not studied the Operator's Manual for your crane carefully, this can lead to fatal accidents and serious damage.



Thus you should:

1. Study the entire Operator's Manual carefully
2. Use the crane only after having done so
3. Follow the directions for use, operation and maintenance of the crane exactly

3-2. Knowledge required

▲ DANGER

Only persons with the requisite knowledge and experience with cranes may use the crane. Never operate the crane when you are sick, tired out, under the influence of medicines, alcohol or other drugs

1. Take the delivery instructions from your HIAB dealer.

Or be instructed by an experienced person from your own company.

2. Only then should you operate your crane. Ensure that you meet the statutory requirements of the country in which you will use the crane (for example, certificate, obligatory safety-helmet)

▲ DANGER

Carry out yourself only the maintenance work prescribed in the Operation and Maintenance for your type of crane.

All other maintenance work should be done in a HIAB service workshop.

▲ Deal yourself only with the faults described in the Operation and Maintenance for your type of crane. All other work to rectify faults may only be performed by personnel in a HIAB service workshop.



3-3. Operating conditions

You may use the crane only under the following conditions :
in the open air, or in spaces with sufficient ventilation

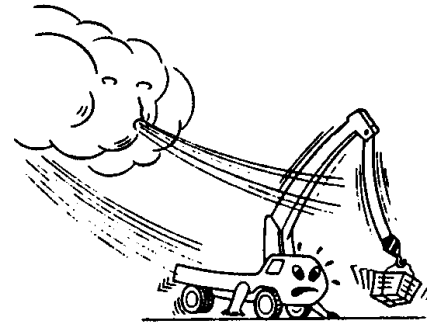
▲ DANGER

If you use the crane in a confined space you should suffocate from the exhaust gases from the vehicle.

With wind speed less than 13.3 m/sec [approx. 29.7 mph]
Consult the wind speed table, Appendix 1.

▲ DANGER

Never use the crane in a high wind or storm.
When wind speeds exceed 13.3 m/sec
the crane will behave unpredictably.



▲ DANGER

Never use the crane during a thunder-storm.

▲ DANGER

At temperature below 0°C (32 °F) additional
precautions are needed when starting crane
operations. See the instructions in the Operation
and Maintenance for your type of crane.

On a flat surface. The maximum
permissible slope for your crane is
in the technical Data.

▲ DANGER

The crane can move uncontrollably
on a slope of more than
the maximum permissible one.



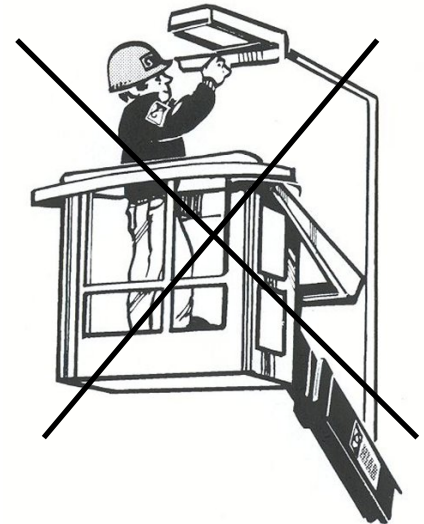
3-4. Use and maximum load

Use your crane only for loading and unloading cargo.

▲ DANGER

Never lift people with the crane!

People may only be lifted by cranes that are approved for it and equipped with a special platform for lifting people for work at heights.



▲ WARNING

Never use your crane for long periods continuously. (unless it has been made suitable for this purpose). Your crane is not a production crane and is not intended for continuous use. Continuous use can shorten its life considerably.

!NOTE

The load plate and the load diagram on your crane show what are the maximum loads you can lift in the operating reach of your crane.

▲ DANGER

Never change the settings of the safety valves.

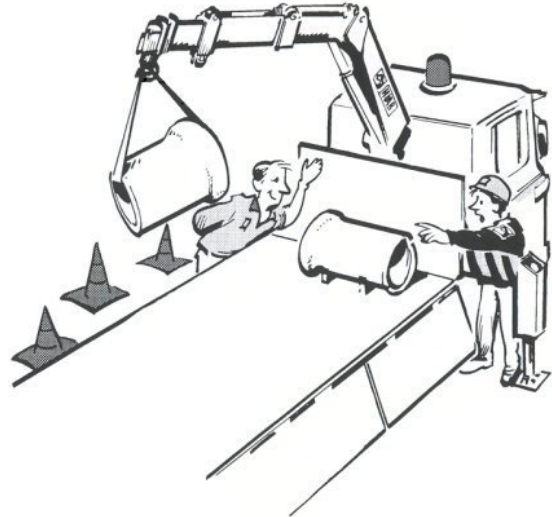
3-5. Preparation for use

▲ DANGER

Ensure that there are no unauthorized persons within the operating range of your crane!

! TIP

Mark out the working range, with for example, cones.



! TIP

Put on your vehicle warning lights.

▲ DANGER

If a part of the crane comes in contact with an electricity line, you will certainly be electrocuted!

▲ Maintain the following minimum distances between the crane and overhead electricity lines unless otherwise prescribed by national rules.



Minimum distance between crane and overhead electricity lines		
Voltage(V)	Minimum distance to an insulated conductor	Minimum distance to an uninsulated conductor
< 500 V	0.5 m	2 m
500 ~ 40000 V	1.5 m	4 m
> 40000 V	2.0 m	6 m
Voltages are found :		
Up to 500V :	to buildings	
500 ~ 40000 V :	trams, trains	
over 40000V :	power transmission	

▲ DANGER

Put on the vehicle parking brake!
Otherwise the vehicle can begin to slide during lifting.

▲ Never use the outrigger legs as a parking brake. Then too, the vehicle could start to slide.

▲ Put on a safety helmet.
(legally compulsory in some countries!)

▲ Check that the ground is sufficiently flat.
Use chocks under the wheels on a gentle slope.



- ▲ Check that the ground is sufficiently firm. Check also that the ground is not undermined. Look out for sewers, cellars, excavations etc.
The outrigger legs must not be able to sink in.

▲ DANGER

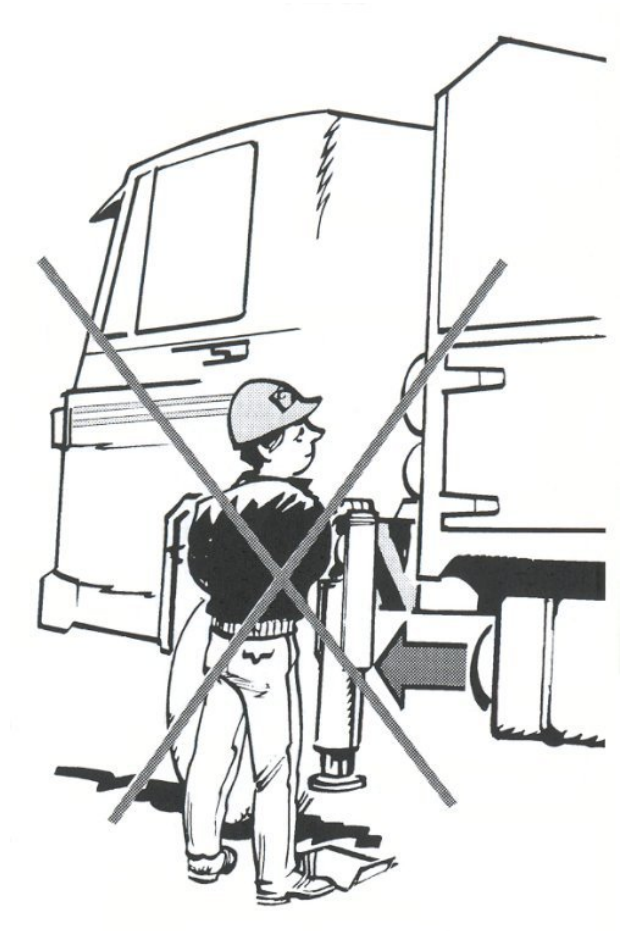
On a slippery surface, for example because of frost :
Place chocks under the wheels.

- ▲ Check that the add-on lifting accessories are in good order!

Add-on lifting accessories are sometimes fitted on the crane (winch, jib) or placed between the boom tip and the load (grapple, rotator).
Separate lifting accessories are connected to the standard load hook (slings, chains, shackles etc).

▲ WARNING

Check that there are no obstacles within the working range.



3-6. Use of the crane

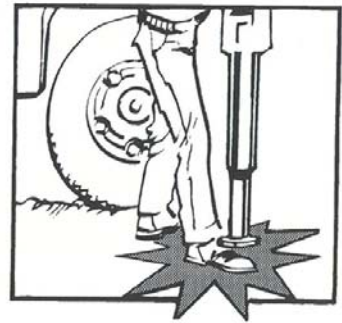
3-6-1. Starting crane operation

▲ DANGER

Ensure that you can see the outrigger legs when you are operating them.

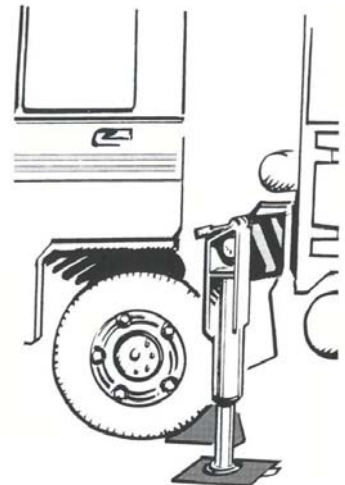
Operate the outrigger legs from the side of the legs concerned.

Do not stand in front of the hydraulically-operated outrigger legs when you are operating them!



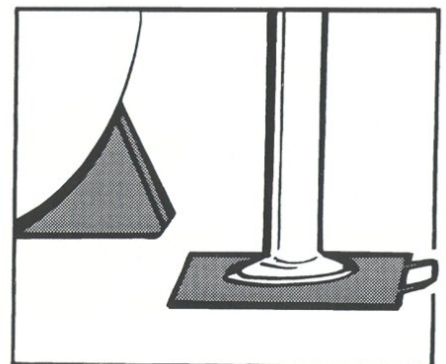
▲ WARNING

Be careful that you do not lower the outrigger legs onto your foot.



▲ DANGER

Where the ground is not firm enough, use support plates under the outrigger legs for additional support. Use support plates that are large and firm enough for your crane. Check that the support plate does not sink into the ground. Check that the support plate does not bend under load.



- ▲ Do not lower the outrigger legs on the edge of an embankment, soft shoulder, slope etc. Lower the outrigger legs only on a flat surface.
- ▲ Slide the outrigger extension beams on both sides of the vehicle out completely. Then lower the outrigger legs for support.
- ▲ If outrigger extension beams are not slide out completely, then the crane can tip over.

▲ WARNING

Place the outrigger legs with low force on the ground.

Only for cranes with 2 outrigger legs :

Do not raise the vehicle! If you raise the vehicle with the outrigger legs this can damage the outrigger legs.

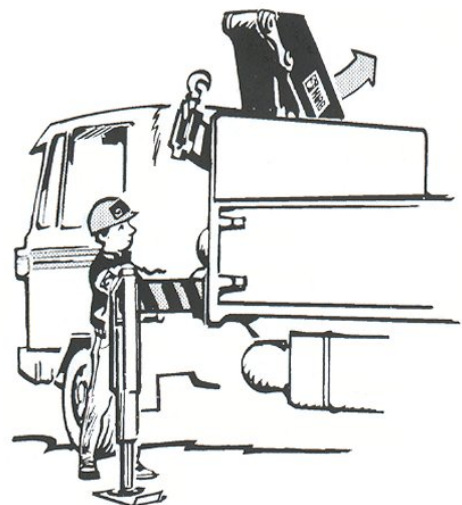
▲ DANGER

Never operate the outrigger legs if the crane has a load.

▲ DANGER

Unfold the crane according to the directions in the operation and maintenance for your crane.

- ▲ Never unfold the loader from the position shown in fig. 5

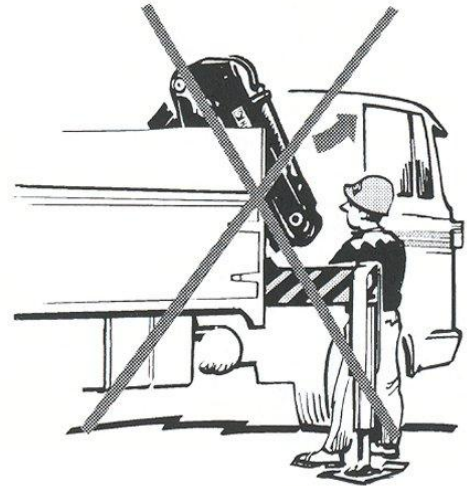


3-6-2. Crane operation

5

▲ DANGER

Keep checking that there is no unauthorized person within the operating reach of the crane!



▲ Make certain that you can always see the load! If your view of the loads is not adequate, have someone else give you signals. Appendix 2 contains a list of signals. Make certain that you and the person Assisting you knows these signals.

▲ Watch out for the safety of the person giving the signals!

▲ Never move the vehicle if you have a freely-suspended load on the crane.

▲ Never walk or stand under a suspended load! When operating never stand below the boom system or load!



▲ WARNING

Never slew at full speed to the final position. This will damage the slewing system.

▲ WARNING

Never push a load along the ground or the vehicle load space with the extension boom. This can cause damage, that will need expensive repairs, to the extension cylinder.

▲ Never use the extension boom as a jack. This could damage the slewing bearings and the connection between the crane column and the crane base.

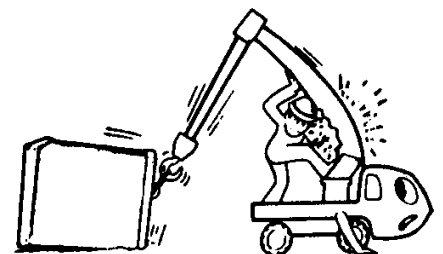
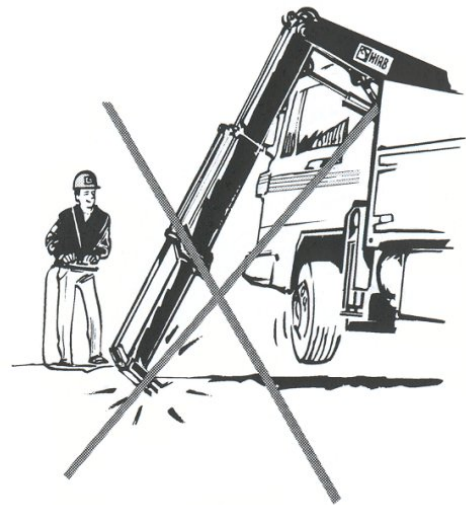
▲ Always lift the load from the ground before you start to slew. Do not tow the load over the ground. This can damage the boom system.

▲ If you are working with loads in tight openings (for example, windows) :

Check that the boom system can move freely upward and downward.

The boom system will bend somewhat when loading and unloading the crane.

▲ If the boom system is in a high position (inner boom above 70°) do not allow the boom to lower at full speed. The crane can go into uncontrolled movement.



▲ When loading the vehicle :

Take the load off the outrigger legs by withdrawing them slightly.

The outrigger legs must remain in light contact with the ground.

▲ Never operate the outrigger legs if the crane has a load.

▲ CAUTION

Operate the crane with smooth and gentle lever movements.

▲ If a cylinder is at its end position, free the operating lever.

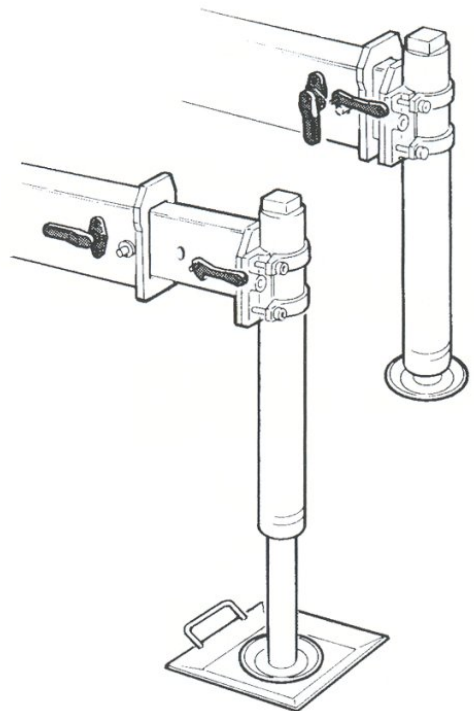
Otherwise you will damage the hydraulic system.

3-6-3. Ending crane operation

▲ DANGER

Always end crane operation as follows :

- Fold up the crane according to the instructions in the Operation and Maintenance for your type of crane.
- Withdraw the outrigger legs and outrigger extension beams.
- Lock the outrigger legs and outrigger extension beams.



- Check that the locking works properly.
- Disengage the P.T.O or power supply after the work.
If you drive with the P.T.O or power supply engaged,
this will cause serious damage to the P.T.O/gearbox combination.
- Only after doing so should you drive away with the vehicle.

▲ DANGER

After use, always place the crane in the transport position!

If you can't park the crane and lifting accessories in the transport positions, but have to park it on the load space or over the load ;
secure the boom and the lifting accessories to prevent any movement
of the crane and the lifting accessories.



3-7. Driving with the crane

▲ DANGER

Never drive the vehicle if there is a load suspended from the crane.

▲ Before you start to drive away :

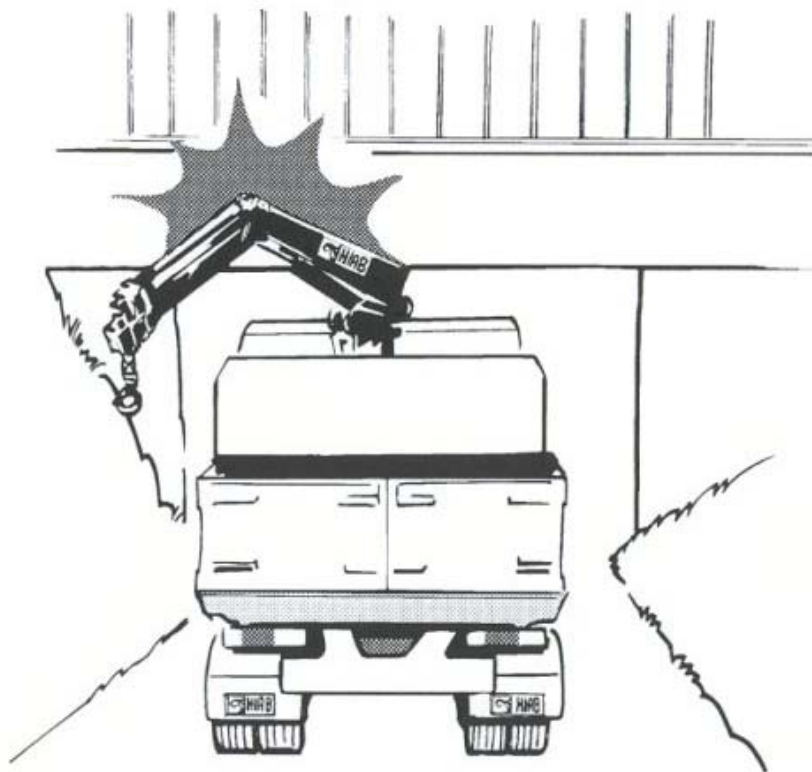
Check that you have no pump flow to the crane control valve.

The PTO or power supply must be disengaged.

▲ Take care that the width and height of the crane in the transport position

can produce no danger because of the headroom of bridges, tunnels etc.

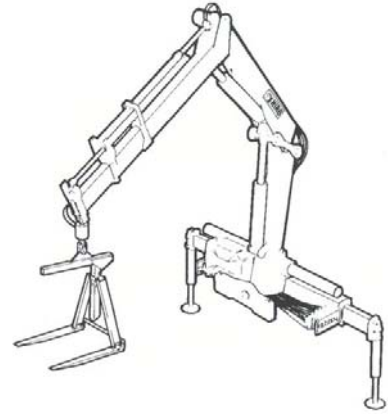
▲ Take care with overhead power lines! Be sure that a part of the crane never comes in contact with overhead power lines.



3-8. Use of lifting equipment

Use only lifting accessories (winch, grapple, rotator) which are suitable for your crane.
Get in touch with a HIAB dealer.

▲ Never attempt to install add-on lifting accessories yourself! Add-on lifting accessories may only be installed by an authorized HIAB dealer.



▲ When using lifting accessories :

Follow the instructions supplied with the equipment! Watch out for hazards!

Never try to adjust lifting accessories when you are working on the crane!

▲ DANGER

After fitting of the lifting accessories :

1. Check that the lifting accessories is securely fixed.
2. Only then should you use your crane.



▲ WARNING

When connecting and disconnecting lifting accessories :

- Clean the couplings.
- Dirt will damage the hydraulic system.
- Take care that your fingers are not trapped.

3-9. Use of demountable cranes

▲ DANGER

Ensure that there are no unauthorized persons in the immediate vicinity of the crane. When putting the crane onto or off the vehicle people can suffer fatal crushing injuries!

▲ After setting up : Check that the crane is properly locked!

▲ WARNING

Be cautious when putting the crane onto and off the vehicle. Putting it on and off roughly can seriously damage the crane or the vehicle.



3-10. Maintenance and service

▲ DANGER

Only carry out the maintenance yourself that is prescribed in the Operation and Maintenance for your type of crane.

All other maintenance work may only be carried out by HIAB service workshop!

▲ Do not do any welding work on the crane yourself! Welding work on the crane may only be carried out by, or in close consultation with, a HIAB service workshop.



▲ WARNING

Before welding on the vehicle

1. Disconnect the MINUS (-) cable,
2. Disconnect the PLUS (+) cable.
3. Place the earth connection as close as possible to the welding spot.
4. After this the welding work can be carried out.

After welding on the vehicle

1. Connect the PLUS (+) cable
2. Connect the MINUS (-) cable

▲ DANGER

Do not drill into the crane yourself.

Drilling work on the crane may only be carried out by,
or in close consultation with, a HIAB service workshop.

▲ Never try to reinstall the crane.

Only a HIAB Dealer. may reinstall the crane.

▲ WARNING

Carry out the periodic maintenance according to
the Operation and Maintenance for your type of
crane.



!NOTE

Always use original HIAB parts and tools.

▲ WARNING

Take the crane at least once a year
to a HIAB service workshop for
inspection and maintenance.

▲ Maintain lifting accessories
according to the supplier's
instructions.

▲ Clean your crane and accessories regularly, but :

- Do not use any aggressive cleaning agents.
- Never use a high-pressure cleaner on the control valves, cylinders, oil tank or on electronic parts. Only the crane surface may be cleaned with a high-pressure cleaner

▲ WARNING

Remember the environments.

- Do not spill oil when topping-up.
- Dispose of waste oil and oil filters to local regulations.

3-11. Faults

▲ DANGER

Ensure that faults in the crane are rectified immediately!

Follow the instructions for faults, in the Operation and Maintenance for your type of crane, exactly.

Never rectified faults yourself that may only be rectified by a HIAB service workshop.

▲ DANGER

If there is an oil leakage from the hydraulic system :

Keep well away from an oil leak.

The oil spraying out can cause serious injuries.

The oil in the hydraulic system is at a high pressure.

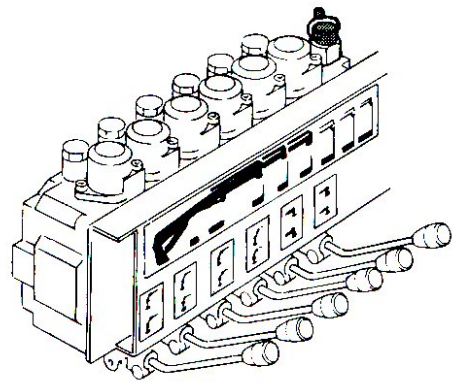
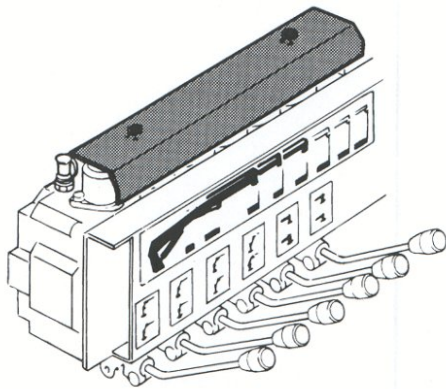
1. Disengage the P.T.O or power supply.
2. For further instructions see the Operation and Maintenance for your type of crane.

3-12. Warranty

!NOTE

HIAB only provides a warranty if :

- The instructions for maintenance and service have been followed, and
- Original HIAB parts have been used, and
- All the seals are still intact and
- The maintenance schedule in your Crane logbook is filled in up-to-date.

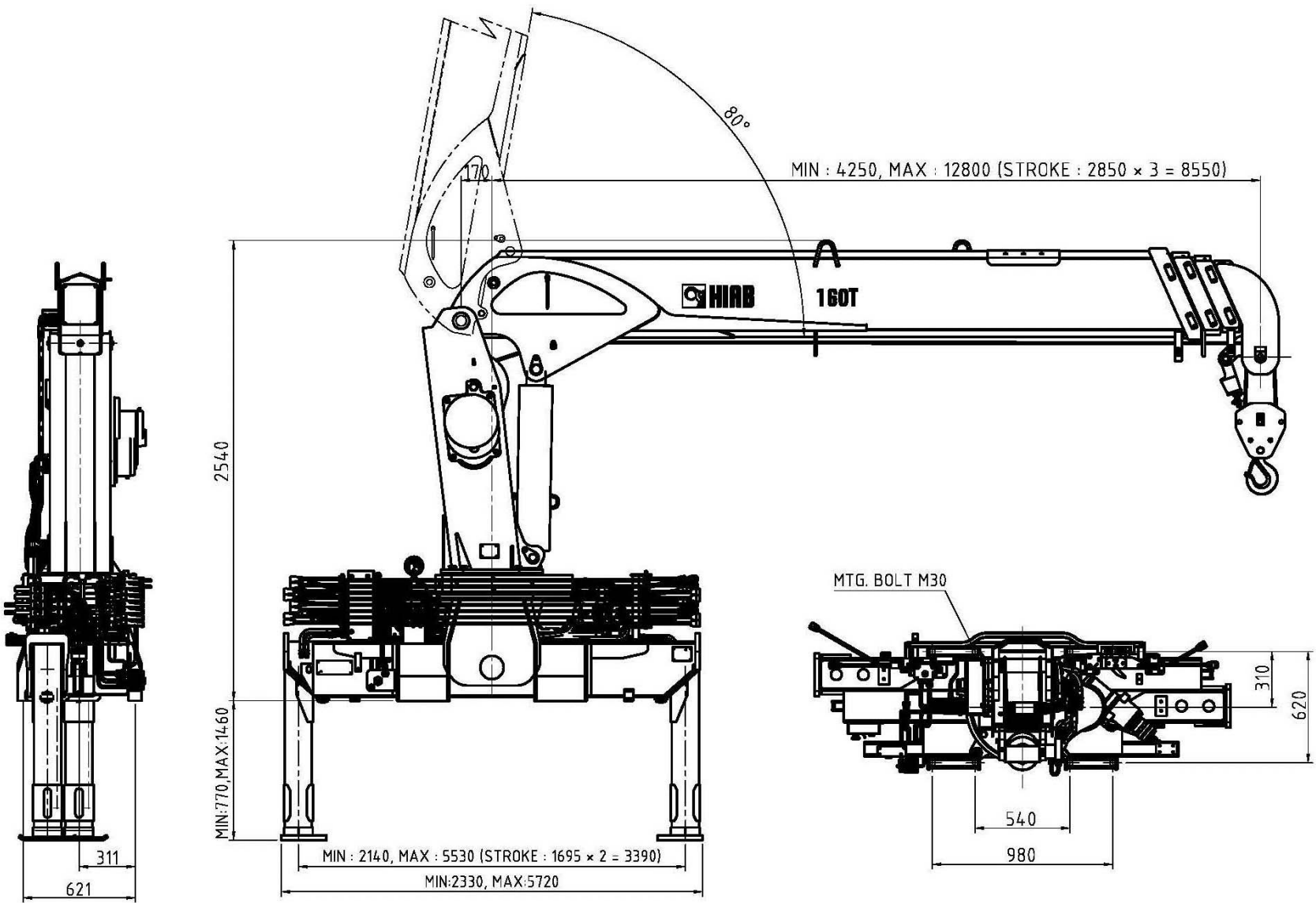


4. Technical description

4 - 1 Specification

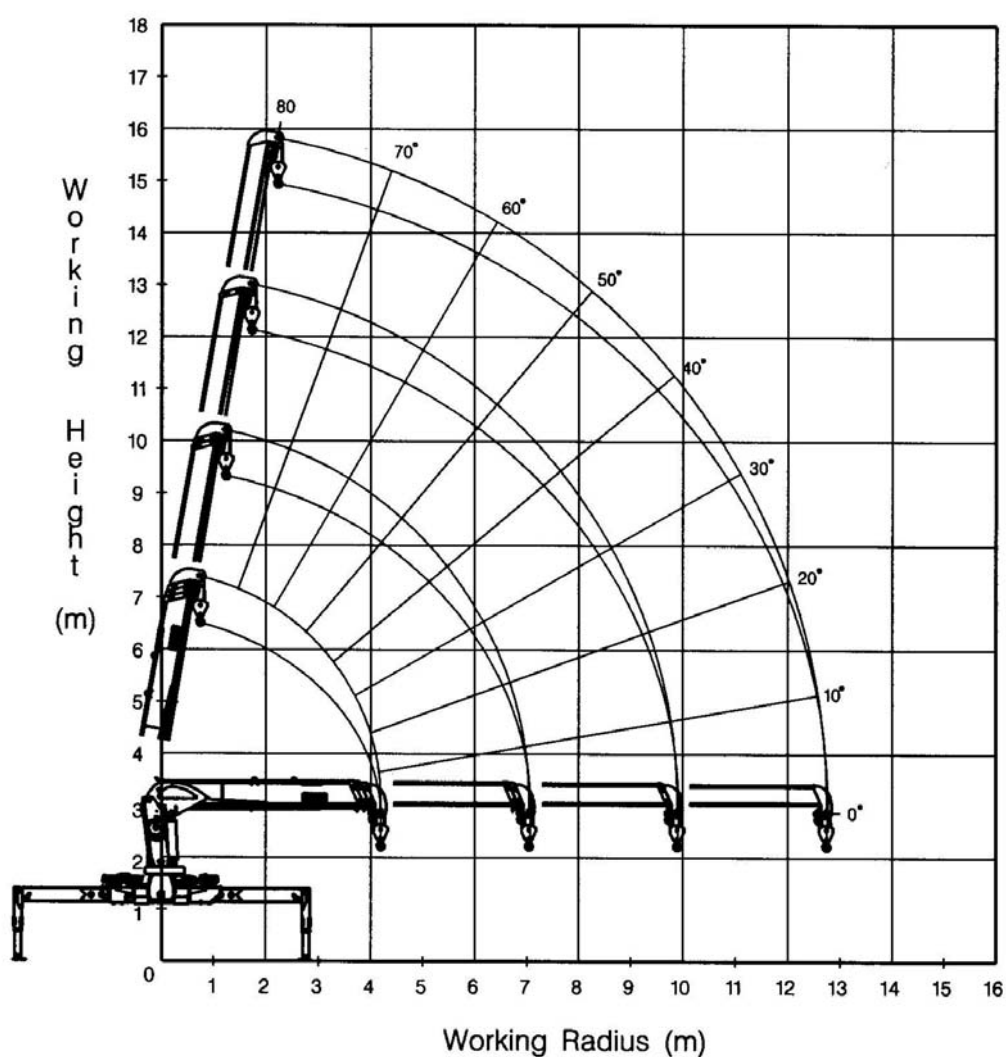
Description		Specification	Remarks
Lifting Capacity		15.5 ton-m	
Working Radius		12.8 m	
Outreach - Lifting Capacity		4.25 m - 3,110 kg 7.10 m - 1,600 kg 9.95 m - 1,040 kg 12.80 m - 740 kg	
Working Height		14.0 m	
Recommended oil flow		56 ℓ / min	
Working pressure		200 bar	
Boom	Type	Hex. 4 sections	
	Lifting angle/speed	0~80 deg / 13 sec	
	Extension speed	11.4 m / 34 sec	
Winch	Type	Spur gear reduction type with mech. Brake	
	Lifting capacity	6500 kg (4 falls)	
	Speed	min. 9.5 m/min ~ max. 12 m/min	
Slewing	Type	Worm gear reduction type with hy'd brake	
	Angle	360 deg continue	
	Speed	2.0 rpm	
Outrigger Span		5.7 m	
Safety devices		Counter balance valve Pilot check valve Pressure relief valve Hydraulic swing brake Hook latch Aut winch brake	
Weight		2,650 kg	with outrigger

4 - 2 Outline view

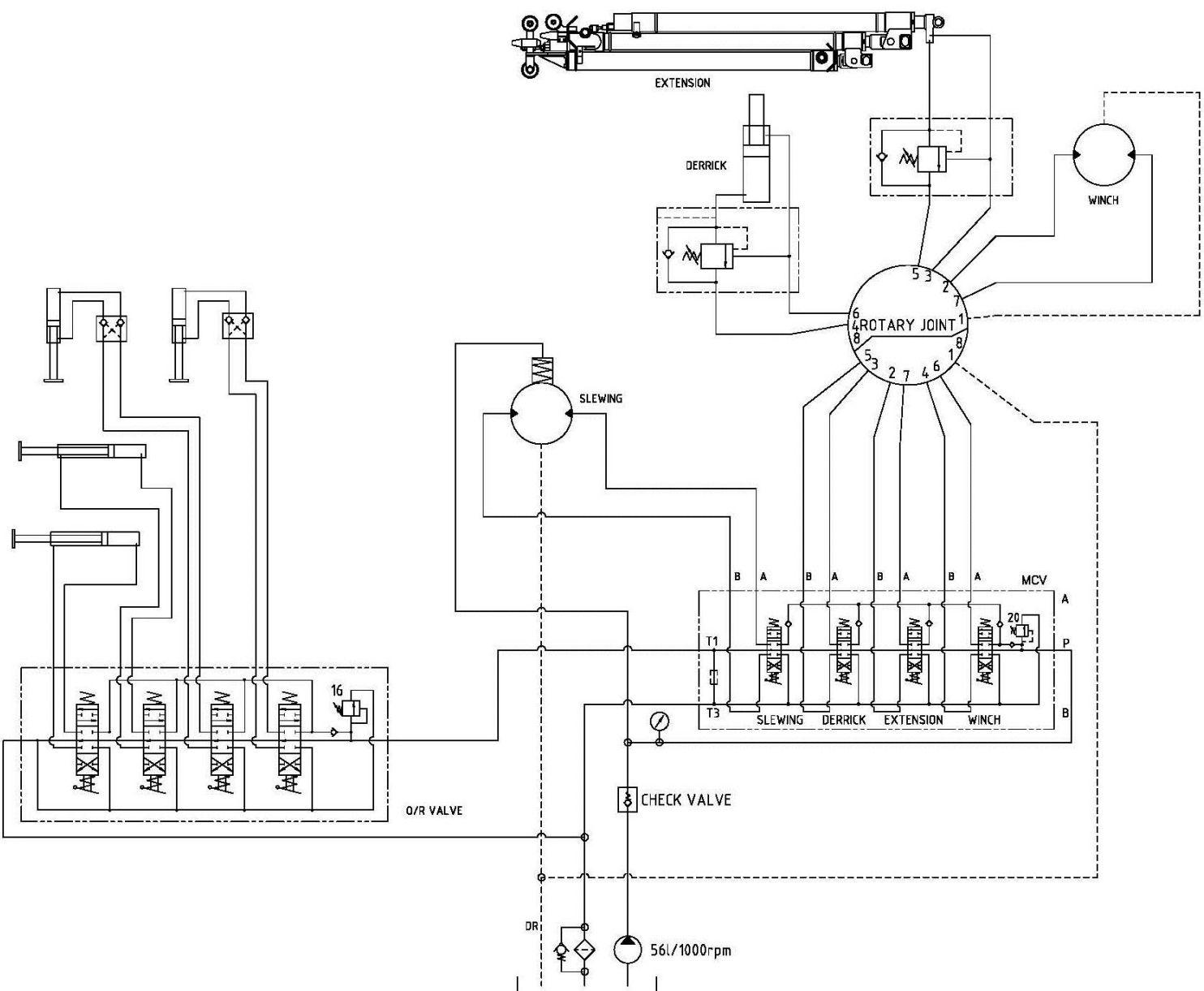


4 - 3 Lifting capacity table & working radius

boom angle	1 st boom		2 nd boom		3 rd boom		4 th boom	
	R(m)	Q(kg)	R(m)	Q(kg)	R(m)	Q(kg)	R(m)	Q(kg)
80 deg	0.60	6,500	1.09	6,000	1.59	4,500	2.08	2,400
70 deg	1.34	6,500	2.31	4,640	3.29	3,220	4.26	2,400
60 deg	2.04	6,090	3.46	3,410	4.89	2,340	6.31	1,760
50 deg	2.67	5,070	4.50	2,790	6.33	1,890	8.16	1,410
40 deg	3.21	4,460	5.40	2,420	7.58	1,620	9.76	1,200
30 deg	3.65	4,040	6.12	2,160	8.59	1,440	11.06	1,060
20 deg	3.98	3,710	6.66	1,960	9.33	1,300	12.01	950
10 deg	4.18	3,420	6.98	1,780	9.79	1,170	12.60	850
0 deg	4.25	3,110	7.10	1,600	9.95	1,040	12.80	740



4 - 4 Hydraulic diagram

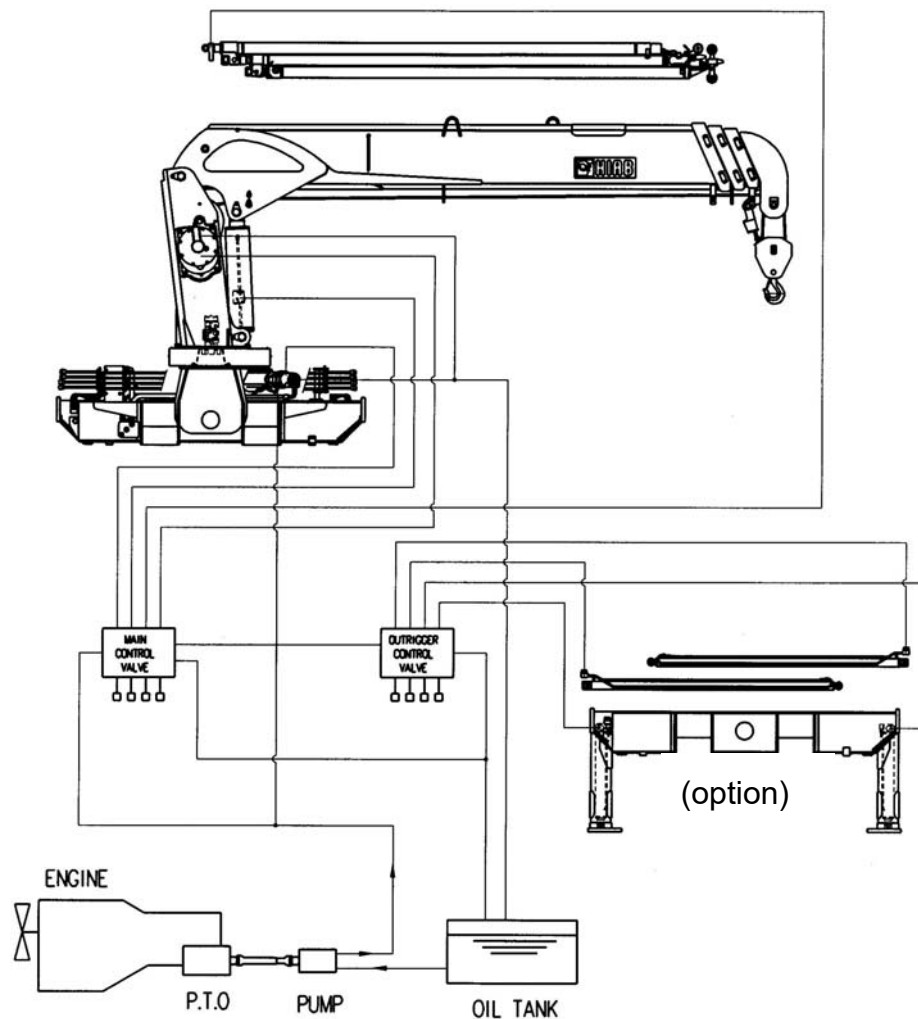


5. General description

5 - 1 Basic operation principle

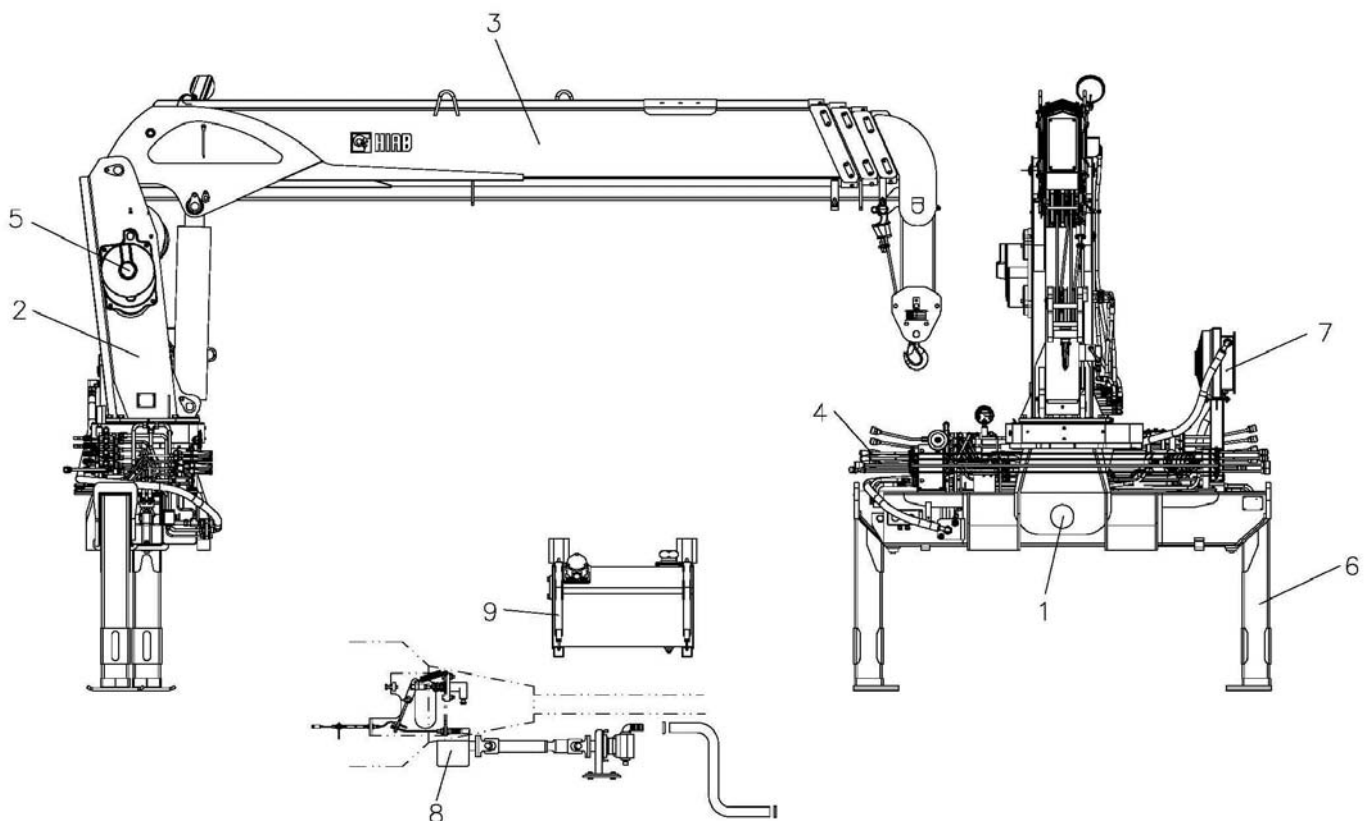
Hiab 160T-4 crane consists of mainly steel structures and hydraulic components.

The required power source for crane operation comes, via the chassis' power take off (PTO), from the hydraulic pump giving high pressure hydraulic oil. This high pressure oil operates the cylinders and motors through the control valve and returns to the oil tank.



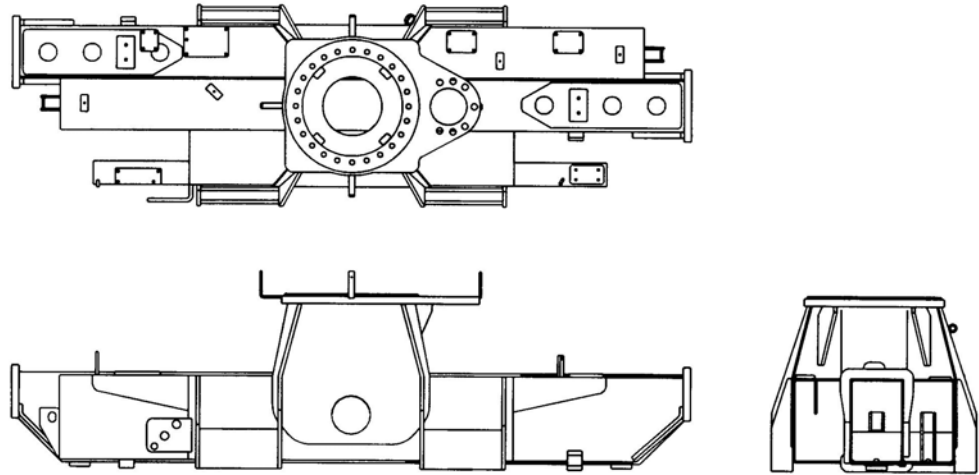
5 - 2 Main components

- 1) Base
- 2) Column
- 3) Boom system
- 4) Control valve
- 5) Winch
- 6) Outrigger system
- 7) Oil cooler
- 8) PTO & Pump
- 9) Oil Tank

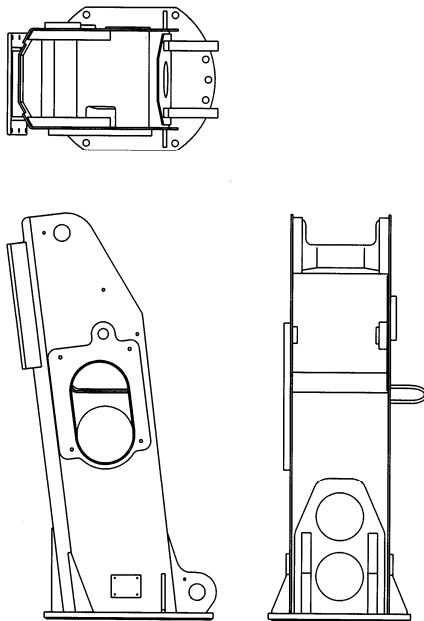


5 - 3 Base

The base is connected with the column structure by slewing bearing, which makes the crane rotate, and is mounted on the chassis frame with strong anchor bolts.



5 - 4 Column

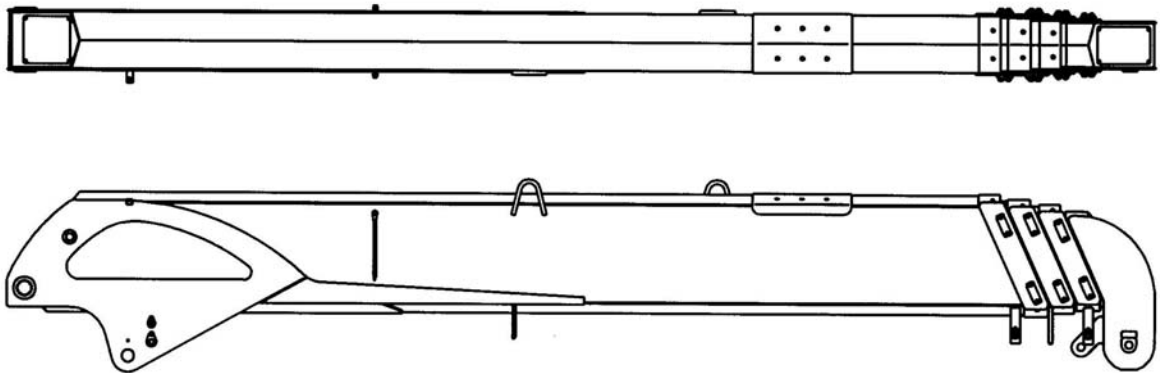


The column is of a closed welded box design and connected with the base. The column supports the boom system via derrick cylinder. The winch system is mounted in the column structure.

5 - 5 Boom system

The boom system consists of main boom, four extension booms and extension cylinders moving each boom.

The three cylinders are of a double acting type. The counter balance valve of each cylinder makes the cylinders operate at constant speed regardless of the suspended load.



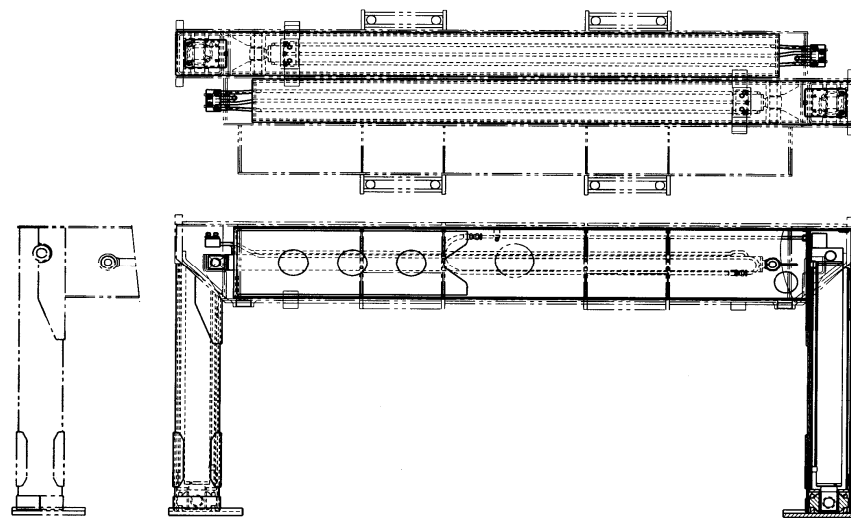
5 - 6 Control Valve

The Hiab 160T stiff boom crane is equipped with two(2) 4-function control valves connected serially.

One is for main controls of the crane (slewing, boom lifting, extension, winch), and the other is for the outrigger control (outrigger beam extending, outrigger jack cylinder). Both valves are equipped with built-in main relief and port relief valves shown in the hydraulic diagram. The relief valves makes that the crane can not lift in excess of the designed load chart.

5 - 7 Outrigger system

The outrigger system consists of outrigger beam and legs which make the chassis stable when the crane is operating. The pilot check valves are installed on leg cylinders to prevent the leg cylinders to go down when driving and from hose failure when operating the crane. Before operating the crane, spread the outrigger and settle the outrigger firmly on the ground to prevent tilting or too much stress on tires and frame of the chassis.



5 - 8 Winch

The winch consists of hydraulic motor, gearbox and winch drum. The winch is installed inside of the column structure. The mechanical brake is included in winch gearbox.

5 - 9 OLP (over load protection) system [option]

The OLP system is a fully hydraulic concept that stops the main hydraulic flow when the pressure in the derrick cylinder has reached the maximum allowable level, preventing aggravating movements.

The safety system resets automatically after a pressure decreasing movement.

Although the safety system is triggered by the pressure of the derrick cylinder only, it prevents all aggravating movement. Only the extension can be moved inwards as an exception, to solve the overload situation. The crane and vehicle have a guaranteed stability.

5 - 10 Anti two block warning and cut-off system [option]

Hiab crane is equipped with an anti two block alarm and cut-off system as an option. If the hook block is too near the boom tip, the alarm and cut-off system will be activated.

- Before operating the crane, turn on the anti two block alarm and cut-off switch.
- If the alarm is activated, do not operate the boom extension and hook lifting, however can operate the boom retraction and hook lowering.
- After operating the crane, turn off the anti two block alarm and cut-off switch.

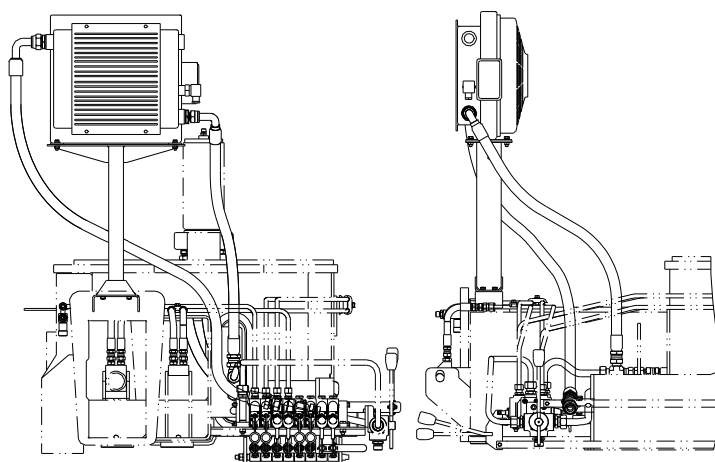
(note) Do not shorten the wire length.

Please be careful when the extending boom or the raising hook, because the alarm system can not be activated when of the electric wire is damaged etc.

If the switch is damaged or electrical wire is cut, do not operate winch up and boom extend.

5 - 11 Oil cooler [option]

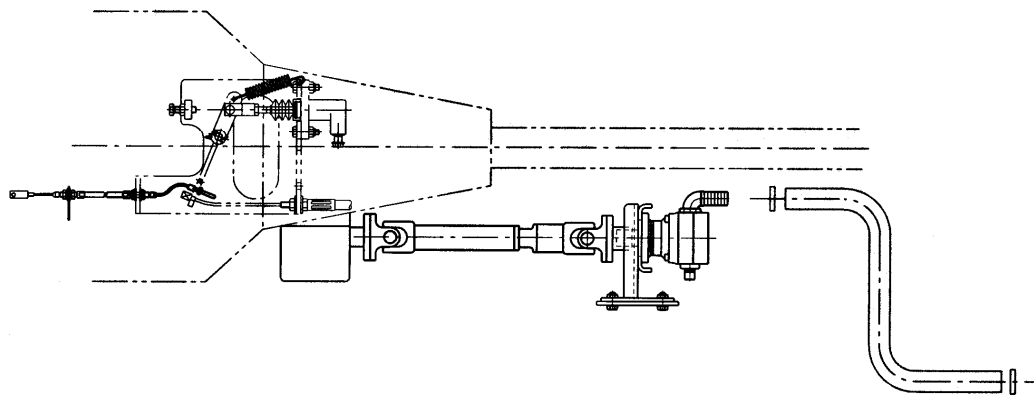
A equipment that is mounted on the base structure and cools the elevated oil temperature down with water or air.



5 - 12 PTO & pump [option]

The PTO is mounted the gearbox of the chassis. This rotating power rotates the pump creating high hydraulic oil pressure. The cylinders and motors are activated by this high pressure oil.

The PTO is designed according to the chassis, so you must consult the chassis manufacturer when you select the PTO.



5 - 13 Oil Tank [option]

The Hiab 160T crane is equipped with an oil tank for separate mounting.

The oil tank consists of tank body itself, return filter, suction strainer, air breather, oil level gauge and drain plug which can be used for drainage of the water or some dirt accumulated in the tank bottom.

6. Operation

6 - 1 Before operating the crane

- Secure the hand brake of the chassis
- Check the oil level in the hydraulic oil tank
- Engage the PTO
- Extend the outrigger beam and down the outrigger legs firmly on the ground

6 - 2 Warming up when cold weather

In cold weather, the viscosity of the operating oil could be very high. It can make some damages to the hydraulic components. Please warm up as follows.

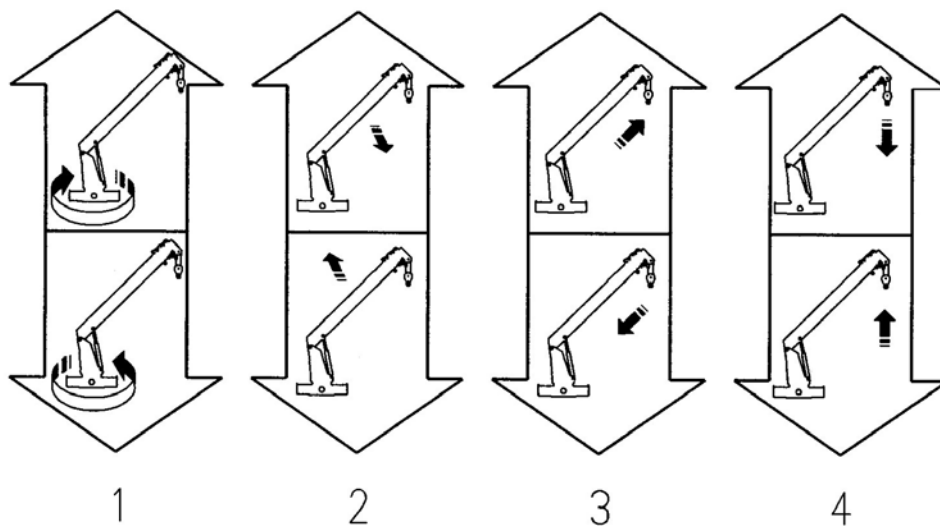
- Engage the PTO and run the hydraulic oil pump at low RPM.
- After 10-15 minutes without movement of the control valve, operate the crane slowly.

6 - 3 Operating of crane

Standard control valve 4 levers for operating crane.

- Before operating the crane, adjust the four (4) way valve lever to main control valve side.
- Please refer to below chart explaining the function of each lever.

NO.	Operated components	Pull	Push
1	Slewing	Left	Right
2	Derrick	Up	Down
3	Extension	Retracting	Extending
4	Winch	Up	Down



6 - 4 Operating the outrigger

The outrigger control valve levers are vertically installed on both sides of the crane.

The first lever : front left side shift cylinder

The second lever : front right side shift cylinder

The third lever : front left jack cylinder

The forth lever : front right jack cylinder

The fifth lever : rear left side shift cylinder

The sixth lever : rear right side shift cylinder

The seventh lever : rear left jack cylinder

The eighth lever : rear right jack cylinder

(All directions here specified when you are looking from rear towards the truck cab)

- Extend the outrigger beam
- Down the outrigger legs firmly on the ground

The stability is up to the length of outrigger extension.

Be careful not to lift up the chassis ' tires.

Please avoid sloped grounds.

the load is clear of the ground.

6 - 5 Boom raising / lowering

- Boom raising

Move the first lever towards the truck cab(up)

- Boom lowering

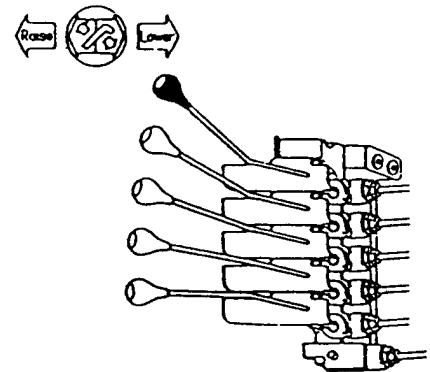
Move the first lever to rear side(down)

- Stopping

Put the first lever in neutral position

(note) Move the lever slowly when loaded.

Fast movement could cause tilting of the chassis or damages to the crane due to dynamic forces.



6 - 6 Hook raising / lowering

- Hook raising

Move the second lever towards the truck cab (up)

- Hook lowering

Move the second lever to rear side(down)

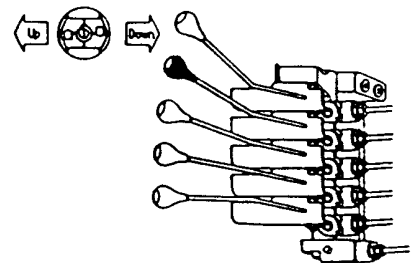
- Stopping

Put the second lever in neutral position

(note) Do not lower more when load touches the ground. If you do so, the wire can get tangled.

When load is lowered, minimum 3 turns of wire should remain on the drum.

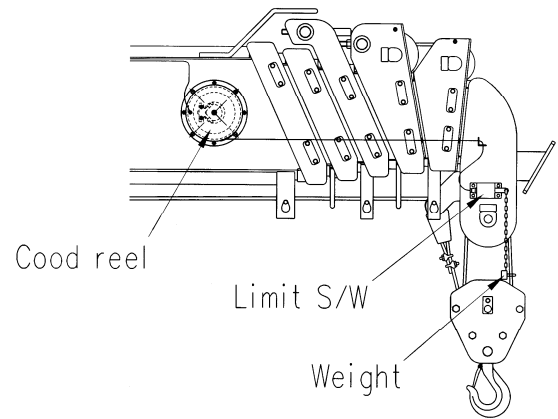
To keep good laying of the wire, wind slowly.



6 - 7 Overwinding alarm

Hiab 160T is equipped with an overwinding alarm system. If the hook block is too near the boom tip, the alarm system will be activated.

- Before operating the crane, turn on the overwinding alarm switch.
- If the alarm is activated, retract the boom or lower the hook.
- After operating the crane, turn off the overwinding alarm switch.



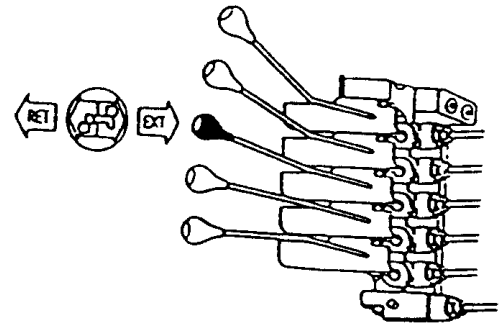
(note) Do not shorten the wire length.

Please be careful when the extending boom or the raising hook, because the alarm system can not be activated when of the electric wire is damaged etc.

6 - 8 Boom extension / retraction

- Boom extension

Move the third lever towards the truck cab(ext)



- Boom retraction

Move the third lever to rear side(ret)

- Stopping

Put the third lever in neutral position

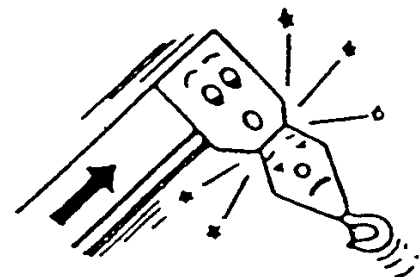
(note) When extending, first 2nd boom is extended and then 3rd boom is extended step by step.

4th are extended simultaneously.

When retracting, operated conversely.

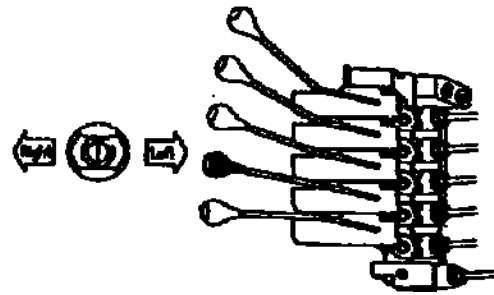
Be carefully aware of the hook movement.

When extending, hook is up and when retracting, hook is down.



6 - 9 Slewing

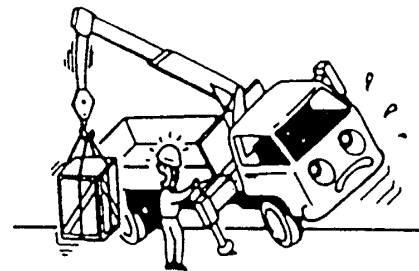
- Counter clockwise
Move the forth lever towards cabin(right)
- Clockwise
Move the forth lever to rear side(left)
- Stopping
Put the forth lever in neutral position



(note) Slew as slow as possible

When you rotate the crane to the truck cab position, be careful about the stability.

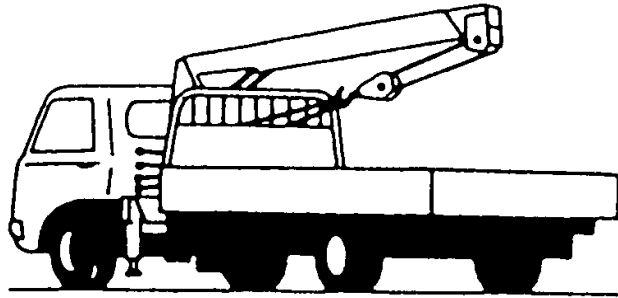
Never slew until the load is clear of the ground.



6 - 10 After operation the crane (for driving)

- Retract the booms fully
- Rotate the boom at the driving position
(front or rear side. Refer the sticker on slewing cover and column base plate)
- Retract the outrigger legs fully
- Retract the outrigger beam fully

- Lower the boom at lowest position
- Settle the hook with the safety rope on the chassis
- Operate (wind) the winch until the safe rope is tensioned
- Put manual accel lever put at neutral position
- Disengage the PTO, then the pump stop to run



(note) Driving with the pump running may cause PTO and pump damages.
It's very dangerous to drive without the hook in a fixed position
Check again that the outrigger system is fully retracted and locked.

7. Safety instruction when operating

- Press the clutch pedal fully and then engage or disengage the PTO
- When cold weather, please warm up the operating oil for 10 - 15 minutes
- Operate the crane on even ground condition
- Before operating the hook, turn on the overwinding alarm switch
- Be careful not to touch the hook block on the ground, which could cause the wire to get tangled
- Do not drag the load using the crane which may cause damage to the crane
- Do not exceed the lifting capacity shown in the load table
- Slew as slow as possible
- When load and unload in underground, check the remaining wire length on the winch drum. Minimum 3 turns.
- Do not leave from the operating position when load is suspended
- Do not be allowed that unauthorized persons is in the working area
- Make sure that the hook is firmly fixed and the PTO disconnected before driving
- You should consult your dealer and/or trained person if you find some problems

- Operate the crane smoothly

If operated fast, the life cycle of the crane can be reduced and/or there can be some component troubles.

At starting/finishing point, the movement should be reduced.

- Before operating the crane, make sure the outrigger system is extended and firmly settled on the ground. If the ground is not rigid, use auxiliary support plate under the support leg plate.

(note) In case low rpm of the engine, boom can

tremble and/or do not move. In this case a little increase of the engine rpm can make it possible.

The oil temperature should be kept below 80° C.

High oil temperature can cause damages to the hydraulic components.

From 45°C to 55°C is the best operating temperature for the crane.

8. Preventive service & Maintenance

8 - 1 Daily Inspection

Where?	What?
Power take off	● Abnormal noise and/or heat? ● Engaging movement is smooth or not?
Drive shaft	Loosened or not?
Oil tank	● Oil level? ● Leakage?
Slewing system	● Leakage? ● Greasing condition? ● No loosened bolts? ● Abnormal noise?
Derrick cylinder	● Shaft condition? ● No loosened bolt? ● Leakage? ● No movement when stopped?
Boom system	● No loosened bolt? ● Any crack? ● Smooth extending/retracting?
Winch gear box	● Oil leakage? ● Abnormal noise?
Pump	● Mounting condition? ● Leakage? ● Abnormal noise?
Hook	● Rotate when loaded? ● Any damages?
Wire rope	Any damage?
Control lever	● Smooth movement? ● Return neutral position well when release the lever?
Outrigger system	● Any crack? ● Leakage of the cylinder? ● Any internal leakage?
Control valve	● Any leakage? ● Smooth movement?
Overwinding alarm	Alarm when activated?
Hook latch	Enough spring tension?
Pipes & hoses	Any leakage?
Base	Mounting bolt condition?

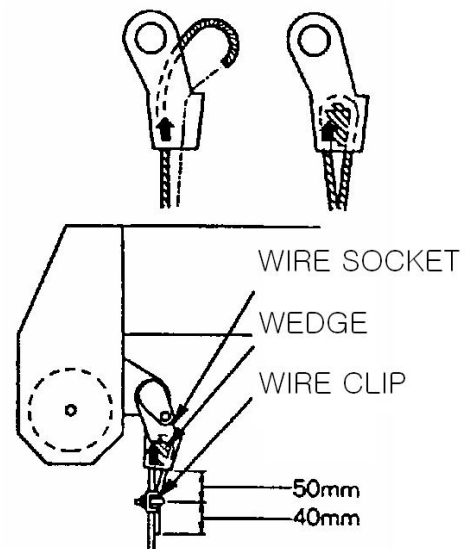
8 - 2 Slewing bearing & mounting bolts

If there is some abnormal noise from the slewing system and/or the mounting parts when operating the crane or driving, please consult your Hiab dealer or service center. That kind of noise could come from loosened condition of the mounting bolts. At least, mounting bolts of slewing bearing located inside of the column should be checked every 2 years.

8 - 3 Wire rope

1) How to fit the wire rope

Pass the end of a wire rope into the wire socket along the arrow mark on the socket. Do not forget to attach the wedge and wire clip. When attaching the wire socket to the boom, make sure that the arrow mark on the face of the wire socket faces outwards as shown the in drawing.



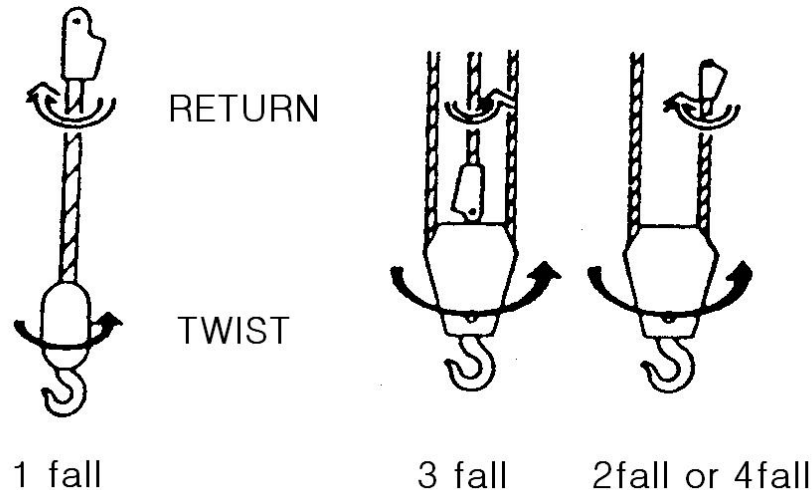
2) Adjusting twisted wire rope

Wire rope tends to turn in the untwisting direction under tension. When the hook carries two or more wire rope slings, they can easily turn, especially when they are still new. Wire rope becomes steady with use.

The wire rope adjustment procedure is:

- Extend the boom fully
- Set the boom to an angle of about 65 deg
- Hoist a load so that the wire ropes will be tense
- Check the wire ropes for the number of turns
- Remove the wire socket, and turn it in the wire rope untwisting direction by the number of wire rope turns multiplied by the number of wire rope slings.
- After attaching the wire rope socket back in position, wind up and down the wire ropes all the way and repeat it a few times to check that they are back to normal. If there is still a twist remaining, repeat the above process.

- Turn the wire socket opposite to the wire rope twisting direction, except in the case of 3-rope sling in which case the wire socket should be turned in the same direction.



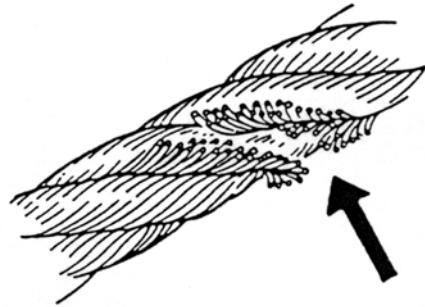
3) How to look after the winch rope

The winch rope must be regularly cleaned and greased for maximum life, bearing in mind that the rope is subject to wear and to heavy stresses which set a limit of its service life. It is good practice to replace the rope at regular intervals by a new one. The gradual wear and unravelling of the rope is difficult to detect and to gauge.

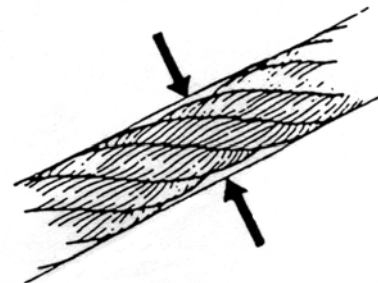
However, the following safety rules applying to winch ropes can be followed.

The winch rope must be changed as soon as any of the following occur;

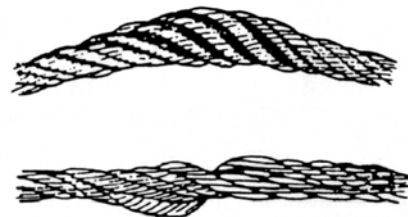
- When 5 % of wires in any length of ten rope diameters are broken, worn or corroded



- When visible changes occur in the diameter (more than 7 %) of the winch in comparison with the normal stretching and diameter reduction of a newly fitted rope



- The rope develops kinks or a heavy spiral twist



4) Lubricating a winch wire

when the rope is used for a long period it becomes dry, which gives rise to rapid wear and to rust attacks.

Accordingly, it is necessary to lubricate or grease the rope at regular intervals.

Lubrication must be carried out as follows:

- Clean dust and dirt off the rope and wipe it dry
- Apply an appropriate amount of a recommended lubricant with a brush, rag or similar implement. Be sure that only a rope lubricant is used.

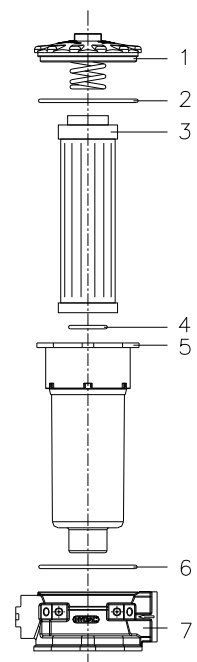
8 - 4 Changing a filter insert

A clean hydraulic system gives better operating economy. The oil filter insert should therefore be changed for the first time after 50 running hours and from then on every 500 running hours, subject to a minimum of at least two changes a year.

For best results, always fit a genuine HIAB insert.

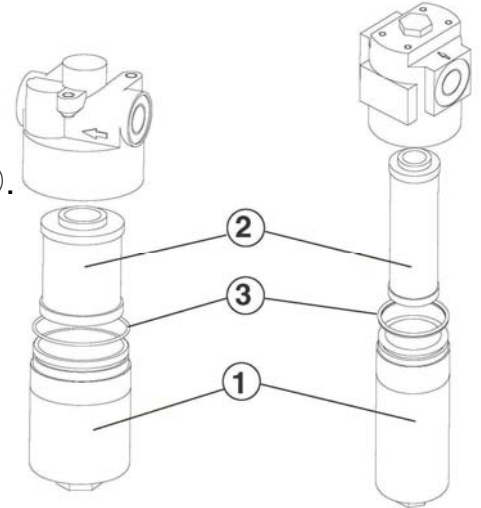
Replacing the return oil filter insert

- Clean the immediate surroundings of the return filter.
- Turn the cover ① to free it while pressing on it
- Pull the return filter out of the oil tank
- Remove the filter insert ③
- Insert a new filter insert
- Fit new sealing rings ②, ④
- Replace the return filter in the tank
- Attach the cover on the return filter



Replacing the filter insert in the pilot filter and in the high pressure filter

- Clean the immediate surroundings of the pressure filter.
- Dismantle the filter housing ①.
- Remove the filter insert ②.
- Clean the filter housing.
- Fit a new filter insert. Also fit new sealings ③.
- Reassemble the filter housing.



▲ WARNING

Dirt will damage the hydraulic system

8 - 5 Inspection points when operating

- Is there any leakage?
- Does the crane operate smoothly?
- Is there any abnormal noise?
- How about the oil temperature?

If the oil temperature is higher than 75 deg C, then stop the crane operation

8 - 6 Inspection after operation

- Clean & smooth for next use?
- Any crack & abnormal noise?
- Any leakage in connections?

8 - 7 Inspection & maintenance after one month in operation

- Check the pressure settings and lead seals
- Check for oil leakage
- Check and advance screwed connections
- Check and advance hose and pipe couplings
- Check catches and other locking devices
- Check the functions and levers of the control valve
- Check hook, ropes and any other lifting gear that is used
- Check that all the prescribed notices are in place
- Carry out a visual inspection of structural parts to detect deformation and cracks etc.

- Cleaning, replacement of filters
- Check oil levels and lubrication in accordance with the lubrication chart in the manual
- Put the crane through test-running and test-loading, and listen for any suspicious noises

8 - 8 Inspection & maintenance after 6 months in operation

- Clean the crane
- Check pressure settings and lead seals
- Check for oil leakage
- Check and advance screwed connections
- Check and advance hose and pipe couplings
- Check catches and other locking devices
- Check the functions and levers of control valve
- Check hook, ropes and any other lifting gear that is used
- Check that all the prescribed notices are in place
- Carry out a visual inspection of structural parts to detect deformation and cracks etc.
- Clean or replace the filters
- Check oil levels and lubrication in accordance with the lubrication chart in the manual
- Put the crane through test-running and test-loading, and listen for any suspicious noises
- Do a security check on the mountings of the pump and anchor bolts
- Clean or change the hydraulic oil

9. Lubrication

9 - 1 Hydraulic oil changing

Check the level in the tank every day. The oil level must be visible in the sight glass. At this check, the crane must be in parked position. Top up with hydraulic oil if necessary.

Hydraulic oil becomes contaminated in time, and malfunctions may begin to appear in the hydraulic system. You should therefore change the hydraulic oil at least once a year.

After changing the oil, you must operate all the hydraulic functions of the crane through their full range. The reason for this is to expel any air that is present, which would otherwise destroy the seals in the hydraulic system.

Always bear in mind that trouble can be minimized by having a clean hydraulic system. For this reason, keep the surroundings clean whenever you have to work inside the hydraulic system.

When changing the oil, change the filters at the same time.

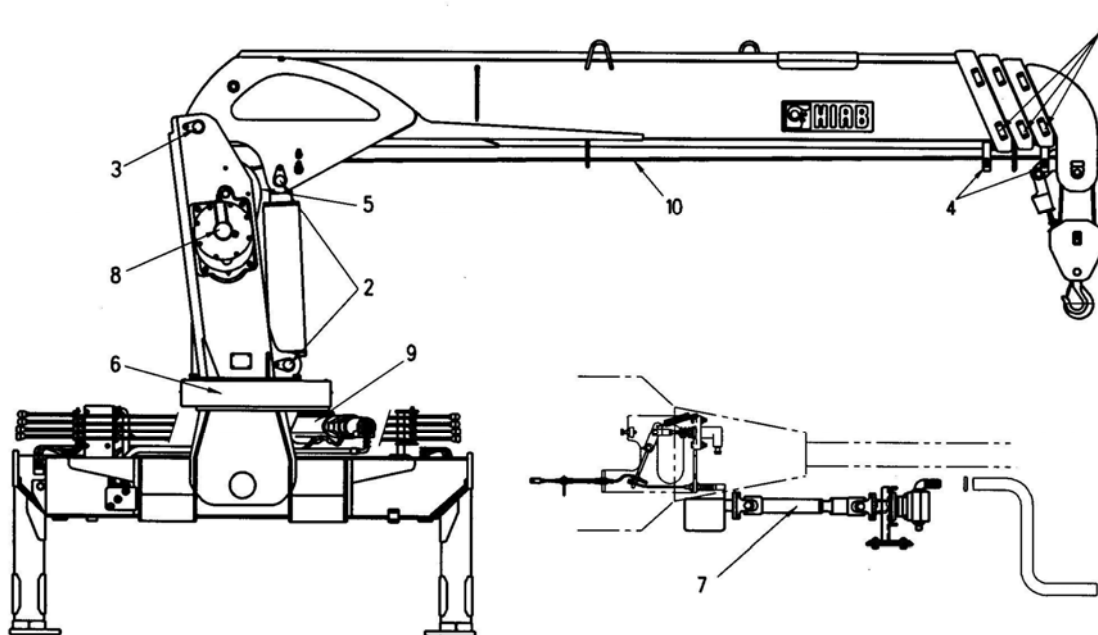
The hydraulic system and the hydraulic fluid are matched in respect of lubricating performance, effect on the seals and other materials, and non-inflammability. For this reason, do not mix different types of hydraulic fluid, such as mineral oils, synthetic fluids and water-based fluids, and never adulterate your hydraulic fluid with diesel oil or alcohol-based products.

Recommendable oil(or equivalents)

- Shell Tellus oil T 46
- Mobil DTE 15M
- Esso Unavis N 46
- Castrol Hyspin AWH-M 46
- Total Equivis ZS 46

9 - 2 Lubrication chart

Good performance and trouble-free operation of the crane depends a great deal on service. For this reason, always follow the recommendation on the lubrication chart.



Pos	Interval	Point	Rec. Lubricant	Q'ty
1	Every day	Boom pads	Shell Alvania EP grease 2 (or equivalents)	6
2		Derrick cylinder shaft		2
3		Boom + Column shaft		1
4		Wire rope roller		2
5	Every week	Winch drum gear	Shell Alvania EP grease 2 (or equivalents)	1
6		Slewing bearing		2
7		Drive shaft		1
8	Every month	Winch gearbox	Shell Spirax HD Gear oil	1
9		Slewing gearbox	Shell Spirax HD Gear oil	1
10		Wire rope	Shell Alvania EP grease 2 (or equivalents)	1

10. Fault Tracing

Fault tracing is mainly based on organized routine for finding a certain functional disorder, its reason and how to correct it.

The following charts contain some faults which may arise during operation, how they appear and which steps to be taken.

Most important, however, is that the crane is checked and necessary adjustments made before the crane is taken into operation.

Some common damages and disturbances:

- The crane is incorrectly used in a way for which it is not designed
- The crane and/or its accessories have not been properly maintained
- The crane is operated against obstacles

Fault	Possible reason	Measure
Overwinding alarm system will not be activated	● Low battery voltage ● Fault in buzzer ● wire disconnect	● Change the battery ● Change the buzzer ● Connect
Still running the engine, but the crane will not operate	● PTO disconnected ● Lack of operating oil ● Low setting of the relief valve ● Damage in pump	● Check the PTO ● Top up the oil ● Check the relief valve setting pressure ● Repair or change pump
Noise in power take off	● Damages in bearing and/or gear ● Damage in drive shaft	● Repair or change Change the drive shaft
Noise in pump	● Lack of oil ● Air leaking in suction pipe ● Loosening of mounting bolt of the pump ● Dirt in strainer ● Drive shaft alignment is not good ● Worn out universal joint ● Damage in pump casing	● Top up the oil ● Tightening ● Tightening ● Clean the strainer ● Realign ● Change ● Change

Boom do not extend	● Dirt in counter balance valve or damage in seal ● Internal leakage in cylinder ● Sequence valve fault ● Lack of oil	● Clean and/or change ● Repair ● Clean/repair ● Top up the oil
Boom do not retract	● Dirties in counter balance valve or damage in seal	Clean and/or change
Boom automatically retract when loaded	● Internal leakage in cylinder ● Some leakage at connection parts ● Damage in check valve	● Repair ● Repair ● Clean and change the seal
Some noise in slewing system and/or insufficient torque	● Damage of slewing gear ● Damage of the bearing in the gearbox ● Pump faulty ● Insufficient oil in the gearbox ● Motor faulty	● Change ● Change the bearing ● Repair/change pump ● Top up the gear oil ● Change motor
Jerky when slewing	● Too much clearance in slewing gear and pinion ● Worn out of the wormwheel in gearbox ● Loosening of the gearbox mounting bolts	● Adjust the pinion gear ● Change the gearbox ● Tighten the bolts
Stabilizer cylinder do not extend	● Lack of oil ● Dirties in relief valve	● Top up the oil ● Clean and/or repair
Stabilizer cylinder retract when loaded	● Damage in pilot check valve ● Internal leakage in cylinder	● Repair/change ● Repair
Stabilizer cylinder down when driving	● Damage in pilot check valve ● Internal leakage in cylinder ● Leakage from the cylinder	● Repair/change ● Repair ● Repair
Leakage in winch motor	● Drain line is clogged ● Damage of the packing	● Repair ● Repair
2 nd boom will not retract	Sequence valve faulty	Repair
3 rd boom will not extend	Sequence valve faulty	Repair

Appendix 1 : Wind speeds

Wind force	Wind speed averaged over 10 min. at a height of 10 m		characteristics
	Above flat ground		
	m/s	Wind type	
0	0.0 ~ 0.2	Calm	Calm, smoke rises vertically or nearly vertically
1 2	0.3 ~ 1.5 1.6 ~ 3.3	Slight breeze	Wind direction recognizable from smoke plumes, wind begins to be noticeable on the face; leaves begin to rustle and weather vanes can start to move
3 4	3.4 ~ 5.4 5.5 ~ 7.9	Moderate wind	Leaves and twigs in continuous movement, small branches begin to move. Dust and paper begin to move over the ground
5	8.0 ~ 10.7	Fairly strong wind	Small leaved branches make swaying movements : crested waves from on lakes and canals.
6	10.8 ~ 13.8	Strong wind	Large branches moving; you hear the wind whistling In the telephone wires : umbrellas can only be held With difficulty.
7	13.9 ~ 17.1	Severe wind	Entire trees moving; the wind causes difficulty when you walk into it.
8	17.2 ~ 20.7	Stormy wind	Twigs break off, walking is difficult.
9	20.8 ~ 24.4	Storm	Causes superficial damage to buildings(chimney pots, roof-tiles, and TV antennae are blown off).
10	24.5 ~ 28.4	Severe storm	Uprooted trees : considerable damage to buildings etc. (occurs infrequently on land).
11	28.5 ~32.6	Very severe storm	Causes extensive damage (occurs very infrequently on land).
12	>32.6	Hurricane	

Appendix 2 : Signals when using a crane

If you cannot see the load properly, someone else should give you signals. This appendix gives a number of the standard signals.

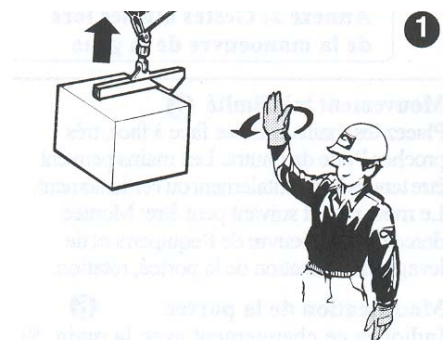
!NOTE

Have the person giving the signal stand facing you as far as possible.

The speed of the hand and arm movements indicate at what speed the crane movement should be made.

Lift

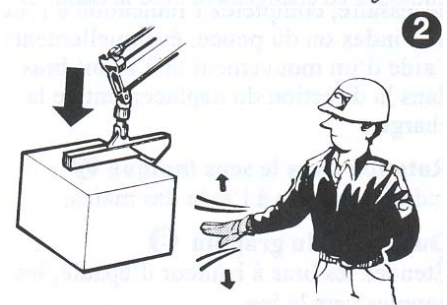
Rotate the fist about the wrist,
with the arm raised at an oblique angle.



Lower

With the arm outstretched, move the hand
energetically up and down.

If necessary, the entire arm may be moved.

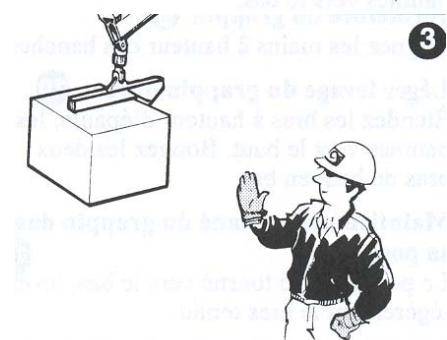


Stop all crane movements

Also : hold the load in position

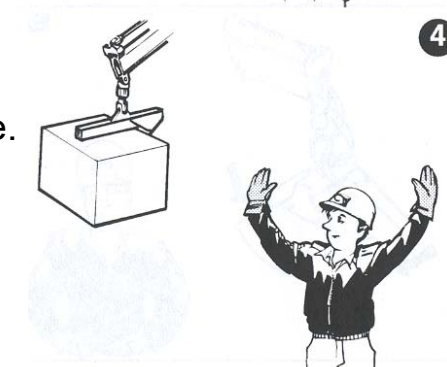
Raise the open hand, with the palm clearly
visible, and the arm to shoulder height.

Keep the hand still.



Emergency stop for all movements by the crane.

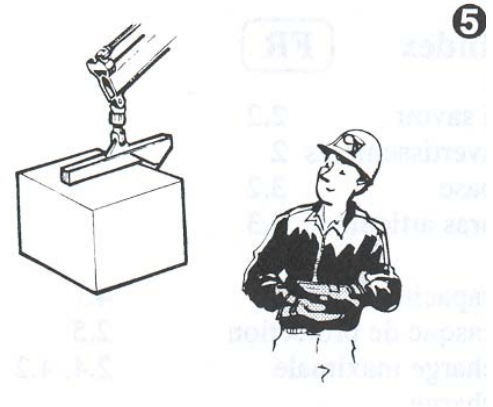
Raise the hands and the arms to an oblique
angle.



Very short movement

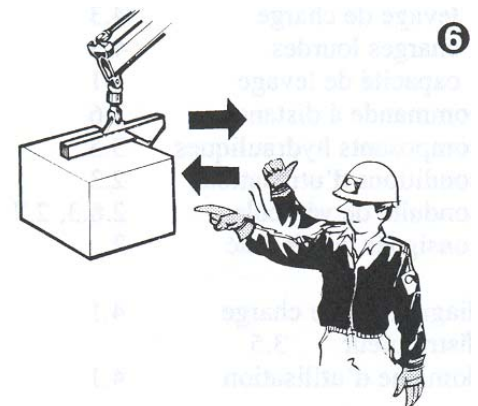
Place the hands a very short distance apart, with the palms facing each other.

The hands may be held either horizontally or vertically. The next movement may be :
Lift, lower, move the lifting gear, change the reach, or turn.



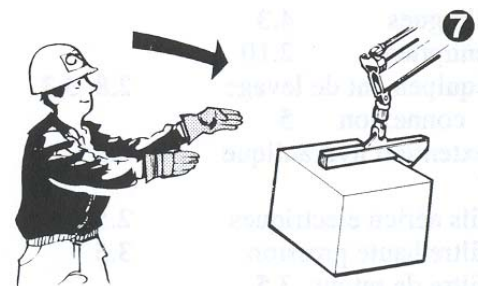
Change the reach

Indicate this by means of the hand, if necessary providing a precise indication with the index finger or the thumb and, if necessary, moving the forearm in the direction in which the load is to be moved.



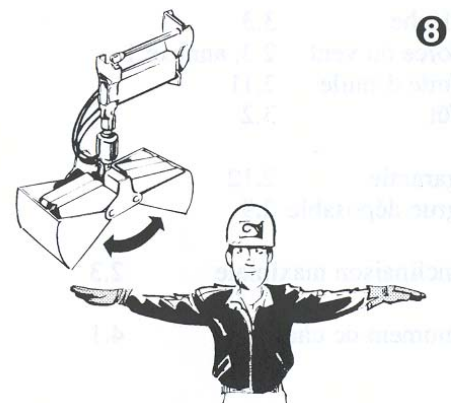
Turn in the direction indicated

Indicate the direction with the hands.



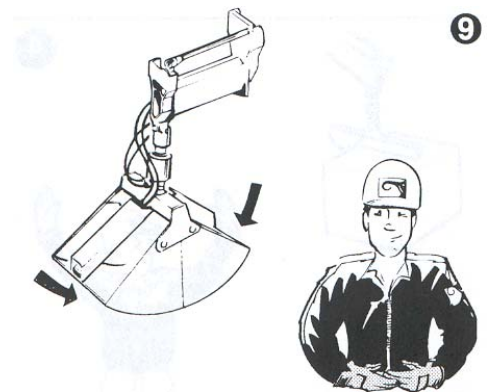
Open the grapple

Extend the arms at shoulder height, with the palms facing downwards.



Close the grapple

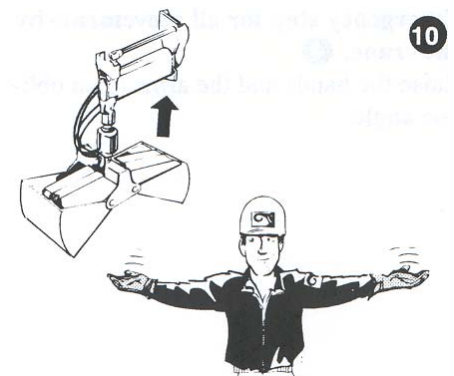
Move both hands close together.



Lift the open grapple a little

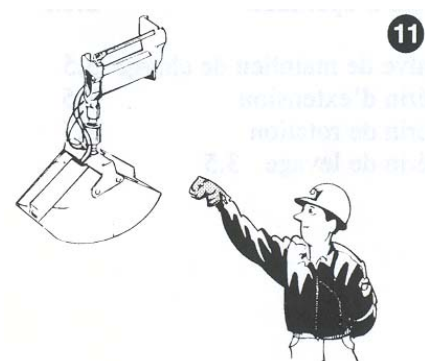
Extend both arms at shoulder height,
with the palms facing upwards.

Make vertical movements with
both arms outstretched.



Keep the grapple in position briefly

Raise the hand slightly drooping,
with the fist clenched.



EN



SPARE PARTS LIST

Cargotec Korea Ltd

Instruction

This spare parts catalogue has several main divisions each embracing a main component group, where you can find easily the necessary part.

Each drawing is followed by the corresponding written material Parts shown in the drawings are numbers by position starting from 1, drawings are numbered as;

(page numbering)

HIAB160T	-	8	:	0
				<u>Edition</u>
		<u>Page</u>		
<u>Model no.</u>				

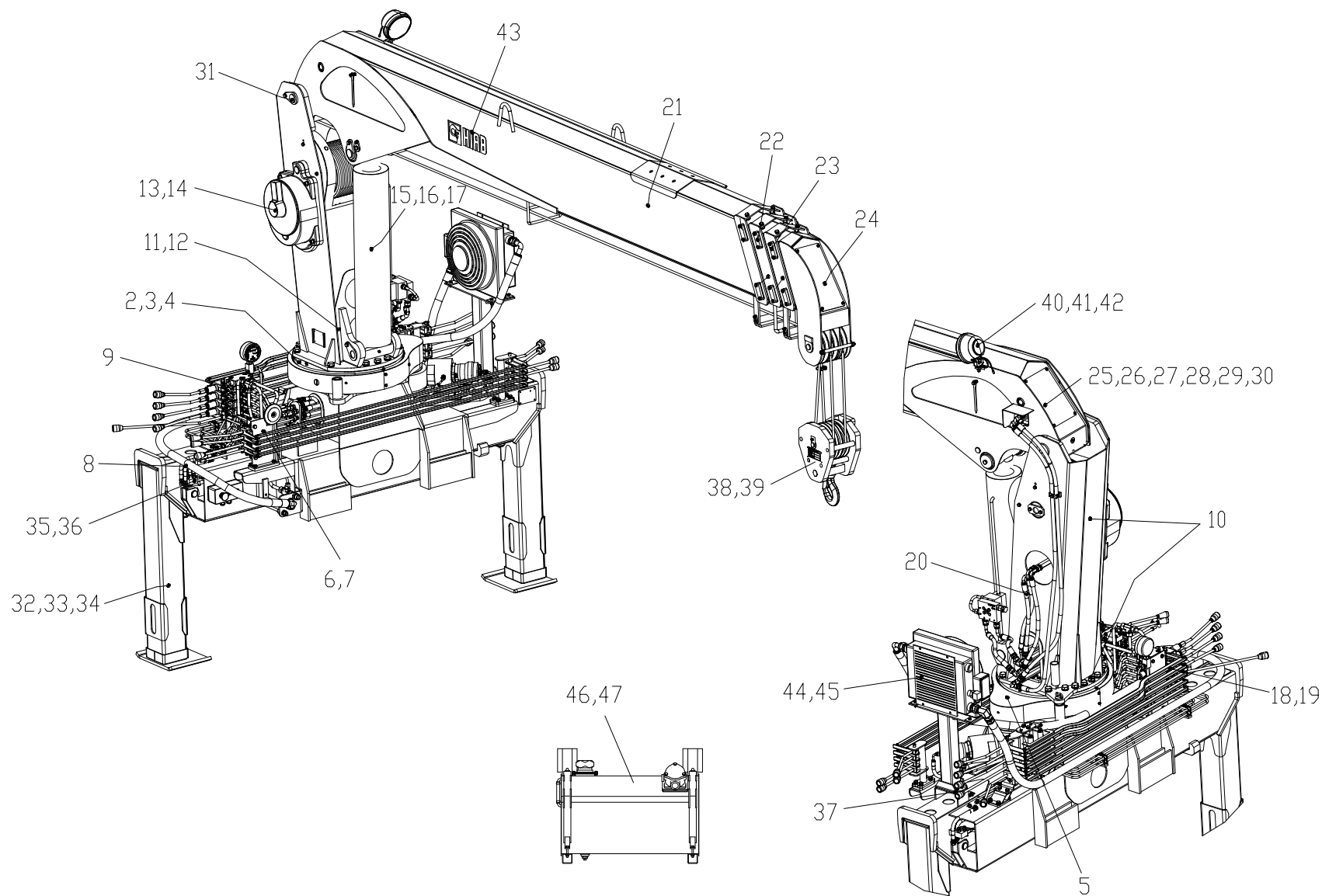
The position number is listed with written part number, description, quantity and specification / remarks.

When ordering spare parts, please inform;

1. Model no.
2. Page numbering
3. Part number & description
4. Quantity
5. Your complete address

This spare parts catalogue can be changed without prior notice will be informed accordingly.

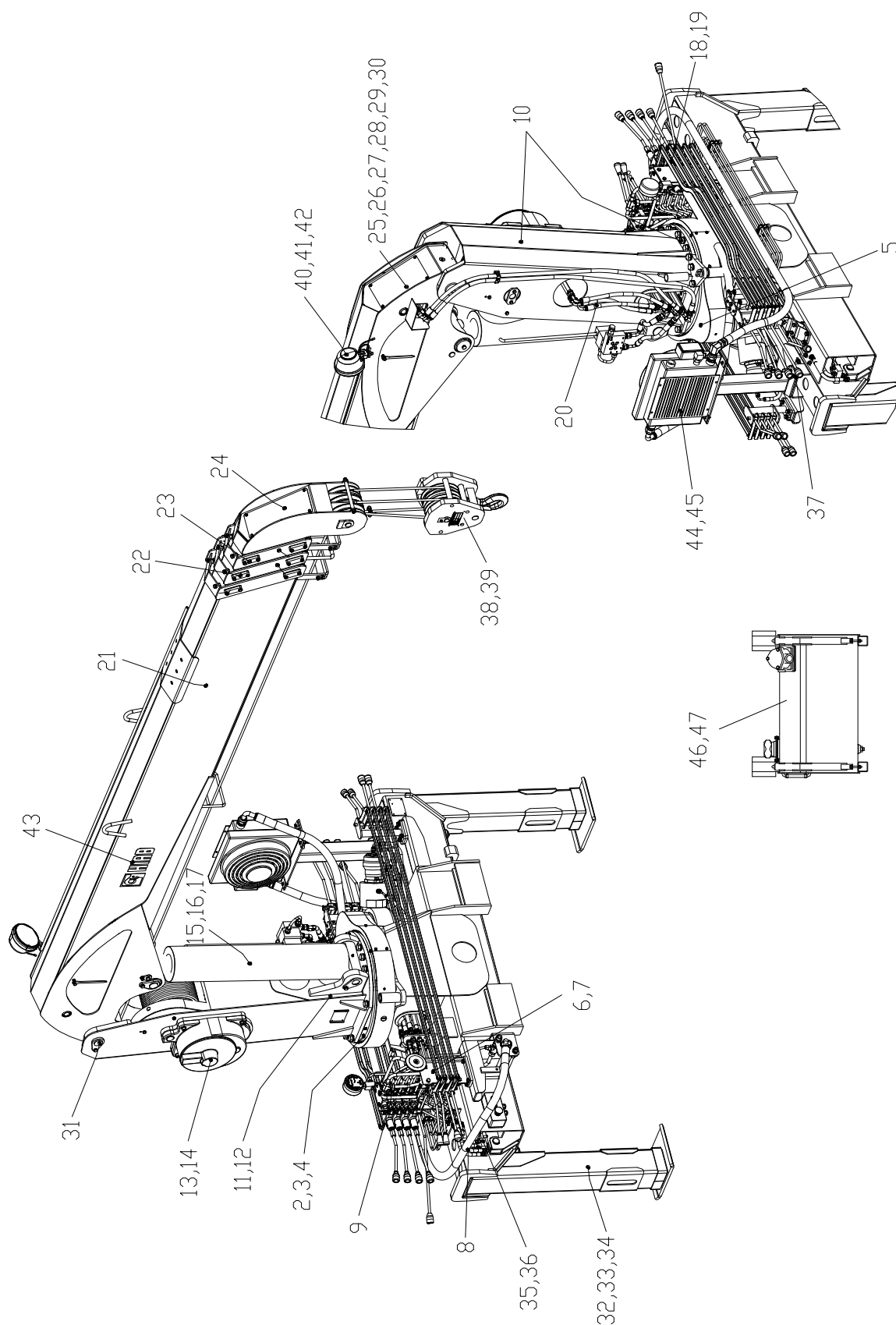
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CONTENTS LIST

CHAP.	DESCRIPTION	RE.	PAGE
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4	SLEWING MOTOR	01	4 - 1 - 1
5	SLEWING COVER ASS'Y	00	5 - 1 - 1
6	OUTRIGGER CONTROL SYSTEM	00	6 - 1 - 1
7	OUTRIGGER CONTROL VALVE	00	7 - 1 - 1
8	OUTRIGGER HOSE & PIPE	00	8 - 1 - 1
9	BASE HOSE & PIPE	00	9 - 1 - 2
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11	ROTARY JOINT CPL.	00	11 - 1 - 1
12	ROTARY JOINT ASS'Y	00	12 - 1 - 1
13	WINCH SYSTEM	00	13 - 1 - 1
14	WINCH REDUCER ASS'Y	00	14 - 1 - 1
15	DERRICK CYLINDER CPL.	01	15 - 1 - 1
16	DERRICK CYLINDER ASS'Y	00	16 - 1 - 1
17	LOAD HOLDING VALVE	01	17 - 1 - 1
18	MAIN CONTROL VALVE SYSTEM	00	18 - 1 - 1
19	MAIN CONTROL VALVE	00	19 - 1 - 1
20	COLUMN HOSE ASS'Y	00	20 - 1 - 1
21	1st BOOM ASS'Y	00	21 - 1 - 1
22	2nd BOOM ASS'Y	00	22 - 1 - 1
23	3rd BOOM ASS'Y	00	23 - 1 - 1
24	4th BOOM ASS'Y	00	24 - 1 - 1
25	EXTENSION CYLINDER CPL (SINGLE)	00	25 - 1 - 1
26	EXTENSION CYLINER ASS'Y	00	26 - 1 - 1
27	1st EXTENSION CYLINER ASS'Y	00	27 - 1 - 2
28	2nd EXTENSION CYLINER ASS'Y	00	28 - 1 - 2
29	3rd EXTENSION CYLINER ASS'Y	00	29 - 1 - 2
30	COUNTER BALANCE VALVE	00	30 - 1 - 1
31	MAIN SHAFT ASS'Y	00	31 - 1 - 1
32	OUTRIGGER ASS'Y	00	32 - 1 - 1
33	OUTRIGGER CYLINDER ASS'Y	00	33 - 1 - 1

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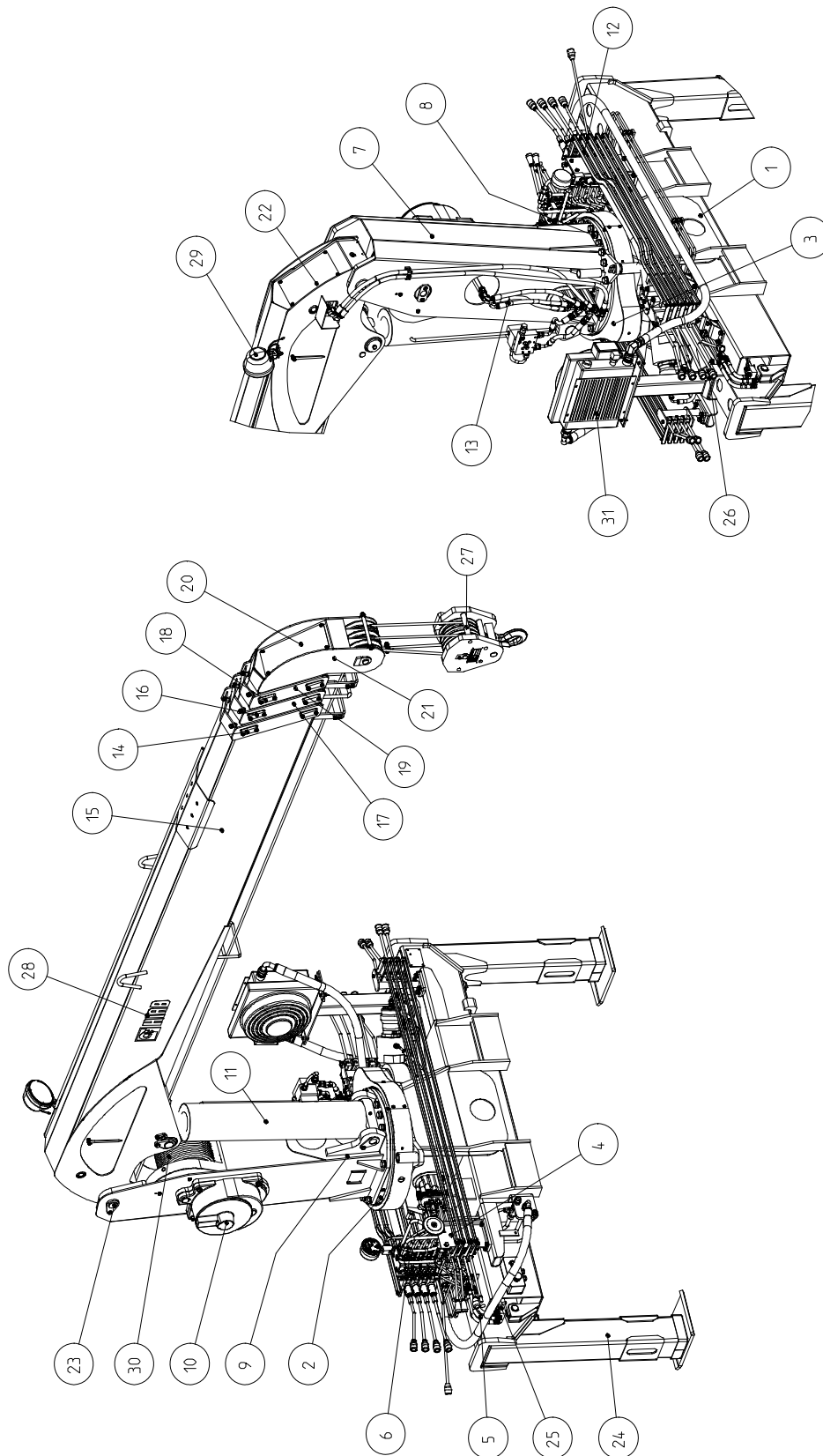
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CHAP.	DESCRIPTION	RE.	PAGE
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38	HOOK ASS'Y	00	38 - 1 - 1
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41	ELECTRIC ASS'Y (MOUNTING PART)	01	41 - 1 - 1
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45	OIL COOLER	00	45 - 1 - 1
46	OIL TANK ASS'Y (OPTION)	00	46 - 1 - 1
47	FILTER, RETURN (OPTION)	00	47 - 1 - 1

SECTION 1.1 - GENERAL ARRANGEMENT

GENERAL ARRANGEMENT

00



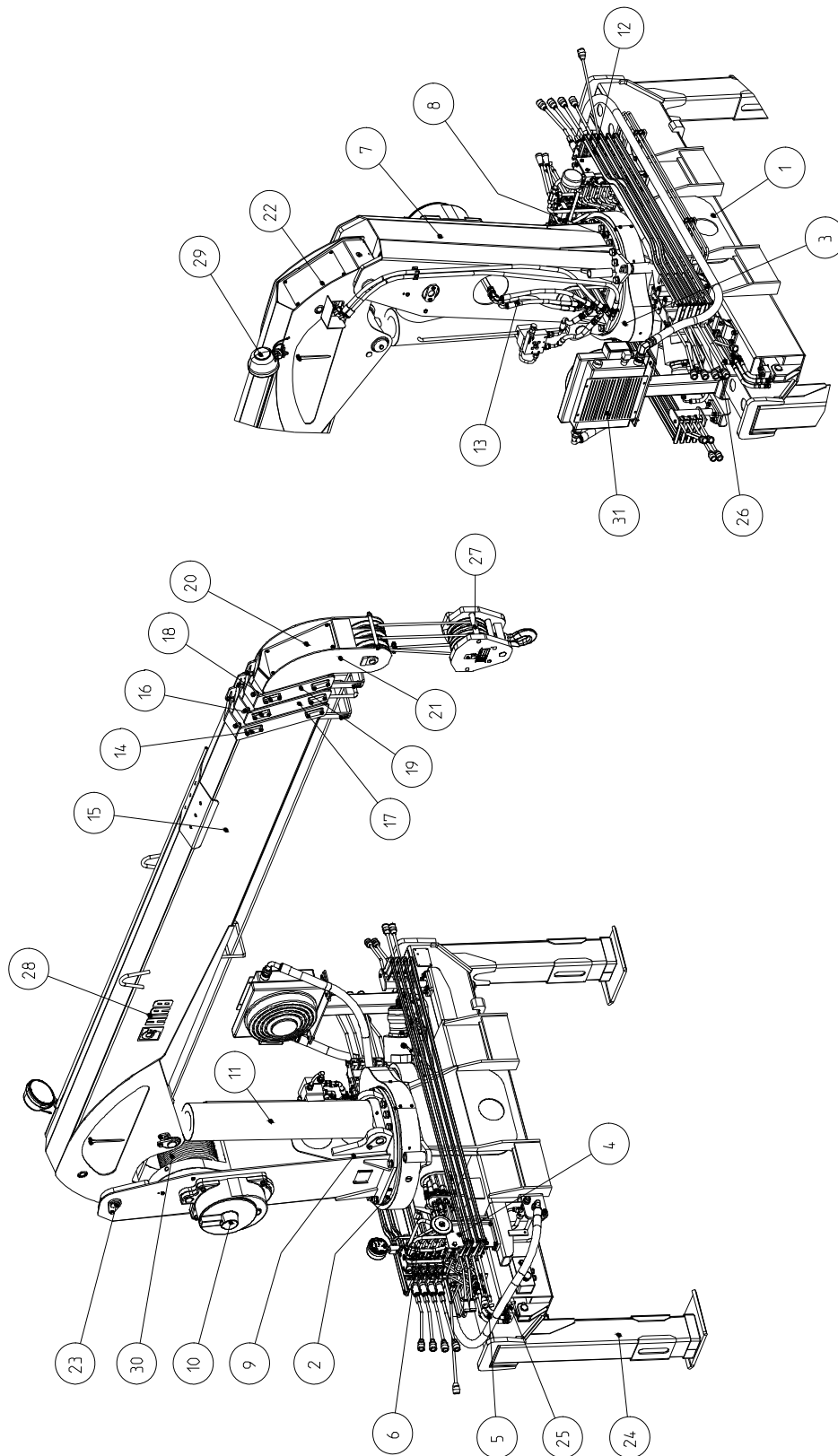
SECTION 1.1 - GENERAL ARRANGEMENT

GENERAL ARRANGEMENT				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	A00-0595	GENERAL ARRANGEMENT	
1	1	A02-4613	BASE STRUCTURE	
2	1	A01-4791	SLEWING SYSTEM	
3	1	A01-4792	SLEWING COVER ASS'Y	
4	1	A01-4813	OUTRIGGER CONTROL SYSTEM	
5	1	A01-4830	OUTRIGGER HOSE & PIPE	
6	1	A01-4831	BASE HOSE & PIPE	
7	1	A02-3819	COLUMN STRUCTURE	
8	1	A01-4836	COLUMN ASS'Y	
9	1	A01-4829	ROTARY JOINT CPL	
10	1	WC-2001	WINCH SYSTEM	
11	1	A01-4800	DERRICK CYLINDER CPL	
12	1	A01-5347	MAIN CONTROL VALVE SYSTEM	
13	1	A01-3671	COLUMN HOSES ASS'Y	
14	1	A01-4802	1st BOOM ASS'Y	
15	1	A02-4079	1st BOOM STRUCTURE	
16	1	A01-4803	2nd BOOM ASS'Y	
17	1	B02-5001	2nd BOOM STRUCTURE	
18	1	A01-5491	3rd BOOM ASS'Y	
19	1	A02-2097	3rd BOOM STRUCTURE	
20	1	A01-5492	4th BOOM ASS'Y	
21	1	A02-4246	4th BOOM STRUCTURE	
22	1	A01-5493	EXTENSION CYLINDER CPL (S)	
23	1	A01-4810	MAIN SHAFT CPL	
24	1	OR1-4002	OUTRIGGER ASS'Y	
25	1	A01-5286	OUTRIGGER EXTENSION CYLINDER KIT	
26	1	A01-5287	OUTRIGGER PAD KIT	
27	1	123 1637	HOOK ASS'Y	
28	1	A01-4750	ARRANGEMENT, STICKER	

SECTION 1.1 - GENERAL ARRANGEMENT

GENERAL ARRANGEMENT

00





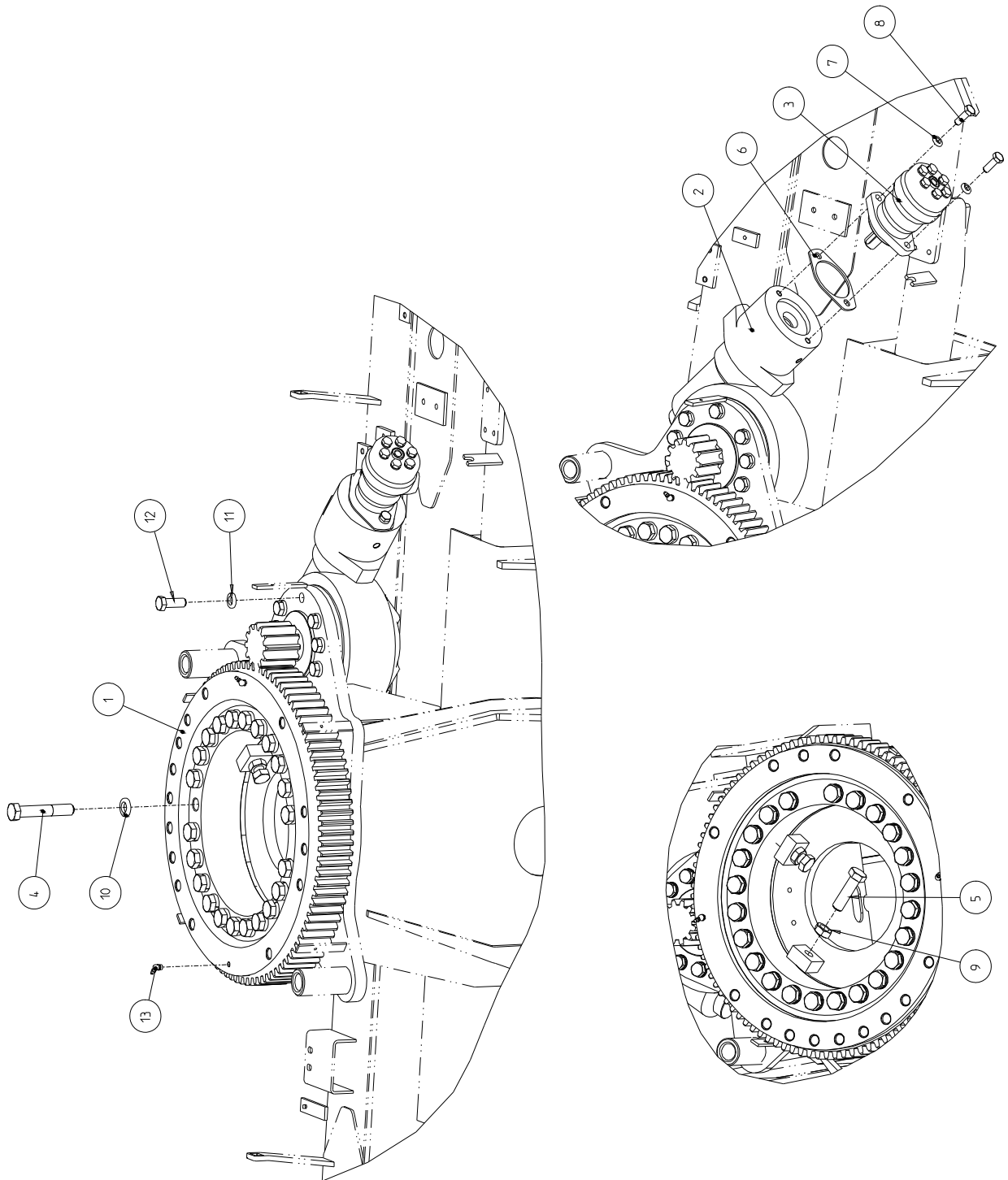
SECTION 1.1 - GENERAL ARRANGEMENT

GENERAL ARRANGEMENT				00
Pos.	Q'ty	Part No.	Description	Remarks
29	1	E00-0127	ELECTRIC DIAGRAM	Φ10×110m
30	1	TA-2004	ROPE, WIRE	
31	1	A01-5277	OIL COOLER MOUNTING	

SECTION 2.1 - SLEWING SYSTEM

SLEWING SYSTEM

01



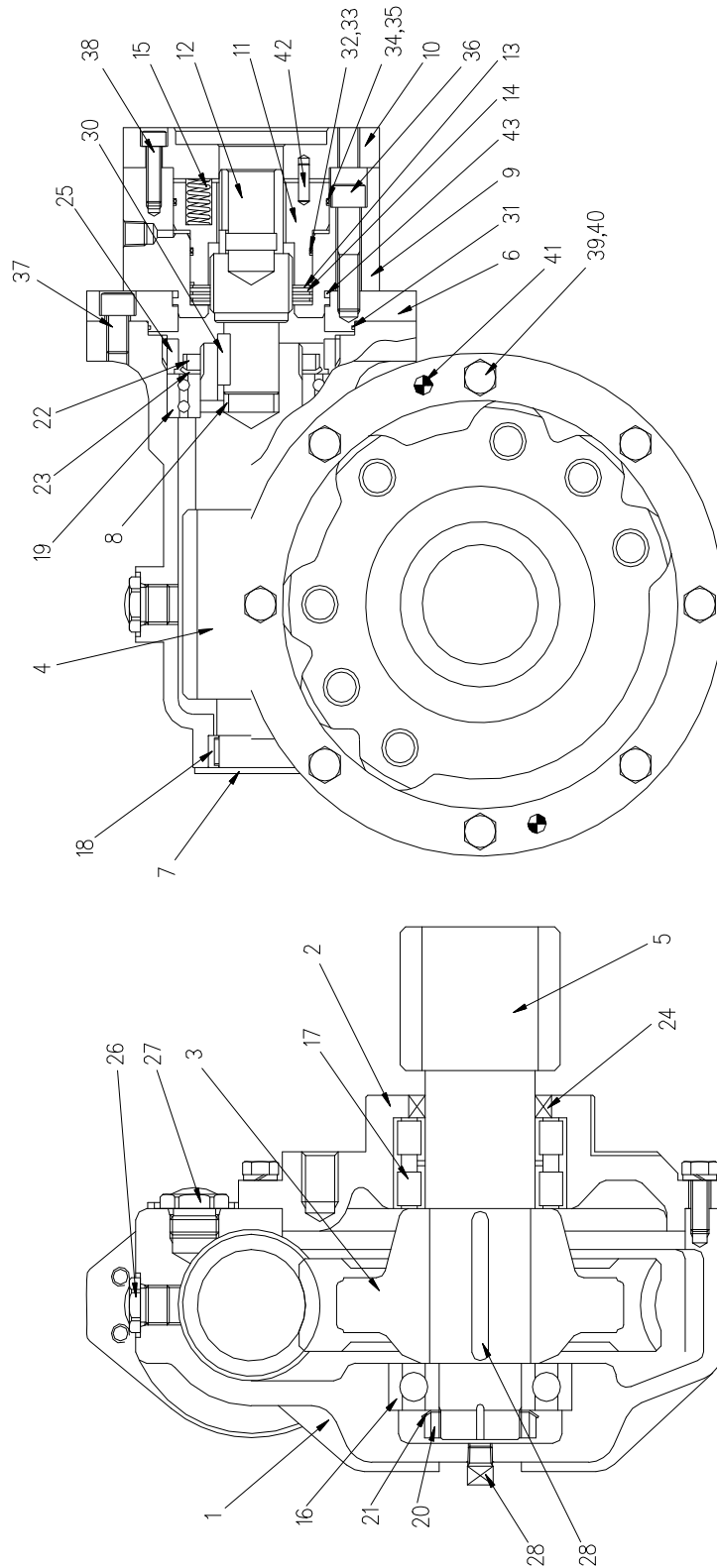
SECTION 2.1 - SLEWING SYSTEM

SLEWING SYSTEM					01
Pos.	Q`ty	Part No.	Description	Remarks	
-	1	A01-4791	SLEWING SYSTEM		
1	1	BC1-3003	BEARING, SLEWING		
2	1	BC1-0003	REDUCER, SLEWING		
3	1	456 0983	MOTOR, HYDRAULIC	MAPN-100CD (Maxima motor)	
4	23	S01-1308	BOLT, HEX. HEAD	BA20×120	
5	2	2112807	BOLT, HEX. HEAD	BA20×70	
6	1	S01-0469	GASKET	t3.2×95×136	
7	2	3612121	WASHER, SPRING	WS12	
8	2	2111537	BOLT, HEX. HEAD	BA12×35	
9	2	3111207	NUT, HEX.	NA20	
10	23	3612201	WASHER, SPRING	WS20	
11	7	3612161	WASHER, SPRING	WS16	
12	7	2112157	BOLT, HEX. HEAD	BA16×45	
13	2	4620702	GREASE NIPPLE		

SECTION 3.1 - SLEWING REDUCER

SLEWING REDUCER

00



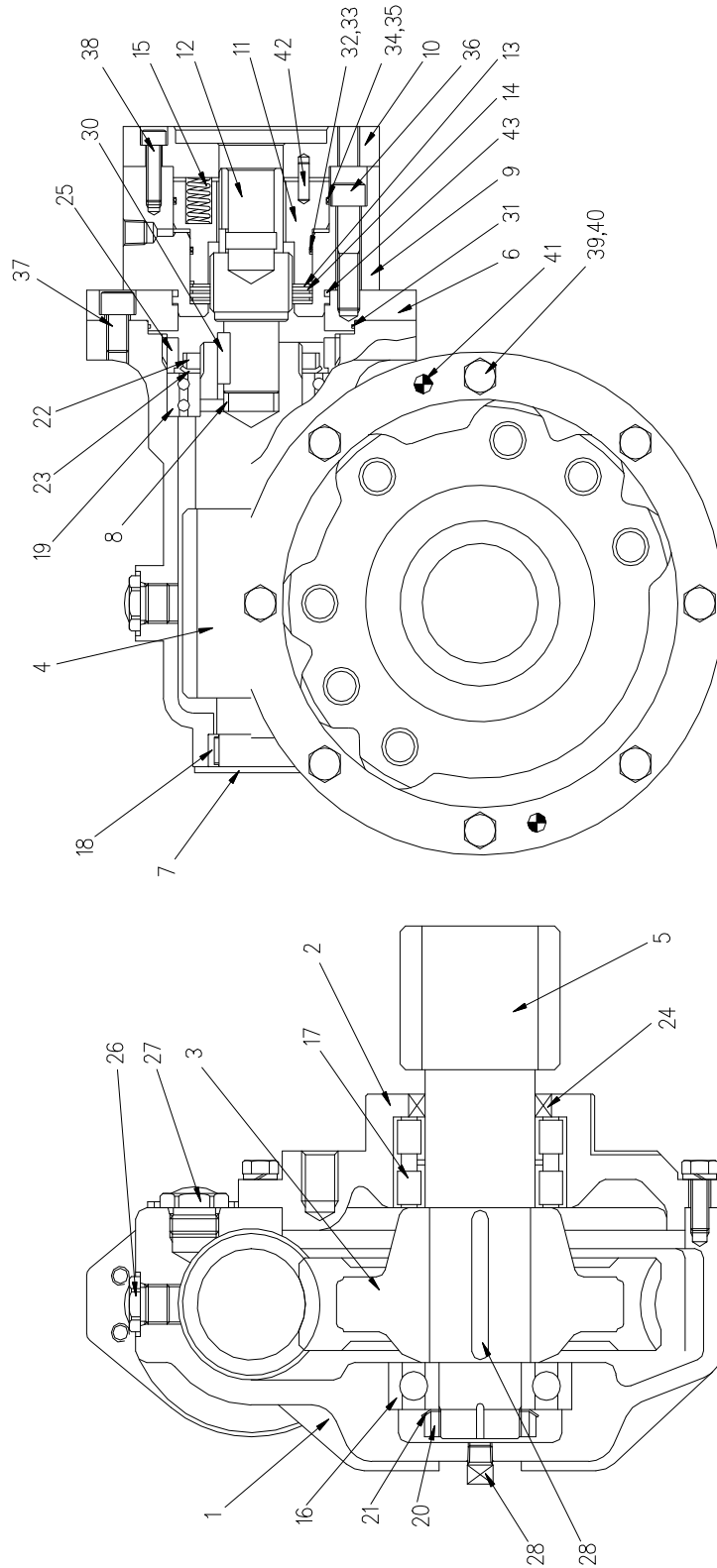
SECTION 3.1 - SLEWING REDUCER

SLEWING REDUCER				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	BC1-0003	SLEWING REDUCER	
1	1	07-07-1001	CASE	
2	1	07-07-1002	COVER	
3	1	07-07-1003	GEAR, WORM	
4	1	07-07-1004	SHAFT, WORM	
5	1	07-07-HA05	SHAFT, PINION	
6	1	07-07-HA06	ADAPTOR	
7	1	07-07-1007	COVER, PLATE	
8	1	07-07-HA08	SPACER	
9	1	07-04-2006	HOUSING, BRAKE	
10	1	07-04-2007	ADAPTOR, MOTOR	
11	1	07-04-2037	PISTON, BRAKE	
12	1	07-04-2909	SHAFT, BRAKE	
13	3	07-04-2033	DISK	
14	3	07-04-2034	PLATE	
15	8	07-04-2018	SPRING	
16	1	4113092	BEARING, RADIAL BALL	6309ZZ
17	1	BC1-0003-1	BEARING, ROLLER	SL04-5012
18	1	BC1-0003-2	BEARING, NEEDLE	RNA4906
19	1	BC1-0003-3	BEARING, BALL	5210
20	1	BC1-0003-4	NUT, BEARING	AN09
21	1	BC1-0003-5	WASHER, BEARING	AW09
22	2	BC1-0003-6	NUT, BEARING	AN10
23	1	BC1-0003-7	WASHER, BEARING	AW10
24	4	BC1-0003-8	OIL SEAL	607812
25	8	07-07-1009	LOCK THREAD	
26	8	BC1-0003-9	GAUGE, OIL	PF1/2
27	2	BC1-0003-10	CAP	PF3/4
28	2	BC1-0003-11	PLUG	PT1/2

SECTION 3.1 - SLEWING REDUCER

SLEWING REDUCER

00



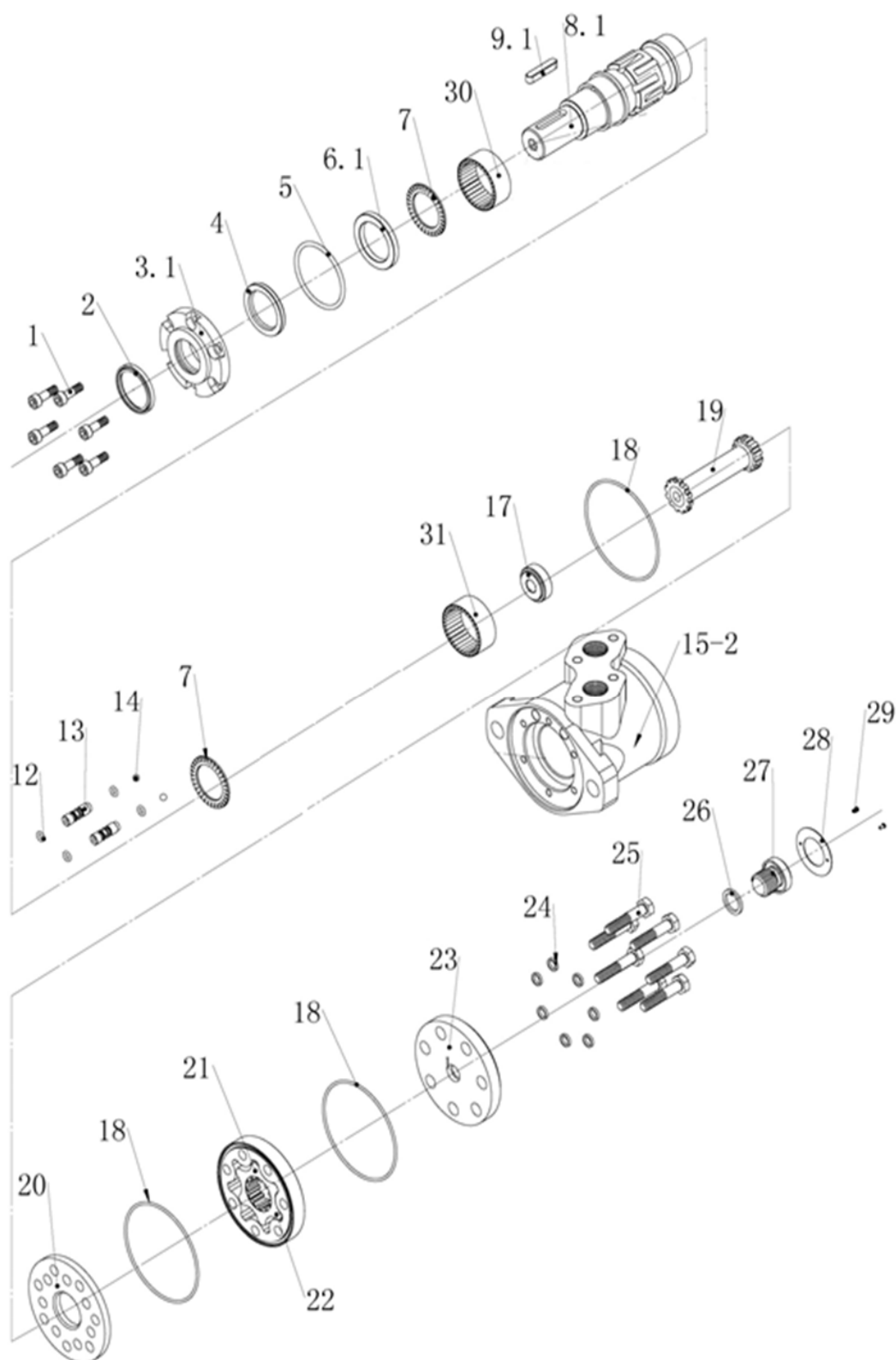
SECTION 3.1 - SLEWING REDUCER

SLEWING REDUCER					00
Pos.	Q'ty	Part No.	Description	Remarks	
29	2	BC1-0003-12	KEY		
30	1	BC1-0003-13	KEY		
31	1	6012392	O-RING	1B G95	
32	1	6012332	O-RING	1B G65	
33	1	6035281	RING, BACK-UP	T2 G65	
34	1	6012352	O-RING	1B G75	
35	1	6035301	RING, BACK-UP	T2 G75	
36	6	2616576	BOLT, HEX. SOCKET HEAD	BB12×55	
37	4	2611486	BOLT, HEX. SOCKET HEAD	BB12×20	
38	6	2610926	BOLT, HEX. SOCKET HEAD	BB8×30	
39	8	2111226	BOLT, HEX. HEAD	BA10×30	
40	1	3612101	WASHER, SPRING	WS10	
41	1	BC1-0003-14	PIN		
42	1	BC1-0003-15	PIN		
43	1	6012362	O-RING	1B G80	

SECTION 4.1 - SLEWING MOTOR

SLEWING MOTOR

01



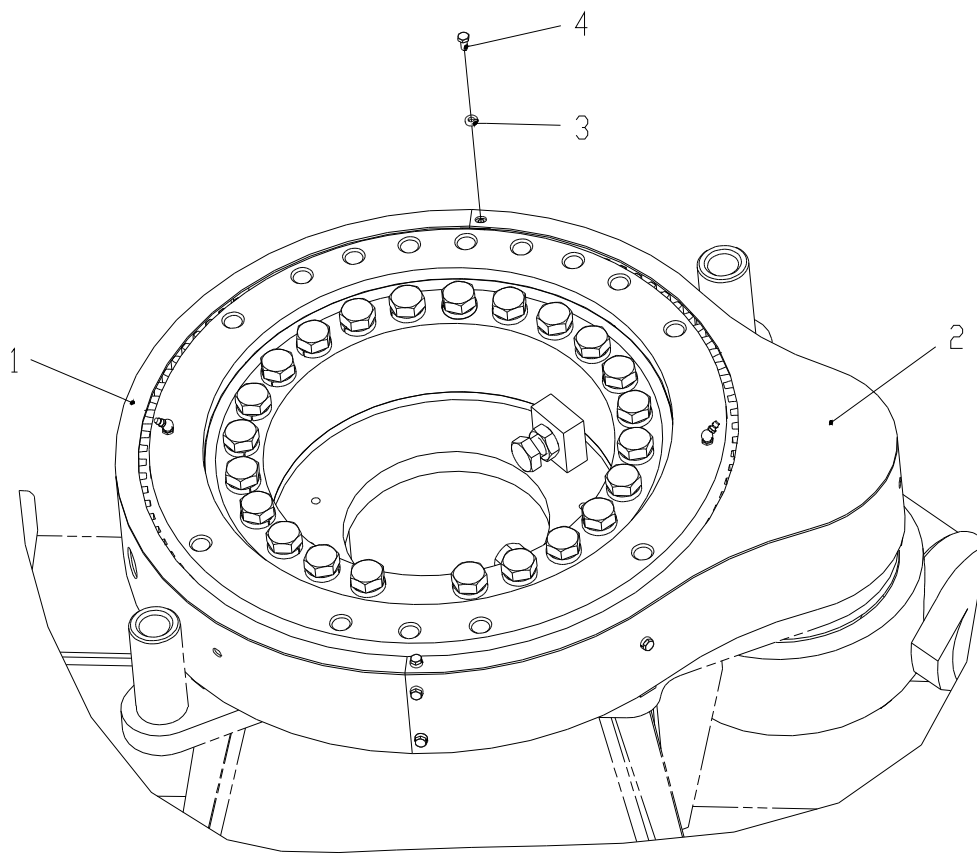
SECTION 4.1 - SLEWING MOTOR

SLEWING MOTOR					01
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	456 0983	MOTOR, HYDRAULIC	MAPN-100CD	
1	6	GB70-85/M6×16-12.9	SCREW	M6×16-12.9	
2	1	MAR-402-2	DUST SEAL	28.5×35×4	*
3.1	1	MAR-302-H	FRONT COVER	Φ44.4	
4	1	MAR-403-2	SHAFT SEAL	28.56×42×5.5	*
5	1	MAR-404	O-RING	55×3.1	*
6.1	1	MAR-303-H	THRUST WASHER	29.5×42.9×8.4	
7	2	BYM(A)-313-G	AXIAL NEEDLE BEARING	29×42×2	
8.1	1	MAR/MARN-301-2-A	SHAFT, CYLINDRICAL	Φ25	
9.1	1	GB1096-79	PARALLEL KEY	8×7×32	
12	4	GB3452.1-92	O-RING	5×1.8G	*
13	2	MAR-306(b)	VALVE, CHECK		
14	2	GB308-77	BALL		
15-2	1	MAP/MAPN -101-2D	HOUSING (2-Φ13.5 Rhomb-flange pilot Φ82.5×8)	2-Φ13.5, Φ82.5×8	
17	1	MAR-308	SPACER	δ=7.5	
18	3	MAP-405	O-RING	82×1.9	*
19	1	MAR-304(b)-86	DIRVE LIKE	L=86mm	
20	1	MAP-305	DISTRIBUTOR PLATE		
21	1	MAP-100-306	STATOR	H=13mm	
22	1	MAP-100-307	ROTOR	H=13mm	
23	1	MAP-309-2	END COVER	G1/4"	
24	7	MAP-202	WASHER	Φ8	
25	7	MAR-310(a)	SCREW	M8×1×45	
26	7	MAR-408-1	WASHER	8.9	
27	1	MAR-314-2	DRAIN PLUG	G1/4"	
28	1	MAR-201	NAME PLATE		
29	2	GB827-86	RIVET	2×4	
30	1	MARN-330-09	BEARING Only for MAPN	8.2×42.9×13	
31	1	MARN-330-02	BEARING Only for MAPN	38.2×42.2×13	
-	1	S01-6112	SEAL KIT (No.2,4,5,12,18)	MARKS *	

SECTION 5.1 - SLEWING COVER ASS'Y

SLEWING COVER ASS'Y

00



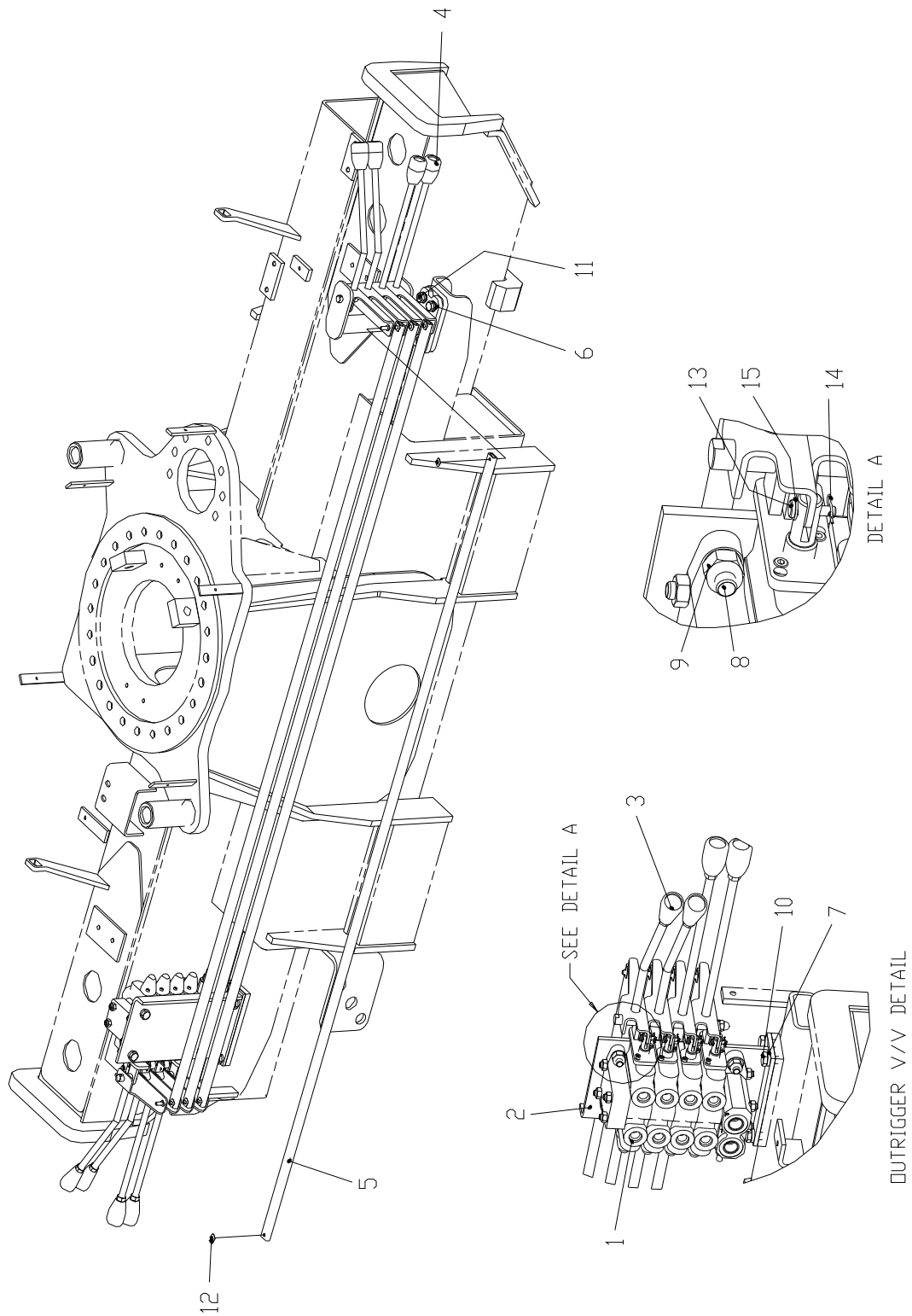
SECTION 5.1 - SLEWING COVER ASS'Y

SLEWING COVER ASS'Y					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	A01-4792	SLEWING COVER ASS'Y		
1	1	BC1-7009	COVER L.H		
2	1	BC1-7013	COVER R.H		
3	11	3612061	WASHER, SPRING	WS6	
4	11	2110447	BOLT, HEX. HEAD	BA6×12	

SECTION 6.1 - OUTRIGGER CONTROL SYSTEM

OUTRIGGER CONTROL SYSTEM

00



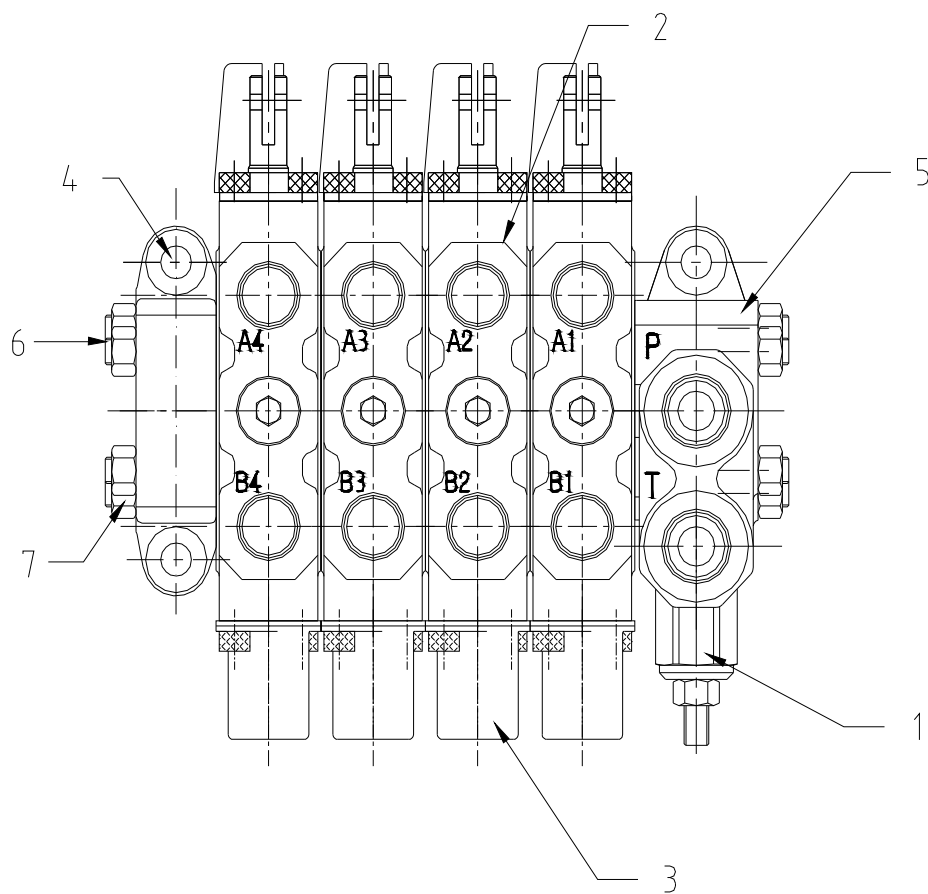
SECTION 6.1 - OUTRIGGER CONTROL SYSTEM

OUTRIGGER CONTROL SYSTEM					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	A01-4813	OUTRIGGER CONTROL SYSTEM		
1	1	H02-1210	VALVE, CONTROL	TX50/4S	
2	1	A05-4013	BRACKT ASS'Y		
3	1	A05-4404	OUTRIGGER LEVER ASS'Y		
4	1	A05-4017	LEVER ASS'Y, OUTRIGGER (2)		
5	4	M05-0967	ROD, LEVER	t2.5×20×10	
6	4	3511101	WASHER, PLAIN	WP10	
7	11	3612101	WASHER, SPRING	WS10	
8	3	2111246	BOLT, HEX. HEAD	BA10×40	
9	3	S01-0006	NUT, SELF LOCKING	M10	
10	4	2111177	BOLT, HEX. HEAD	BA10×18	
11	4	2111206	BOLT, HEX. HEAD	BA10×25	
12	8	S01-0854	WASHER, BACK	SPN06	
13	4	CS-0008	PIN	Φ4.5×20	
14	4	CS-0009	SPRING PIN, SINGLE	Φ1×15	
15	4	3511051	WASHER, PLAIN	WP5	

SECTION 7.1 - OUTRIGGER CONTROL VALVE

OUTRIGGER CONTROL VALVE

00



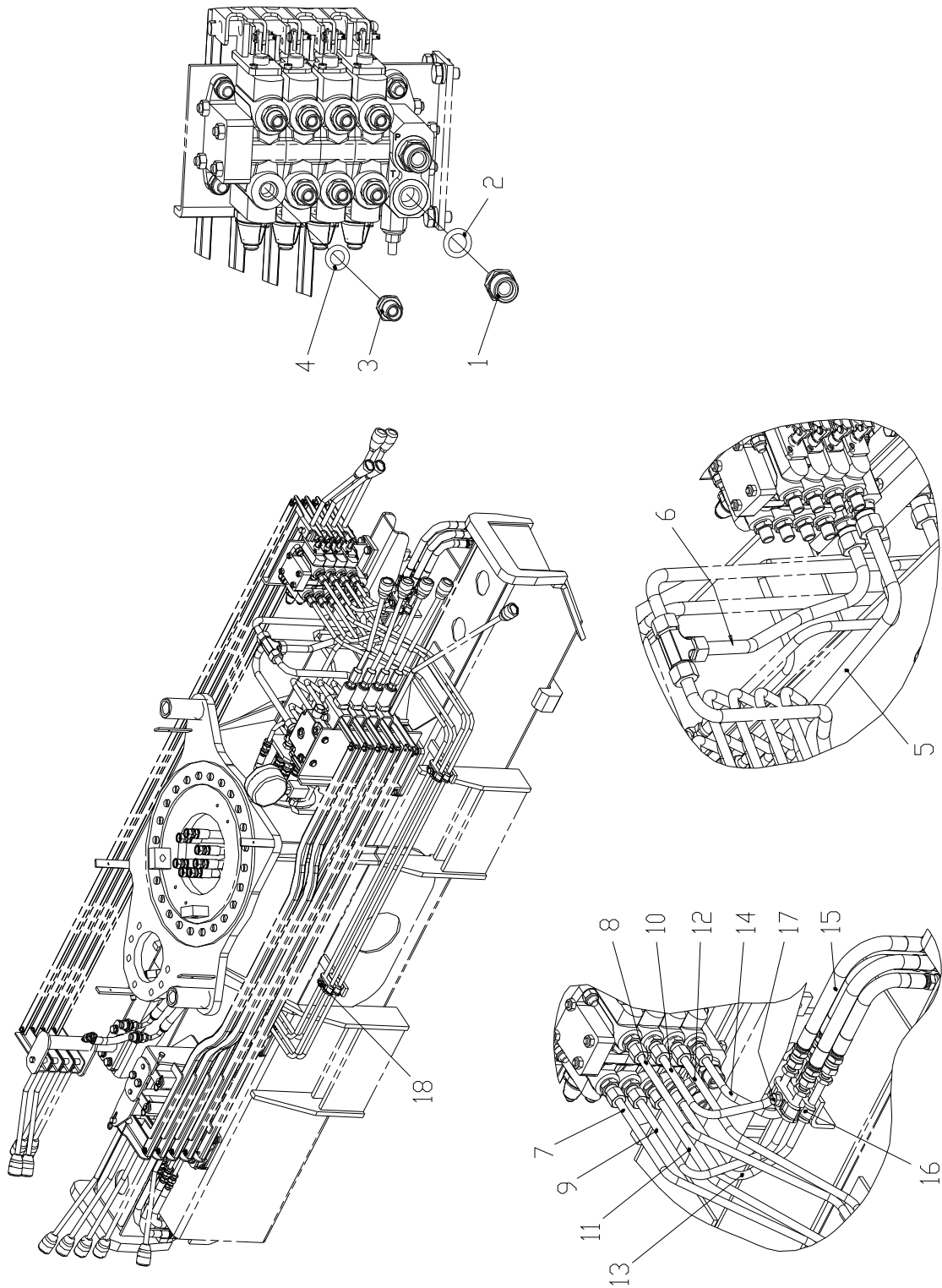
SECTION 7.1 - OUTRIGGER CONTROL VALVE

OUTRIGGER CONTROL VALVE				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	H02-1210	OUTRIGGER CONTROL VALVE	TX50/4S
1	1	H02-1146	BLOCK ASS'Y INLET	
-	1	1024005010	BLOCK INLET	
-	1	2001420055	O-RING	
-	1	1000028020	PLUG PORT	
-	1	2001020011	O-RING	
2	1	3019500030	VALVE, MAIN RELIEF	
3	4	H02-1147	BLOCK ASS'Y WORK	
-	4	1024005020	BLOCK WORK	
-	4	2001420055	O-RING	
-	4	1021025060	POPPET LOAD CHECK	
-	4	2011011012	SPRING LOAD CHECK	
-	4	1021028010	PLUG LOAD CHECK	
-	4	2001020014	O-RING	
-	4	1024010020	CAP RETURN	
-	4	2001010013	O-RING	
-	4	2015340010	SEAL DUST	
-	4	1024009020	BRACKET PLATE LEVER	
-	8	2003212024	BOLT	
-	8	2003212025	BOLT	
4	4	H02-1148	SPOOL ASS'Y WORK	
-	4	1024036011	SPOOL STD	
-	4	2001010013	O-RING	
-	4	2015340010	SEAL DUST	
-	4	1024023010	PLATE SEAL	
-	8	1024031010	SEAT SPRING	
-	4	2011011023	SPRING RETURN	
-	4	1024004010	BOLT SPACER	
5	1	1024005030	BLOCK COVER END	
6	4	1000088390	BOLT TIE	
7	4	2008021008	NUT	

SECTION 8.1 - OUTRIGGER HOSE & PIPE

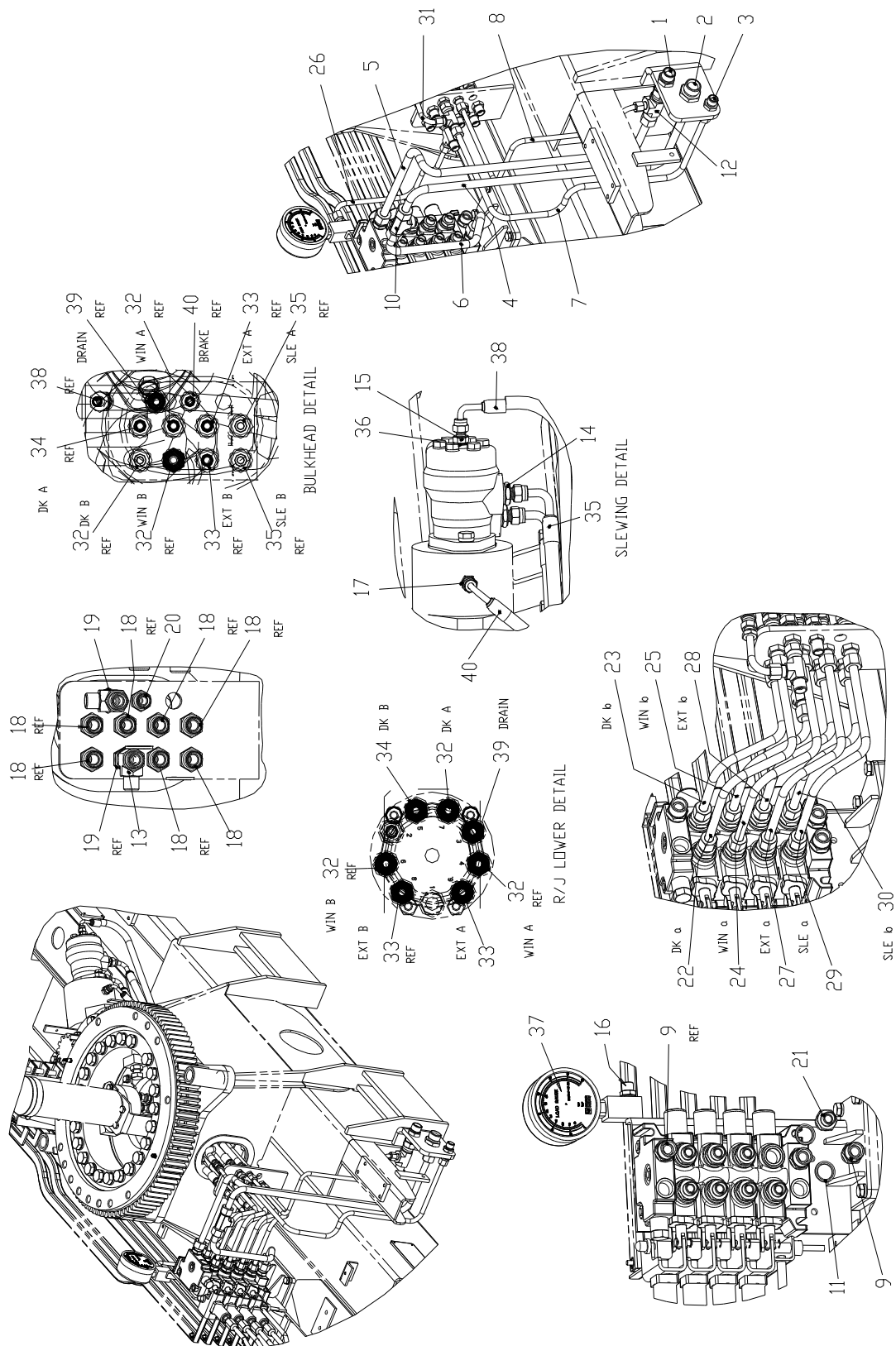
OUTRIGGER HOSE & PIPE

00



SECTION 8.1 - OUTRIGGER HOSE & PIPE

OUTRIGGER HOSE & PIPE					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	A01-4830	OUTRIGGER HOSE & PIPE		
1	2	9003401	NIPPLE, PF PIPE	NTPO16-PF1/2	
2	2	6010212	O-RING	P18	
3	8	9003321	NIPPLE, PF PIPE	NTPO10-PF3/8	
4	8	6010172	O-RING	P14	
5	1	H01-6759	PIPE, HYDRAULIC	Φ16×t2.0	
6	1	H01-6682	PIPE, HYDRAULIC	Φ16×t2.0	
7	1	H01-6691	PIPE, HYDRAULIC	Φ10×t1.5	
8	1	H01-6692	PIPE, HYDRAULIC	Φ10×t1.5	
9	1	H01-7826	PIPE, HYDRAULIC	Φ10×t1.5	
10	1	H01-7827	PIPE, HYDRAULIC	Φ10×t1.5	
11	1	H01-6693	PIPE, HYDRAULIC	Φ10×t1.5	
12	1	H01-6694	PIPE, HYDRAULIC	Φ10×t1.5	
13	1	H01-7828	PIPE, HYDRAULIC	Φ10×t1.5	
14	1	H01-7829	PIPE, HYDRAULIC	Φ10×t1.5	
15	8	7113511	HOSE, H.P (HUCF)	1/4×300-PF1/4	
16	14	S01-4269	CLAMP, PIPE		
17	2	2185966	BOLT, HEX. W/F	BWF8×50	
18	2	2185986	BOLT, HEX. W/F	BWF8×60	



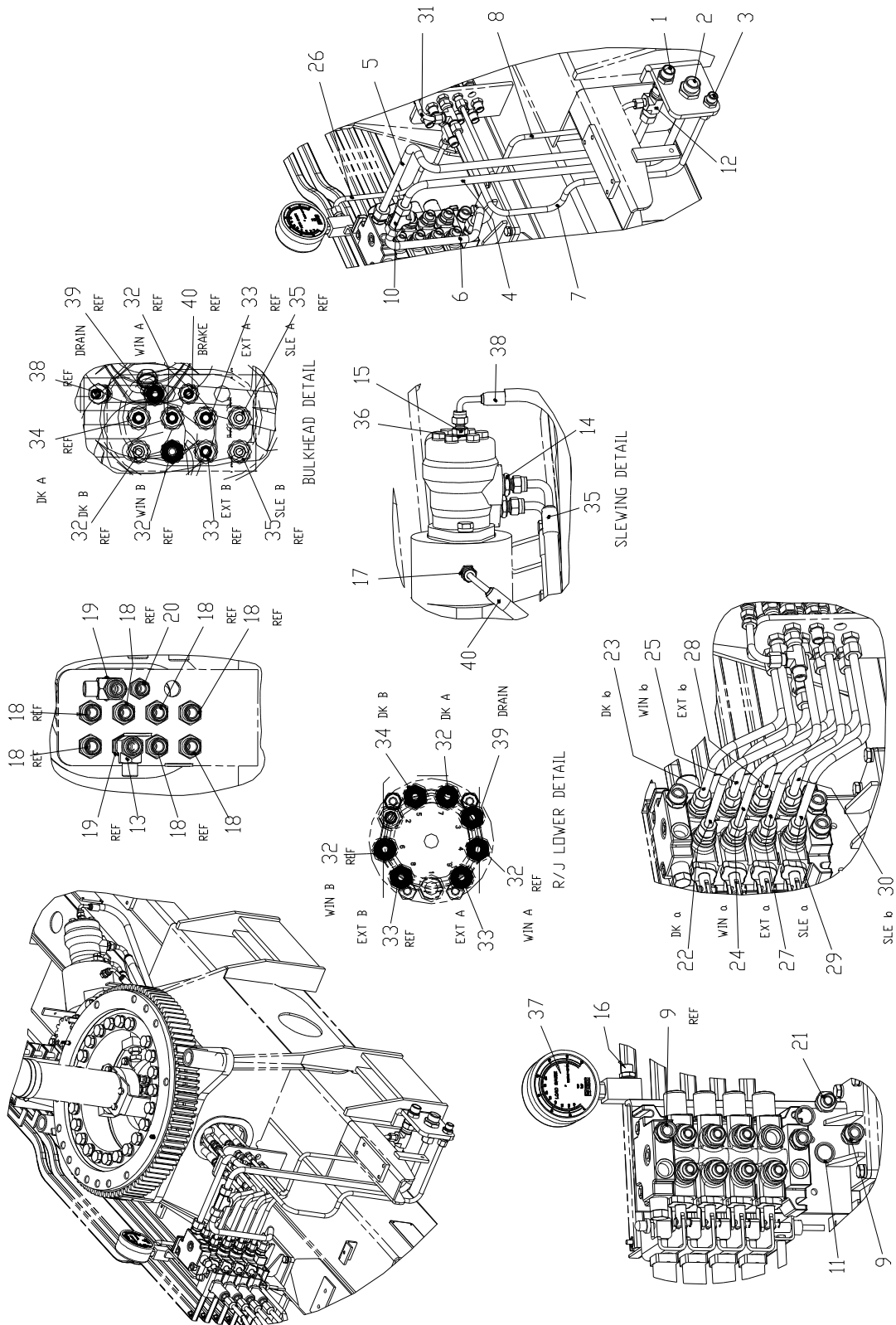
SECTION 9.1 - BASE HOSE & PIPE

BASE HOSE & PIPE					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	A01-4831	BASE HOSE & PIPE		
1	1	S01-3167	UNION, HOSE BULKHEAD	PF3/4×3/4	
2	1	S01-2959	UNION, HOSE BULKHEAD	Φ16×PF1	
3	1	S01-0758	UNION, HOSE BULKHEAD	Φ12×PF1/2	
4	1	H01-6756	PIPE, HYDRAULIC	Φ16×t2.0	
5	1	H01-6757	PIPE, HYDRAULIC	Φ16×t2.0	
6	1	H01-6758	PIPE, HYDRAULIC	Φ16×t2.0	
7	1	H01-2713	PIPE, HYDRAULIC	Φ12×t1.5	
8	1	H01-3163	PIPE, HYDRAULIC	Φ10×t1.5	
9	3	9001401	NIPPLE, PF PIPE	NFP16-PF1/2	
10	1	9022502	TEE	TPU16	
11	13	6800081	WASHER, SEALING	NF1/2	
12	1	S01-2616	TEE, PIPE SWIVEL		
13	2	S01-2614	TEE, PIPE SWIVEL	PF3/8×Φ12×Φ10	
14	2	9270392	UNION, CONICAL	1/2×3/8	
15	1	9270052	UNION, CONICAL	1/4×1/4	
16	1	9002512	NIPPLE, PF PIPE	10-PT1/4	
17	1	9211062	NIPPLE, PT HOSE	PF1/4×PT1/4	
18	7	S01-2612	UNION, HOSE BULKHEAD	Φ12-PF3/8	
19	2	S01-1719	UNION, BULKHEAD	2-PF3/8	
20	1	S01-2613	UNION, HOSE BULKHEAD	Φ10-PF1/4	
21	8	9001361	NIPPLE, PF PIPE	NFP12-PF1/2	
22	1	H01-6754	PIPE, HYDRAULIC	Φ12×t2.0	
23	1	H01-3166	PIPE, HYDRAULIC	Φ12×t1.5	
24	1	H01-3167	PIPE, HYDRAULIC	Φ12×t1.5	
25	1	H01-3168	PIPE, HYDRAULIC	Φ12×t1.5	
26	1	H01-3169	PIPE, HYDRAULIC		
27	1	H01-3170	PIPE, HYDRAULIC	Φ12×t1.5	
28	1	H01-6755	PIPE, HYDRAULIC	Φ12×t2.0	

SECTION 9.1 - BASE HOSE & PIPE

BASE HOSE & PIPE

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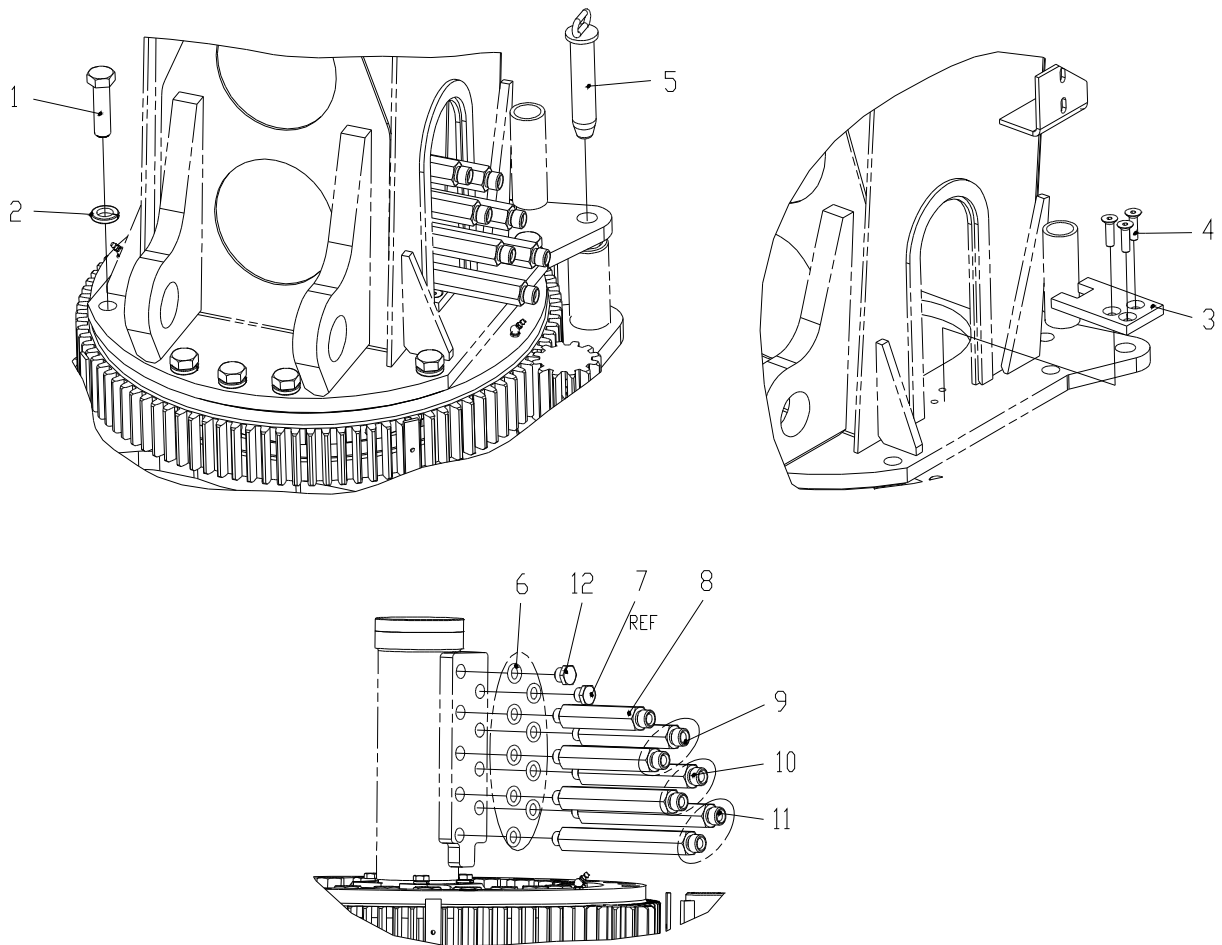
SECTION 9.1 - BASE HOSE & PIPE

BASE HOSE & PIPE					00
Pos.	Q`ty	Part No.	Description	Remarks	
29	1	H01-3172	PIPE, HYDRAULIC	Φ12×t1.5	
30	1	H01-3173	PIPE, HYDRAULIC	Φ12×t1.5	
31	1	H01-2714	PIPE, HYDRAULIC	Φ10×t1.5	
32	3	7116572	HOSE, H.P (HUCF)	3/8×420-PF3/8	
33	2	7116622	HOSE, H.P (HUCF)	3/8×550-PF3/8	
34	1	7116602	HOSE, H.P (HUCF)	3/8×500-PF3/8	
35	2	7170802	HOSE, H.P (HECF)	3/8×1000-PF3/8	
36	1	6800041	WASHER, SEALING	NF1/4	
37	1	H02-0279	GAUGE, LOAD	PT3/8	
38	1	7167911	HOSE, H.P (HECF)	1/4×1280-PF1/4	
39	1	7116672	HOSE, H.P (HUCF)	3/8×680-PF3/8	
40	1	7167783	HOSE, H.P (HECF)	1/4×950-PF1/4	

SECTION 10.1 - COLUMN ASS'Y

COLUMN ASS'Y

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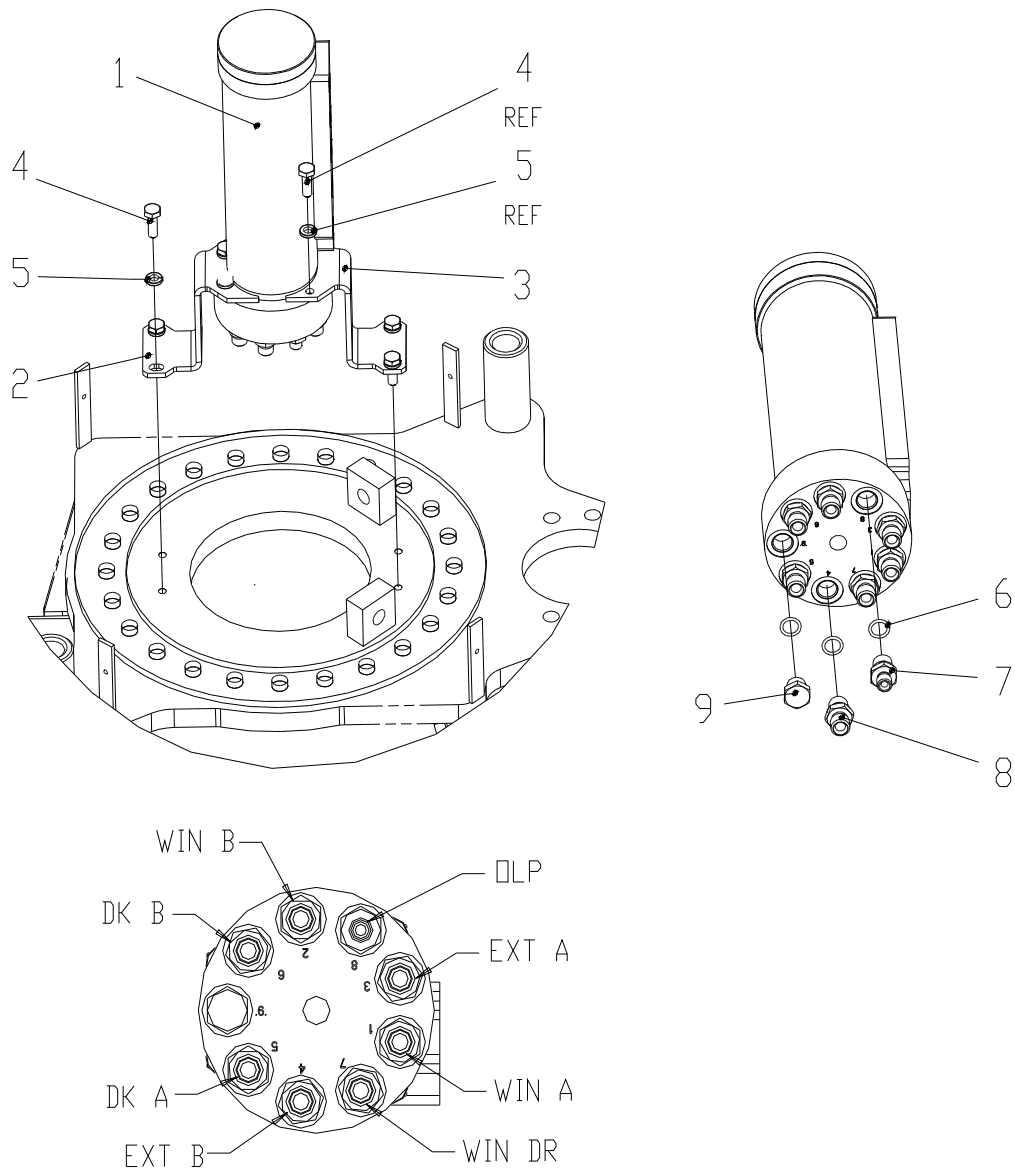
SECTION 10.1 - COLUMN ASS'Y

COLUMN ASS'Y					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	A01-4836	COLUMN ASS'Y		
1	14	S01-1307	BOLT, HEX. HEAD	BA20×70	
2	14	3612201	WASHER, SPRING	WS20	
3	1	M05-2434	BRACKET	t12×70×120	
4	3	BO-0005	BOLT, FALT HEAD	M10×35	
5	1	S01-2086	PIN		
6	8	6010172	O-RING	P14	
7	1	S01-1048	PLUG	PF3/8	
8	1	S01-2239	NIPPLE, LONG		
9	2	BSP-1034	NIPPLE, LONG		
10	2	BSP-1035	NIPPLE, LONG		
11	2	BSP-1036	NIPPLE, LONG		
12	1	S01-1047	PLUG	PT3/8	

SECTION 11.1 - ROTARY JOINT CPL.

ROTARY JOINT CPL.

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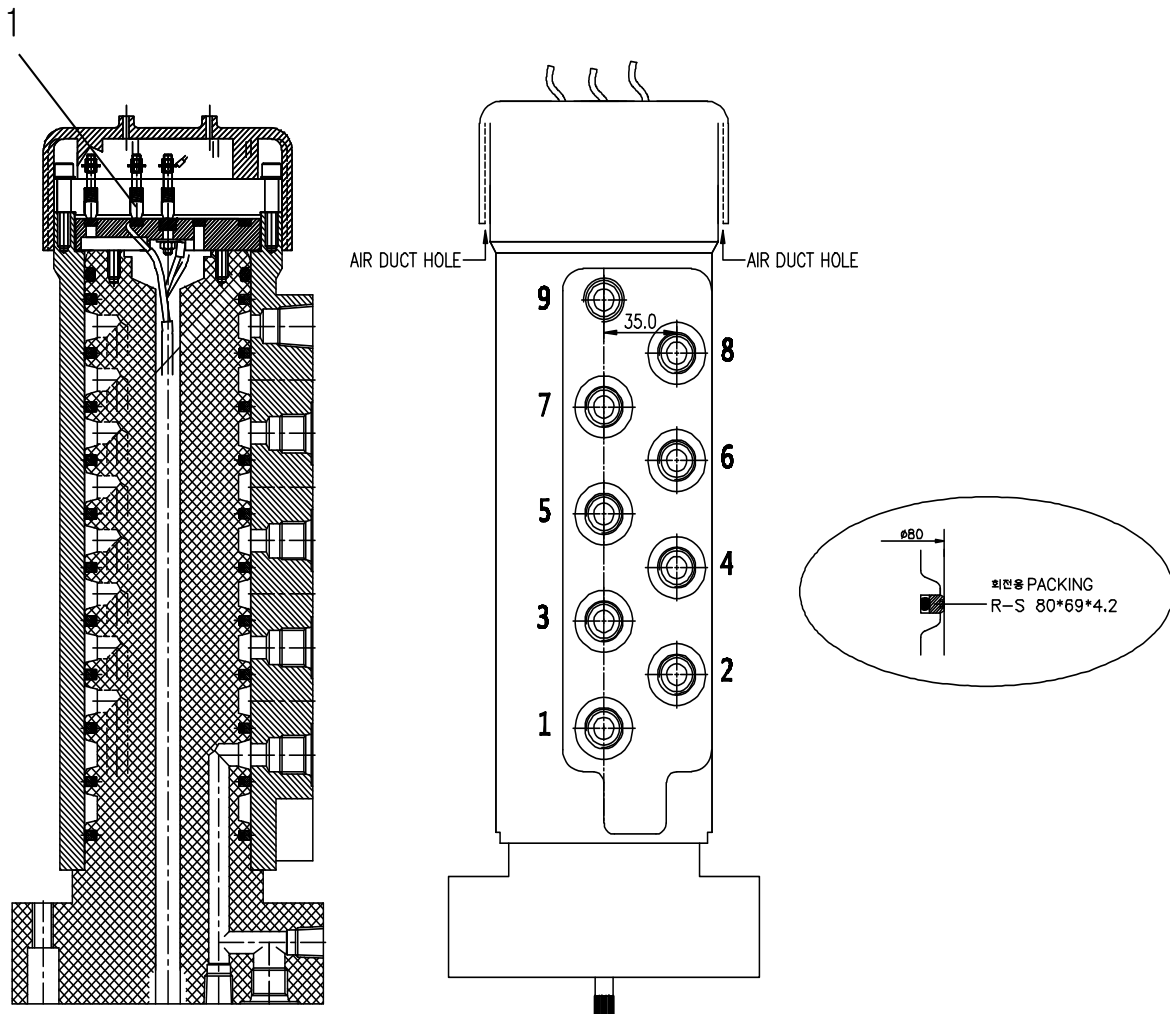
SECTION 11.1 - ROTARY JOINT CPL.

ROTARY JOINT CPL.					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	A01-4829	ROTARY JOINT CPL.		
1	1	A02-1692	ROTARY JOINT ASS'Y	9 PORT	
2	1	W01-3715	BRACKET	t6×100×231	
3	1	W01-3716	BRACKET	t6×100×231	
4	8	2111226	BOLT, HEX. HEAD	BA10×30	
5	8	3612101	WASHER, SPRING	WS10	
6	8	6010172	O-RING	P14	
7	1	S01-0948	NIPPLE, PF HOSE	PF1/4(C)×PF3/8(O)	
8	7	9381032	NIPPLE, PF HOSE	NFCO3/8×3/8	
9	1	S01-1047	PLUG	PF3/8	

SECTION 12.1 - ROTARY JOINT ASS'Y

ROTARY JOINT ASS'Y

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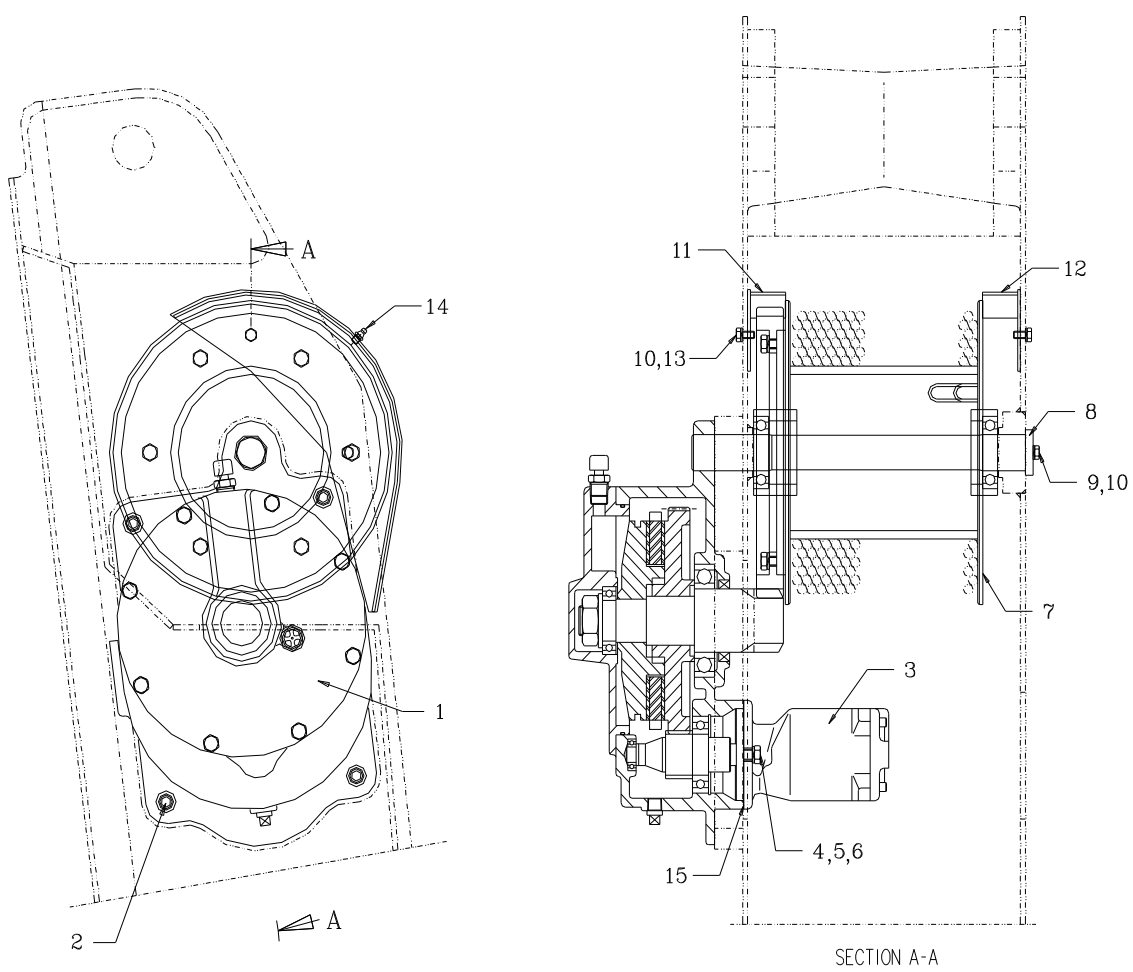
SECTION 12.1 - ROTARY JOINT ASS'Y

ROTARY JOINT ASS'Y					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	A02-1692	ROTARY JOINT ASS'Y	9 PORT	
1	1	S01-4451	SLIP LING ASSY		
-	1	S01-5870	SEAL KIT		

SECTION 13.1 - WINCH SYSTEM

WINCH SYSTEM

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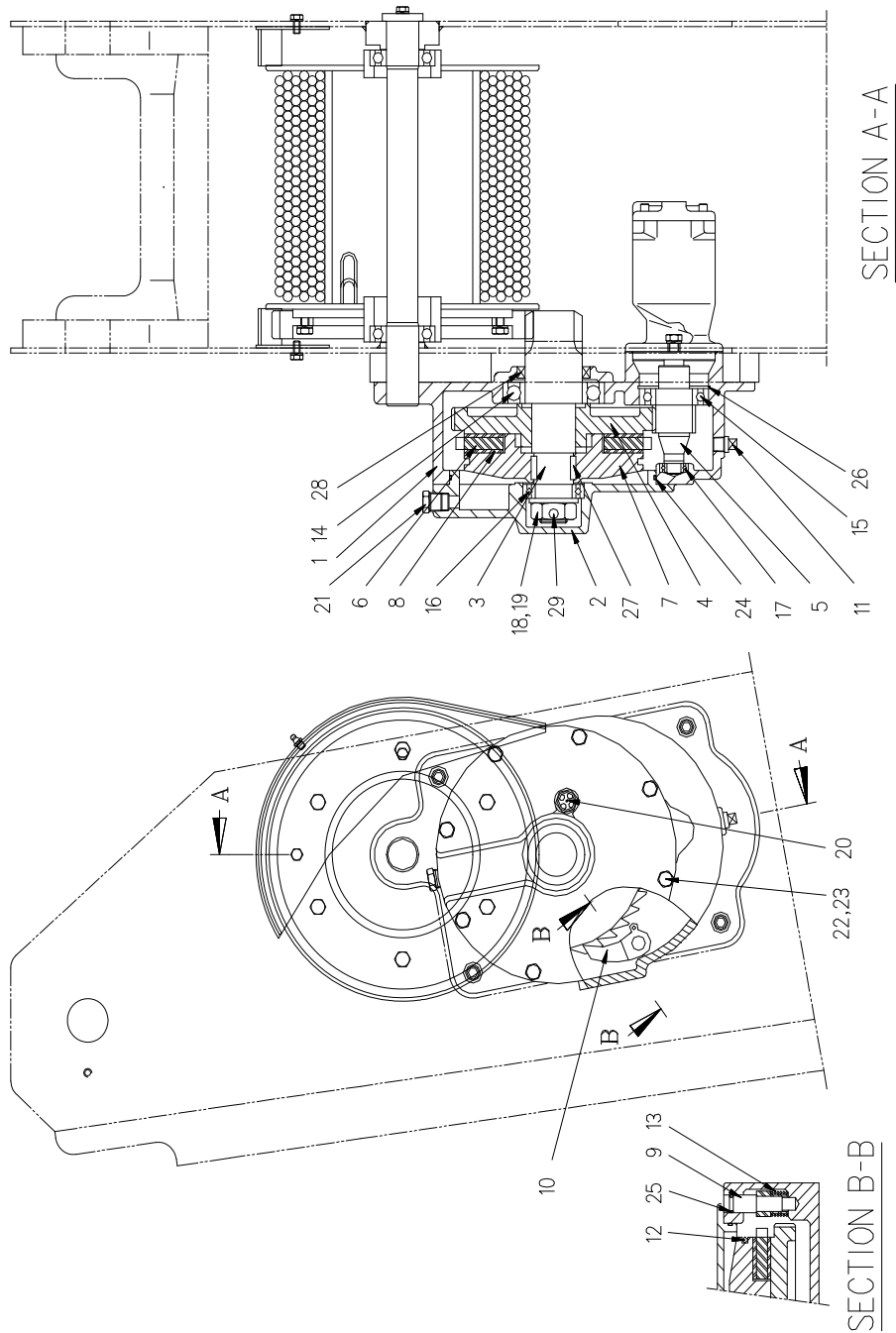
SECTION 13.1 - WINCH SYSTEM

WINCH SYSTEM				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	WC-2001	WINCH SYSTEM	
1	1	H02-0296	REDUCER ASS'Y	
2	4	2611547	BOLT, HEX. SOCKET HEAD	BB12×40
3	1	WC-0003	MOTOR, WINCH	JMF-36-01
-	1	S01-5768	SEAL KIT	
4	2	2111567	BOLT, HEX. HEAD	BA14×50
5	2	3511141	WASHER, PLAIN	WP14
6	2	3612141	WASHER, SPRING	WS14
7	1	WC-2002	DRUM ASS'Y	
8	1	WC-2005	SHAFT	
9	2	2110926	BOLT, HEX. HEAD	BA8×30
10	6	3612081	WASHER, SPRING	WS8
11	1	WC-2006	COVER, LH	
12	1	WC-2008	COVER, RH	
13	4	2110866	BOLT, HEX. HEAD	BA8×16
14	1	4610212	NIPPLE, GREASE	A-MT6×1
15	1	N01-3202	GASKET	t=3.2

SECTION 14.1 - WINCH REDUCER ASS'Y

WINCH REDUCER ASS'Y

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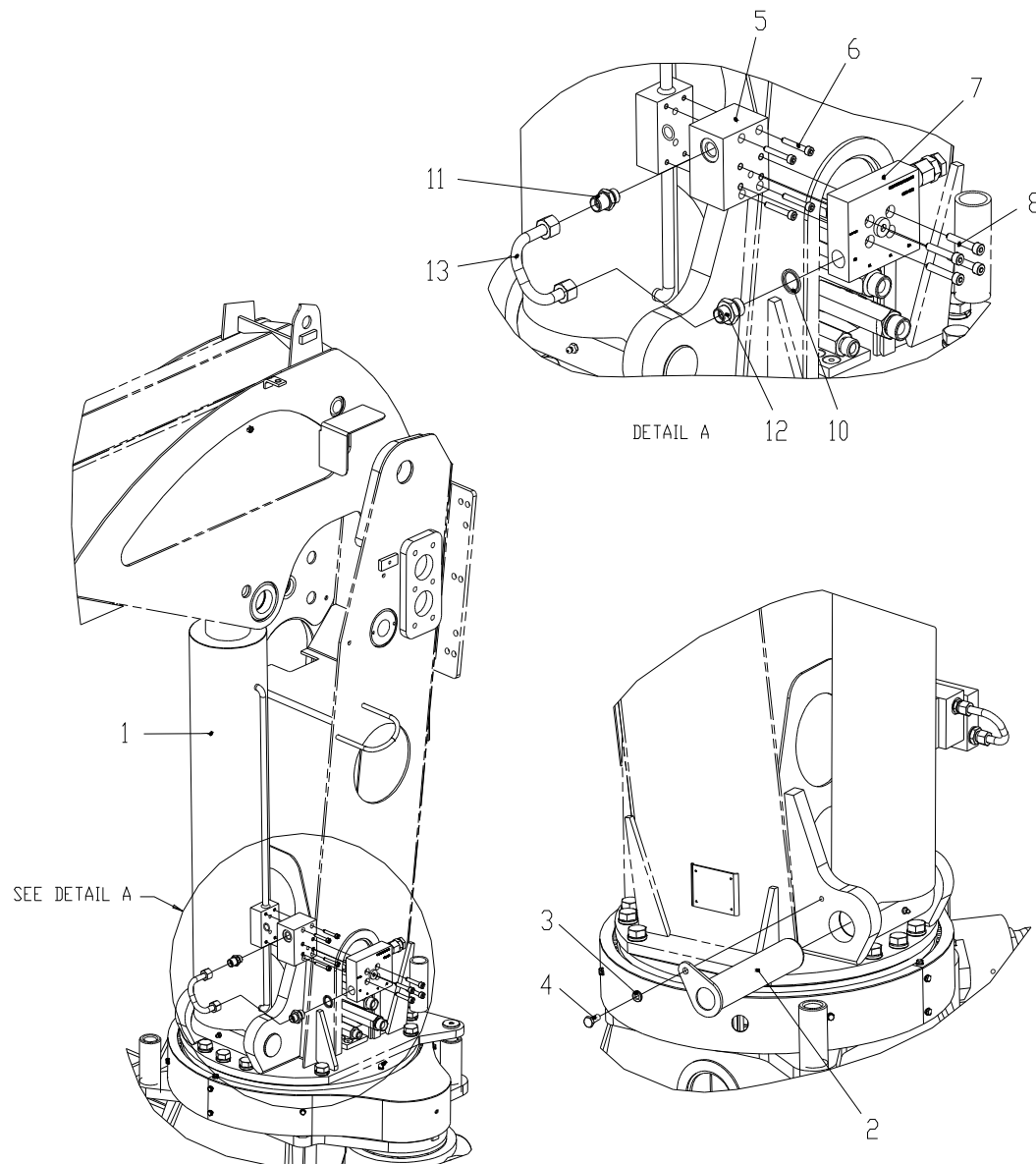
SECTION 14.1 - WINCH REDUCER ASS'Y

WINCH REDUCER ASS'Y				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	H02-0296	WINCH REDUCER ASS'Y	
1	1	DM-06-11-1001-02	CASE	
2	1	DM-06-11-1002	COVER, CASE	
3	1	DM-06-11-1003	GEAR, OUTPUT	
4	1	DM-06-11-1004	GEAR, SPUR	
5	1	DM-06-11-1005	GEAR, INPUT	
6	1	DM-06-11-1006	RATCHET	
7	1	DM-06-11-1007	PLATE, DISK	
8	2	DM-06-11-1008	DISK	
9	1	DM-06-11-1009	SHAFT, HOOK	
10	1	DM-06-11-1010	HOOK	
11	1	DM-06-11-1011	CAP	PT3/8
12	1	DM-06-11-1012	SPRING	
13	1	DM-06-11-1013	SPRING	
14	1	4112144	BEARING, RADIAL BALL	6214ZZ
15	1	4112094	BEARING, RADIAL BALL	6209ZZ
16	1	4110094	BEARING, RADIAL BALL	6009ZZ
17	1	4110034	BEARING, RADIAL BALL	6003ZZ
18	1	DM-06-11-1014	NUT, SLOT HEX.	M33
19	1	DM-06-11-1015	WASHER, PLAIN	M33
20	1	DM-06-11-1016	GAUGE, OIL	PF1/2
21	1	DM-06-11-1017	BRETHET, AIR	
22	8	2111206	BOLT, HEX. HEAD	BA10×25
23	8	3612101	WASHER, SPRING	
24	1	DM-06-11-1018	O-RING	WS10
25	1	6010192	O-RING	273
26	1	4525561	RING, RETAINING	1B P16
27	2	DM-06-11-1019	KEY, PLAIN	RB85
28	1	DM-06-11-1020	SEAL, OIL	
29	1	1311141	PIN, SPLIT	PW5×50

SECTION 15.1 - DERRICK CYLINDER CPL.

DERRICK CYLINDER CPL.

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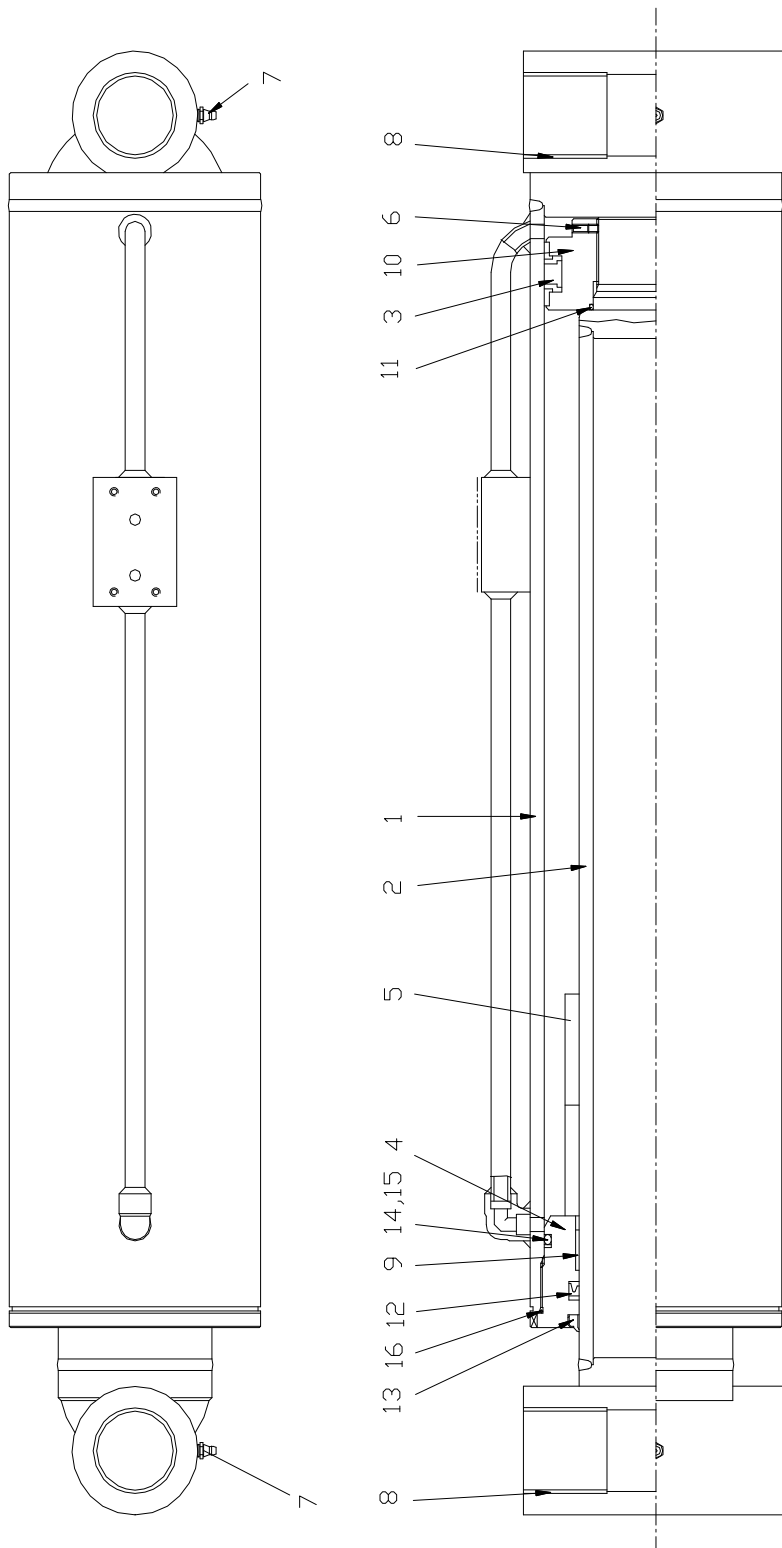
SECTION 15.1 - DERRICK CYLINDER CPL.

DERRICK CYLINDER CPL.					01
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	A01-4800	DERRICK CYLINDER CPL.		
1	1	H01-5022	DERRICK CYLINDER ASS'Y		
2	1	BC1-3005	SHAFT	Φ55×260	
3	1	3612101	WASHER, SPRING	WS10	
4	1	2111206	BOLT, HEX. HEAD	BA10×25	
5	1	M01-1423	BLOCK		
6	4	2615547	BOLT, HEX. SOCKET HEAD	BB6×40	
7	1	456 4702	LOAD HOLDING VALVE	SINGLE	
8	4	2615957	BOLT, HEX. SOCKET HEAD	BB8×45	
9	1	6010172	O-RING	P14	
10	1	6800081	WASHER, SEALING	NF1/2	
11	1	9003351	NIPPLE, PF PIPE	NTPO12-PF3/8	
12	1	9001361	NIPPLE, PF PIPE	Φ12-PF1/2	
13	1	H01-3440	PIPE, HYDRAULIC	Φ12×t2.0	

SECTION 16.1 - DERRICK CYLINDER ASS'Y

DERRICK CYLINDER ASS'Y

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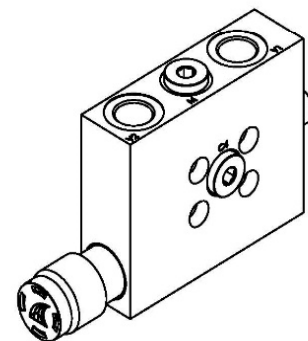
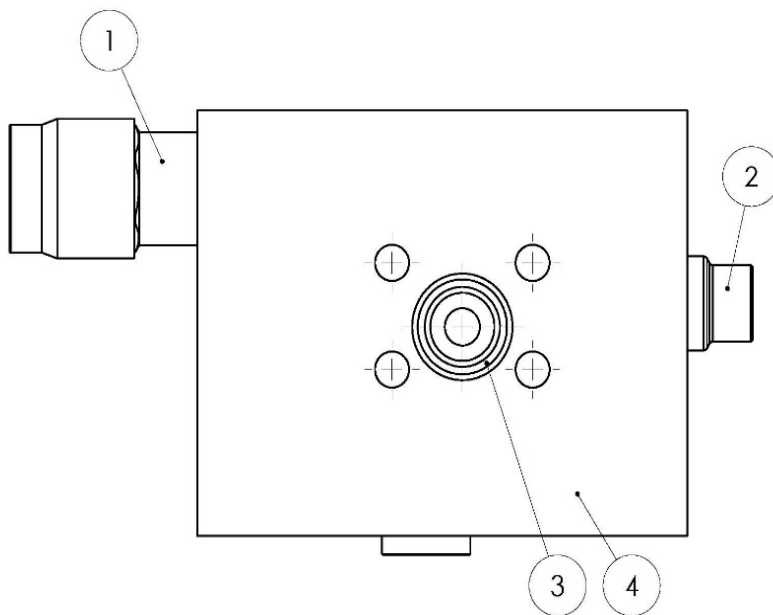
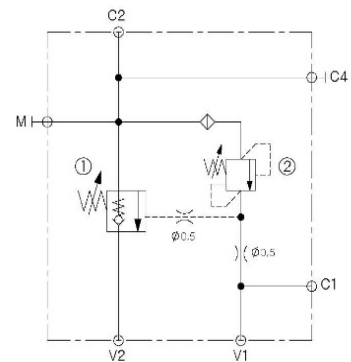
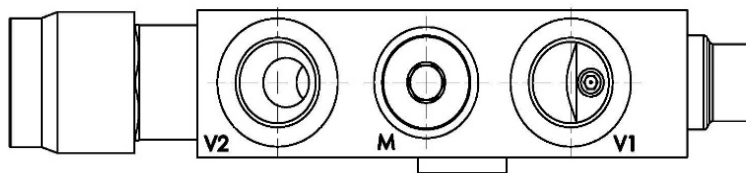
SECTION 16.1 - DERRICK CYLINDER ASS'Y

DERRICK CYLINDER ASS'Y					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	H01-5022	DERRICK CYLINDER ASS'Y	HN89D0000	
1	1	HN89D0010	TUBE ASS'Y		
2	1	HN80D0020	ROD ASS'Y		
3	1	HN01D0030	PISTON		
4	1	SH15DA040-01	ROD COVER		
5	2	HN89D0050	SPACER	Φ105.5×Φ125×100L	
6	1	2350466	SET, SCREW	M6×P1.0	
7	1	4610702	GREASE NIPPLE	A-PT1/8	
8	4	4325451	DX BUSHING	PM5560DX	
9	1	S01-0867	DU BUSHING	MB10530DU	
10	1	-	PISTON PACKING	PCB 1A1600-NCRO	*
11	1	6012412	O-RING	1B G105	*
12	1	S01-2189	ROD PACKING	NI - 300	*
13	1	S01-4110	DUST WIPER	KWU 105	*
14	1	6012502	O-RING	1B G150	*
15	1	6035911	RING, BACK-UP	T3 G150	*
16	1	6012492	O-RING	1B G145	*
-	1	S01-2126	KIT, SEAL	PARTS MARKED *	

SECTION 17.1 - LOAD HOLDING VALVE

LOAD HOLDING VALVE

01



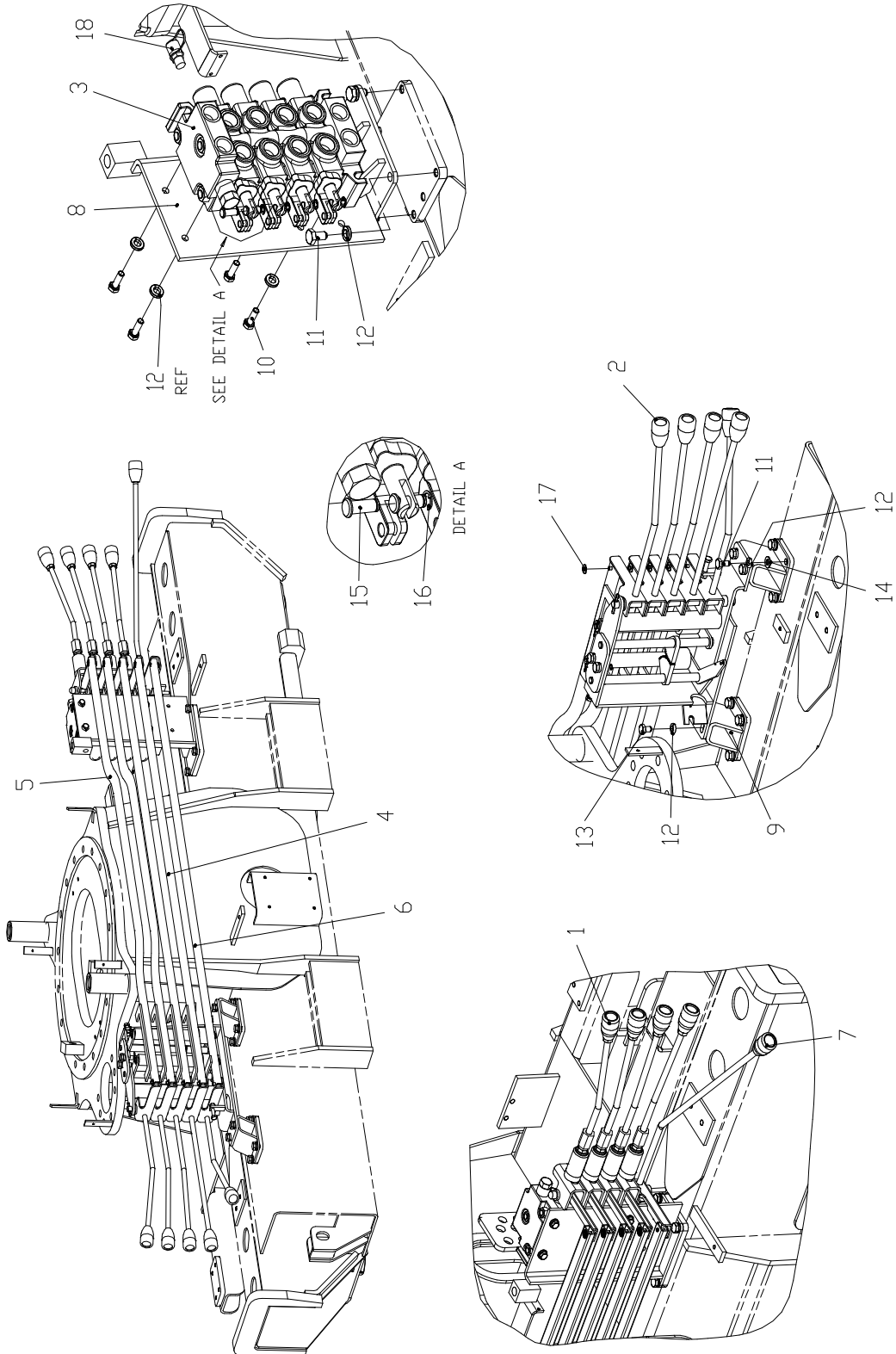
SECTION 17.1 - LOAD HOLDING VALVE

LOAD HOLDING VALVE				01
Pos.	Q'ty	Part No.	Description	Remarks
-	1	456 4702	LOAD HOLDING VALVE	MB000372
1	1	LB000001	MANIFOLD	
2	1	AR700145	OR 18.64 x 3.53 - N70	
3	1	CP000015	CPLN-005-DPNA-PF-005-N460	
4	1	SB000261	SG SBSN-100 1:0 350 bar F11	

SECTION 18.1 - MAIN CONTROL VALVE SYSTEM

MAIN CONTROL VALVE SYSTEM

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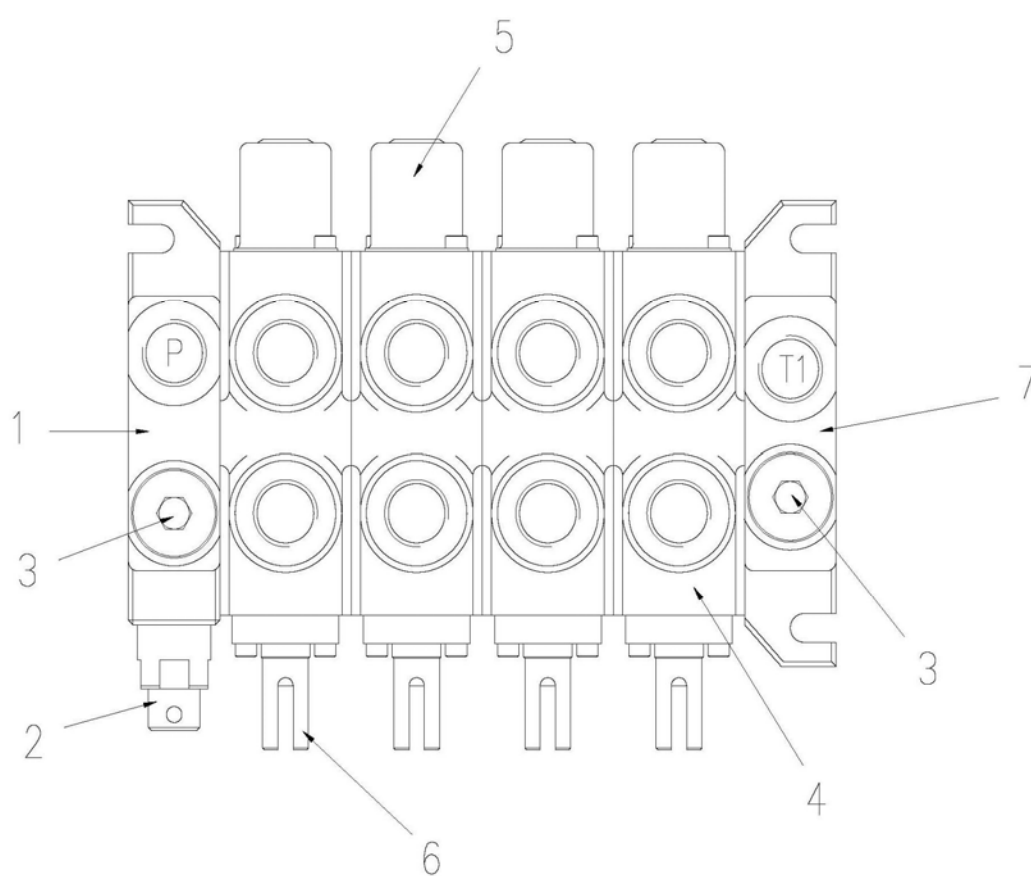
SECTION 18.1 - MAIN CONTROL VALVE SYSTEM

MAIN CONTROL VALVE SYSTEM					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	A01-5347	MAIN CONTROL VALVE SYSTEM		
1	1	A02-1671	LEVER ASS'Y, MAIN (1)		
2	1	A02-2172	LEVER ASS'Y, MAIN (2)		
3	1	H02-0289	MAIN CONTROL VALVE	RS214	
4	2	A05-2669	ROD, LEVER		
5	2	A05-2670	ROD, LEVER		
6	1	A05-2671	ROD, LEVER		
7	1	A05-2722	LEVER ASS'Y		
8	1	A05-2456	BRACKET ASS'Y		
9	2	A05-1967	BRACKET ASS'Y		
10	4	2110907	BOLT, HEX. HEAD	BA8×25	
11	12	2111177	BOLT, HEX. HEAD	BA10×18	
12	20	3612101	WASHER, SPRING	WS10	
13	4	2111156	BOLT, HEX. HEAD	BA10×14	
14	8	3511101	WASHER, PLAIN	WP10	
15	4	S01-0049	PIN	Φ5×19.5	
16	4	4551091	PING, RETAINING	RE6	
17	8	S01-0854	WASHER, BACK	SPN06	
18	1	998 9676	MEASURING NIPPLE	PF1/4	

SECTION 19.1 - MAIN CONTROL VALVE

MAIN CONTROL VALVE

00



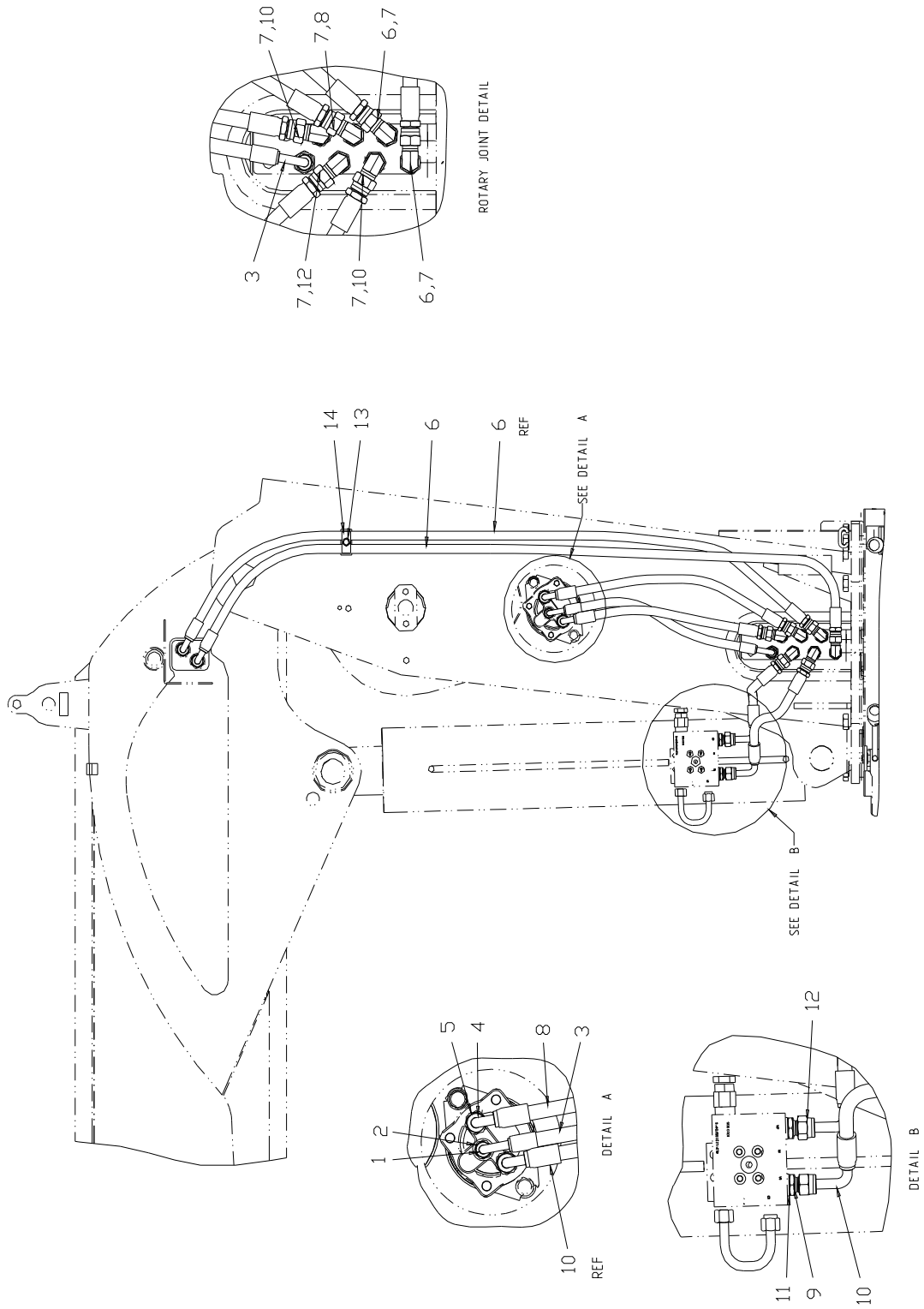
SECTION 19.1 - MAIN CONTROL VALVE

MAIN CONTROL VALVE				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	H02-0289	MAIN CONTROL VALVE	RS214
1	1	2330-260982	P BLOCK	
2	1	2330-210203	VALVE ASSY, MAIN RELIEF	
-	1		SCREW, ADJUSTING	
-	1		O-RING	13.3×2.4
-	1		BUSHING	
-	1		O-RING	17.3×2.4
-	1		SPRING	
-	1		SPACER	
-	1		POPPET	
-	1		RING, BACK-UP	
-	1		O-RING	12.3×2.4
3	2	2330-450059	PLUG ASSY, PORT	PF1/2
-	1		PLUG, PORT	
-	1		O-RING	17.3×2.4
4	4	2330-263162	WORK BLOCK	
5	4	2330-260944	SPOOL CONTROL ASSY	9:10
-	1		SEAL, SPOOL	
-	1		WASHER	
-	2		SCREW	
-	1		CAP	
-	1		GUIDE, SPRING	
-	1		SCREW, SPACING	
-	1		GUIDE, SPRING	
-	1		SPRING	
6	1	2330-266992	3-POSITIONS BRACKET ASSY	9:1
-	1		BRACKET	
-	1		RING, SCRAPER	
-	1		SPOOL SEAL ASSY	
-	1		SCREW	
7	1	2330-260984	T BLOCK	

SECTION 20.1 - COLUMN HOSE ASS'Y

COLUMN HOSE ASS'Y

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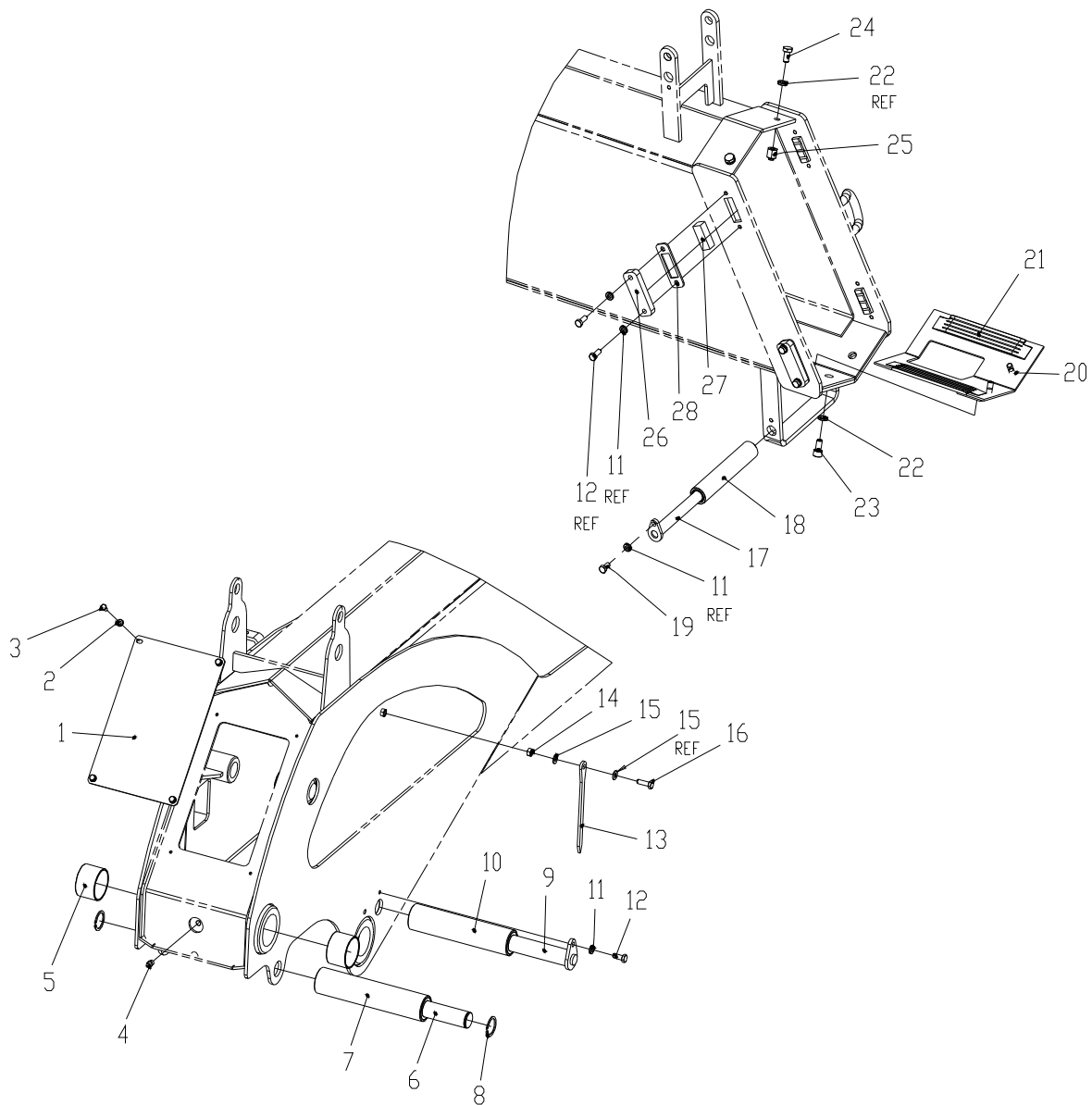
SECTION 20.1 - COLUMN HOSE ASS'Y

COLUMN HOSE ASS'Y				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	A01-3671	COLUMN HOSE ASS'Y	
1	1	N01-3217	NIPPLE, LONG	PF3/8×PF3/8-210
2	1	6010172	O-RING	P14
3	1	H01-7088	HOSE, H.P (HECF)	3/8×550-PF3/8
4	2	N01-3218	NIPPLE, LONG	PF1/2×PF1/2-175
5	2	6010212	O-RING	P18
6	2	7172602	HOSE, H.P (HECF)	3/8×1750-PF1/2
7	6	S01-1579	ELBOW, SWIVEL	2-PF1/2
8	1	7172132	HOSE, H.P (HECF)	3/8×580-PF1/2
9	2	9270072	UNION, CONICAL	1/2×1/2
10	2	7172102	HOSE, H.P (HECF)	3/8×500-PF1/2
11	2	6800081	WASHER, SEALING	NF1/2
12	1	7172072	HOSE, H.P (HECF)	3/8×420-PF1/2
13	1	2185936	BOLT, HEX. W/F	BWF8×35
14	2	TS-0038	CLAMP, HOSE	3/8

SECTION 21.1 - 1st BOOM ASS'Y

1st BOOM ASS'Y

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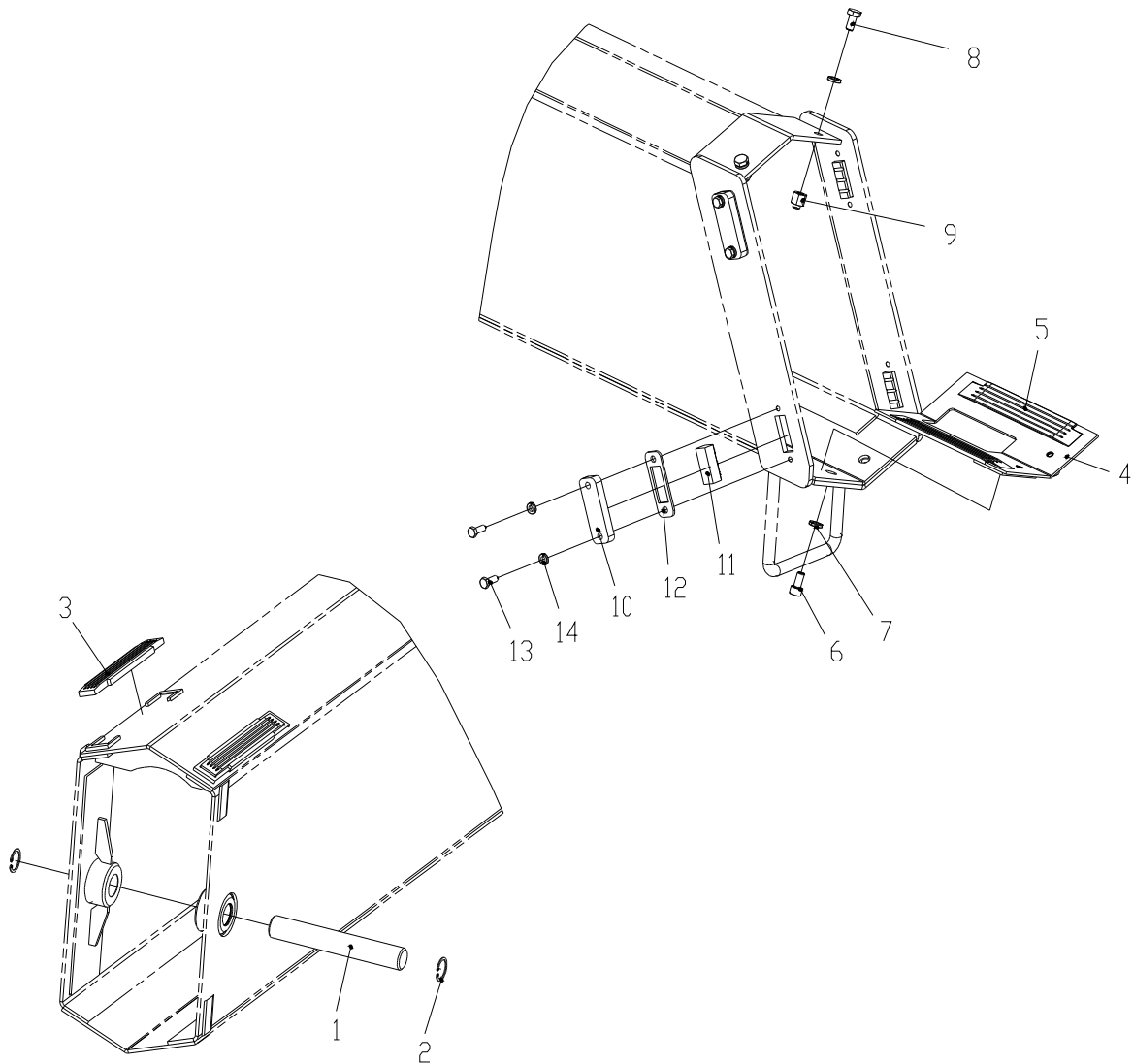
SECTION 21.1 - 1st BOOM ASS'Y

1st BOOM ASS'Y				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	A01-4802	1st BOOM ASS'Y	
1	1	B00-0062	PLATE	t2.3×190×290
2	4	3612061	WASHER, SPRING	WS6
3	4	2110447	BOLT, HEX HEAD	BA6×12
4	1	4610702	GREASE NIPPLE	FNA-PT1/8
5	2	4321251	BUSHING, DU	MB5540DU
6	1	M01-1047	PIN	Φ30×250L
7	1	A05-1814	ROLLER ASS'Y	
8	2	4510291	RING, RETAINING	RS30
9	1	TA-0005	SHAFT	
10	1	A05-1813	ROLLER ASS'Y	Φ48.6×210L
11	10	3612081	WASHER, SPRING	WS8
12	9	2110887	BOLT, HEX HEAD	BA8×20
13	2	W01-3761	INDICATOR, ROD	t4.5
14	2	3111086	NUT, HEX.	NA8
15	4	3511081	WASHER, PLAIN	WP8
16	2	2110906	BOLT, HEX HEAD	BA8×25
17	1	B00-2005	SHAFT	
18	1	B00-2006	ROLLER ASS'Y	Φ30×180
19	1	2110867	BOLT, HEX HEAD	BA8×16
20	1	B00-5052	GUIDE, PAD	
21	2	B00-5055	FRONT LOWER PAD	
22	4	3612101	WASHER, SPRING	WS10
23	2	2611186	BOLT, HEX. SOCKET HEAD	BB10×20
24	2	2111187	BOLT, HEX. HEAD	BA10×20
25	2	B00-5014	NUT, HEX.	M10×16L
26	4	N01-3491	PLATE	32×90×12t
27	4	S01-4209	PAD, SIDE	49×19×18t
28	4	B00-5043	SHIM	90×30×3t

SECTION 22.1 - 2nd BOOM ASS'Y

2nd BOOM ASS'Y

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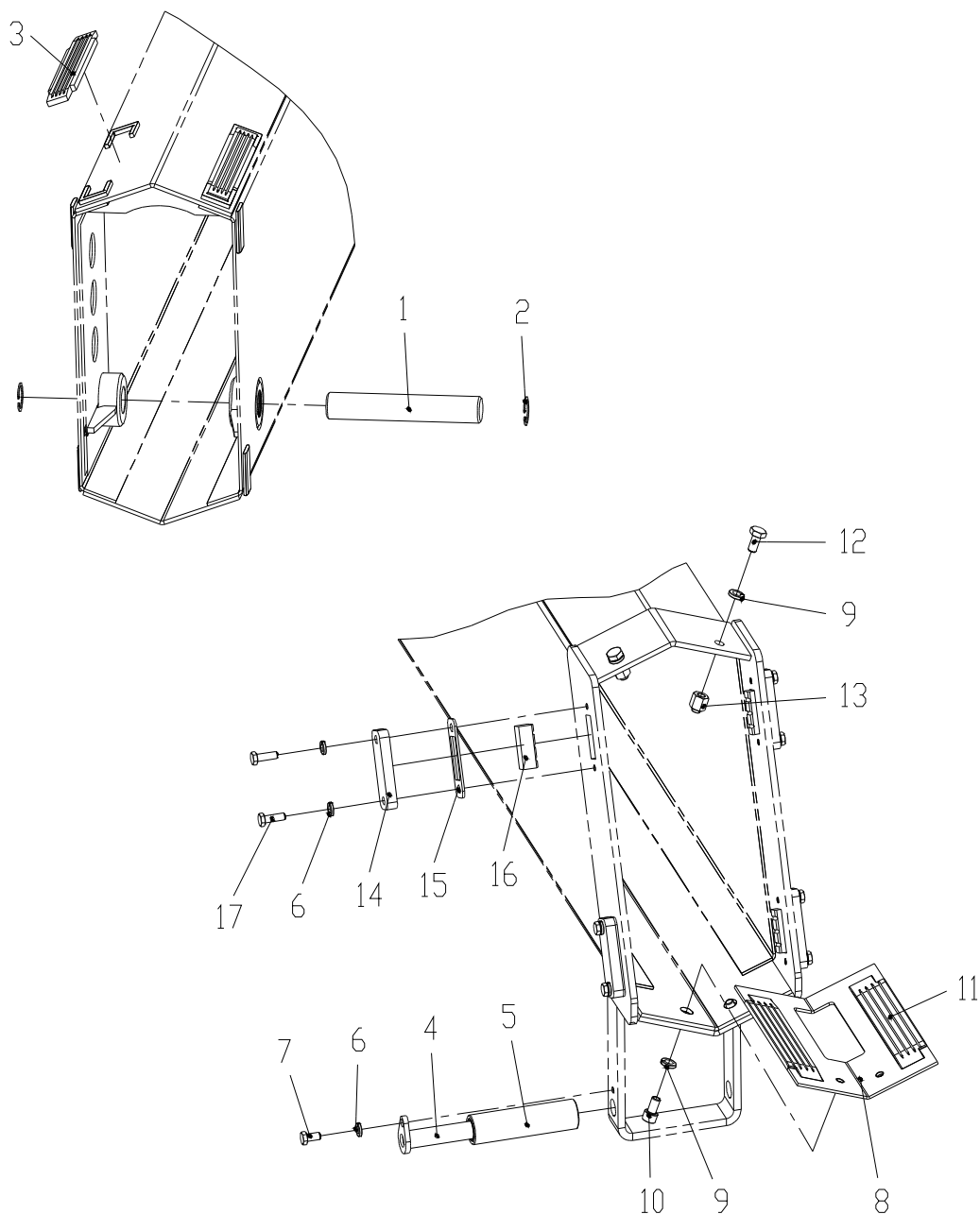
SECTION 22.1 - 2nd BOOM ASS'Y

2nd BOOM ASS'Y				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	A01-4803	2nd BOOM ASS'Y	
1	1	B00-2003	SHAFT	Φ30×200
2	2	4525281	RING, RETAINING	RB30
3	2	B00-5037	PAD, SLIDE	t9×35×220
4	1	B00-5056	GUIDE, PAD	
5	2	B00-5055	FRONT LOWER PAD	
6	2	2611186	BOLT, HEX. SOCKET HEAD	BB10×20
7	4	3612101	WASHER, SPRING	WS10
8	2	2111187	BOLT, HEX. HEAD	BA10×20
9	2	B00-2018	NUT, HEX.	M10×15L
10	4	N01-3491	PLATE	32×90×12t
11	4	S01-4209	PAD, SIDE	49×19×18t
12	4	B00-5043	SHIM	90×30×3t
13	8	2110887	BOLT, HEX. HEAD	BA8×20
14	8	3612081	WASHER, SPRING	WS8

SECTION 23.1 - 3rd BOOM ASS'Y

3rd BOOM ASS'Y

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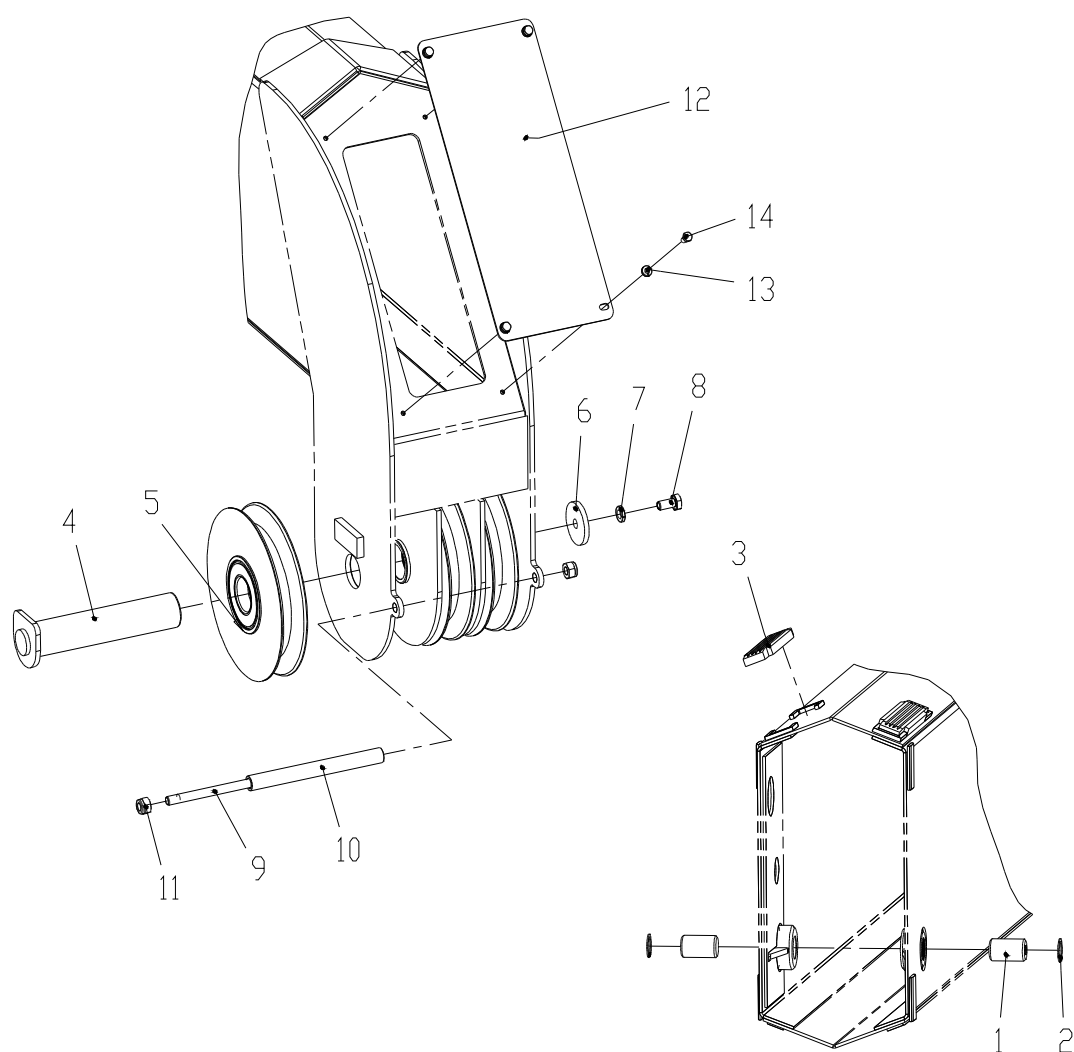
SECTION 23.1 - 3rd BOOM ASS'Y

3rd BOOM ASS'Y				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	A01-5491	3rd BOOM ASS'Y	
1	1	B00-2004	SHAFT	Φ30×182
2	2	4525281	RING, RETAINING	RB30
3	2	B00-5038	PAD, SLIDE	t8×35×220
4	1	S01-5112	SHAFT	Φ20×153
5	1	B00-2009	ROLLER ASS'Y	
6	9	3612081	WASHER, SPRING	WS8
7	1	2110867	BOLT, HEX. HEAD	BA8×16
8	1	B00-5059	GUIDE, PAD	
9	4	3612101	WASHER, SPRING	WS8
10	2	2611176	BOLT, HEX. SOCKET HEAD	BA8×25
11	2	B00-5055	FRONT LOWER PAD	WS10
12	2	2111187	BOLT, HEX. HEAD	BA10×20
13	2	B00-5014	NUT, HEX.	t17.5
14	4	N01-3491	PLATE	T=11.5
15	4	B00-0058	SHIM	T=6
16	4	S01-3121	PAD, SIDE	BA8×20
17	8	2110906	BOLT, HEX. HEAD	t9×35×220

SECTION 24.1 - 4th BOOM ASS'Y

4th BOOM ASS'Y

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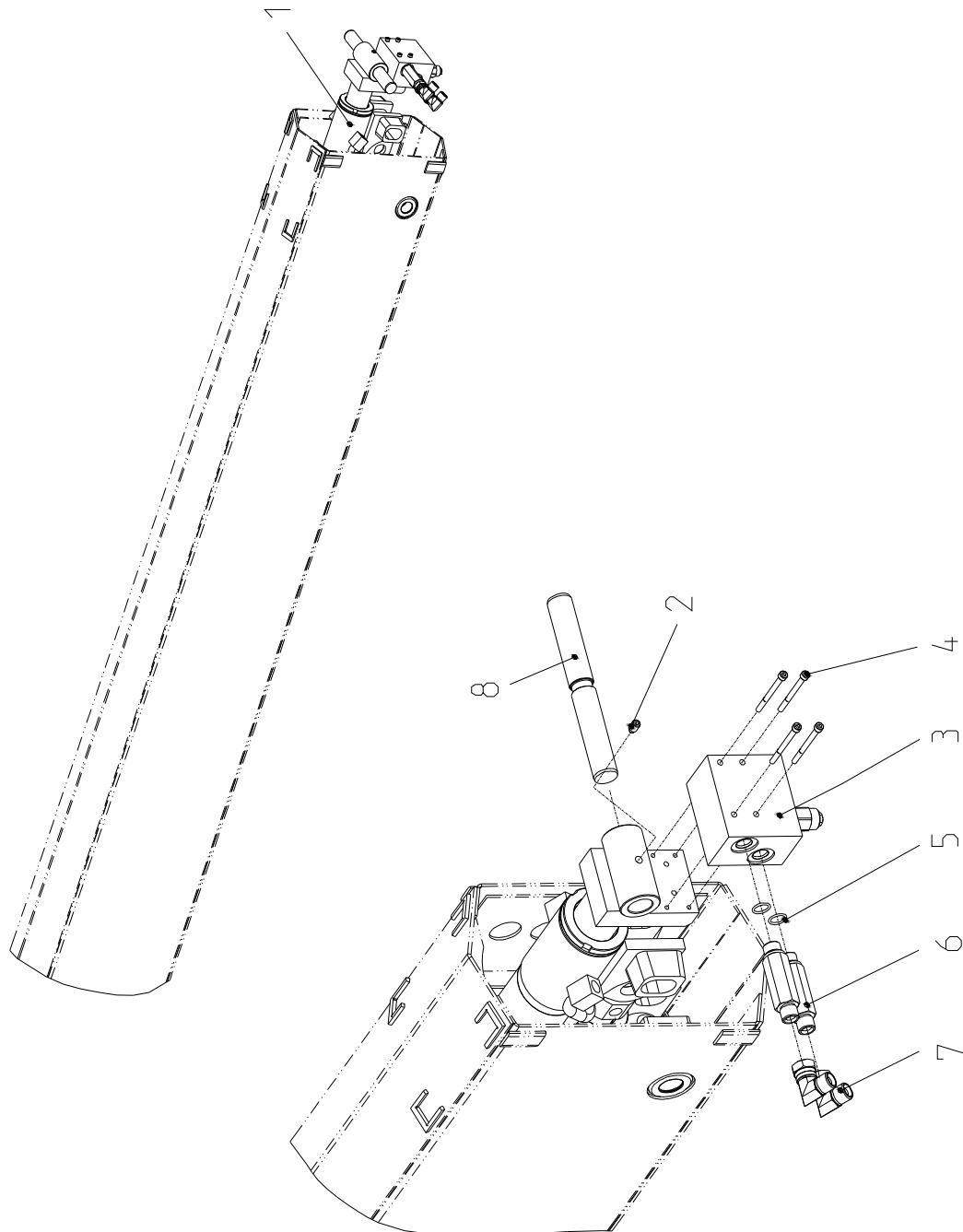
SECTION 24.1 - 4th BOOM ASS'Y

4th BOOM ASS'Y				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	A01-5492	4th BOOM ASS'Y	
1	2	B00-0007	PIN	Φ25×45
2	2	4525251	RING, RETAINING	RB25
3	2	B00-5039	PAD, SLIDE	t10.6×35×220
4	1	B00-7002	SHAFT	Φ40×189
5	3	A05-3387	SHEAVE ASS'Y	
6	1	N01-0543	PLATE	Φ50×6t
7	1	3612101	WASHER, SPRING	WS10
8	1	2111187	BOLT, HEX. HEAD	BA10×20
9	1	M01-1852	BOLT	M10×220L
10	1	M01-1851	PIPE	Φ17.3×t2.8×173
11	2	S01-0006	NUT, SELF LOCKING	M10
12	1	B00-7003	PLATE, COVER	T=2.3
13	4	3612061	WASHER, SPRING	WS6
14	4	2110447	BOLT, HEX. HEAD	BA6×12

SECTION 25.1 - EXTENSION CYLINDER CPL. (SINGLE)

EXTENSION CYLINDER CPL (SINGLE)

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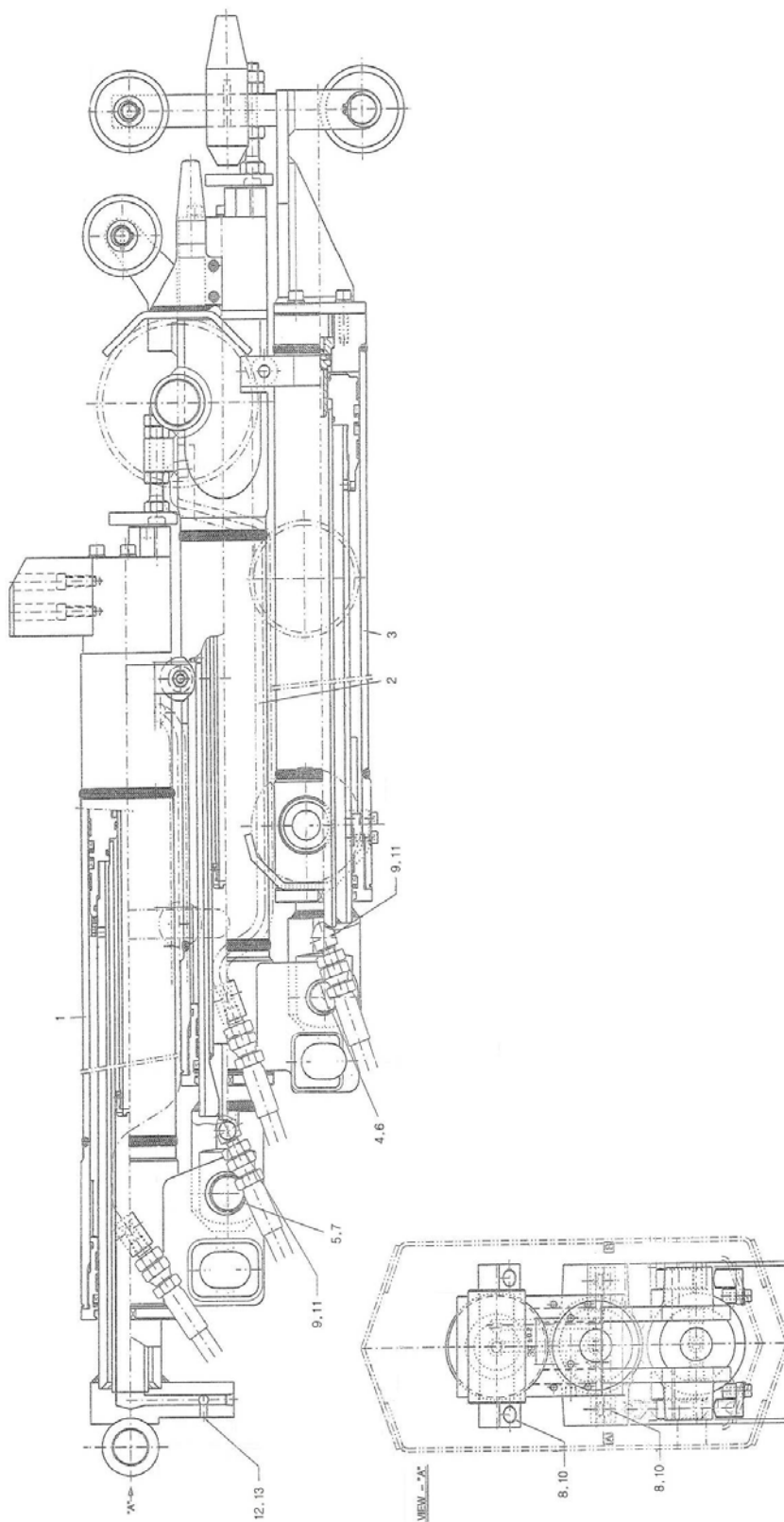
SECTION 25.1 - EXTENSION CYLINDER CPL. (SINGLE)

EXTENSION CYLINDER CPL (SINGLE)					00
Pos.	Q`ty	Part No.	Description	Remarks	
-	1	A01-5493	EXTENSION CYLINDER CPL		
1	1	H01-2963	EXTENSION CYLINDER ASS'Y		
2	1	2350796	SCREW, SET	BV10×16	
3	1	H02-0201	VALVE, COUNTER BALANCE	9706-04D-P09	
4	4	2615617	BOLT, HEX. SOCKET HEAD	BB6×75	
5	2	6010212	O-RING	P18	
6	2	B00-5051	NIPPLE, LONG	PF1/2(O)×PF1/2(C)	
7	2	S01-1579	ELBOW, SWIVEL	2-PF1/2	
8	1	B00-5002	SHAFT	Φ30×240	

SECTION 26.1 - EXTENSION CYLINDER ASS'Y

EXTENSION CYLINDER ASS'Y

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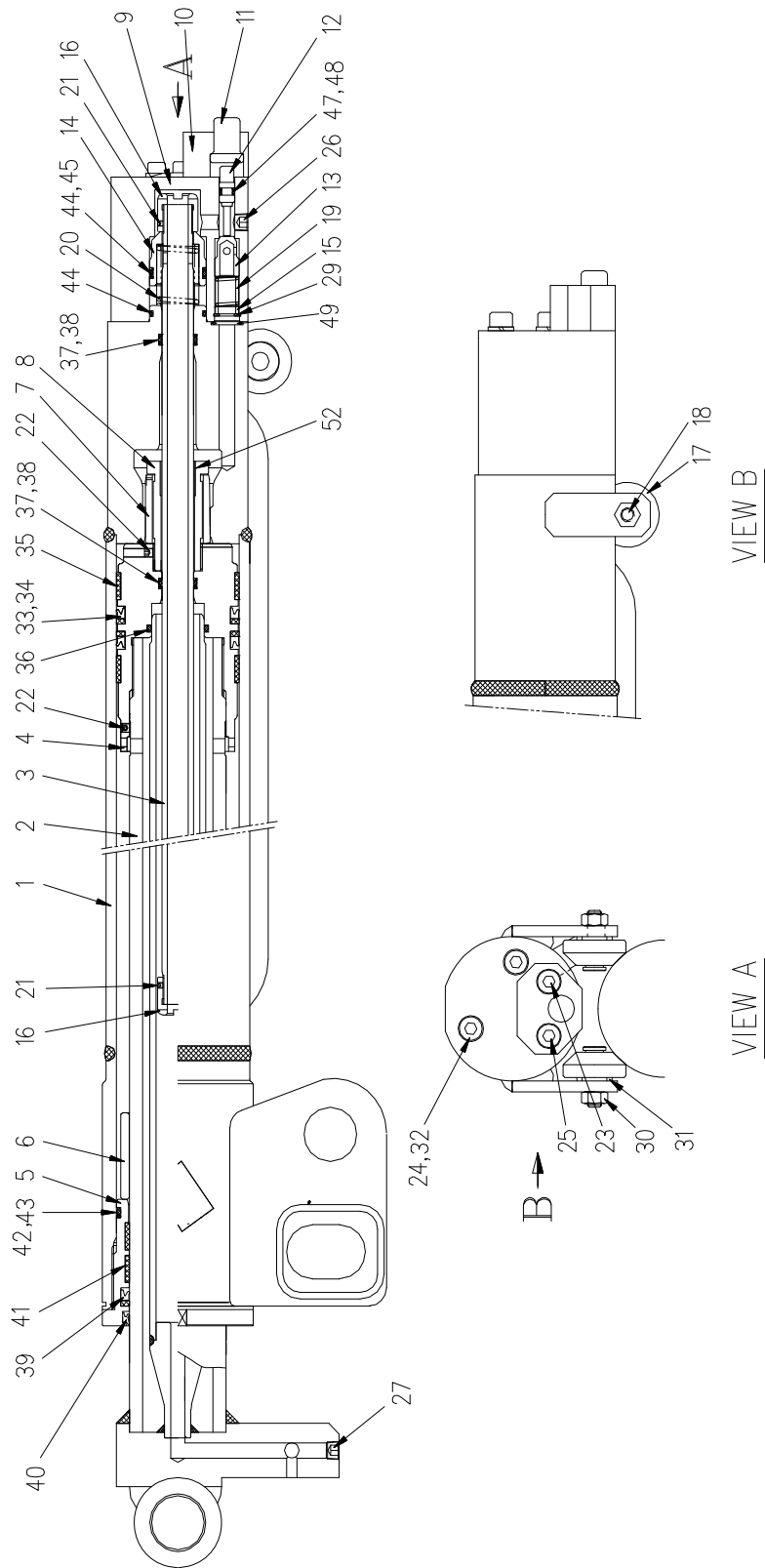
SECTION 26.1 - EXTENSION CYLINDER ASS'Y

EXTENSION CYLINDER ASS'Y					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	H01-2963	EXTENSION CYLINER ASS'Y	HN39T0000-01	
1	1	H01-2963-1	1st EXTENSION CYLINDER ASS'Y	HN39T1000	
2	1	H01-2963-2	2nd EXTENSION CYLINDER ASS'Y	HN39T2000-01	
3	1	H01-2963-3	3rd EXTENSION CYLINDER ASS'Y	HN39T3000-01	
4	1	SS25T0010	PIN	Φ25×73L	
5	1	SS33T0010	PIN	Φ30×73L	
6	2	-	SNAP RING	Φ25	
7	2	-	SNAP RING	Φ30	
8	2	-	HOSE, HYDRAULIC	PF3/8×PF3/8×400L	
9	2	-	HOSE, HYDRAULIC	PF3/8×PF3/8×480L	
10	4	-	ADAPTER-A	ZPC3010-6-6(3/8×3/8)	
11	4	-	ADAPTER-B	ZPC3090-6-6(3/8×3/8)	
12	1	-	DUST COVER	67×85×3T	
13	2	-	SOCKET BOLT	M6×P1.0×10L	
14	2	-	TI CABLE		

SECTION 27.1 - 1st EXTENSION CYLINDER ASS'Y

1st EXTENSION CYLINDER ASS'Y

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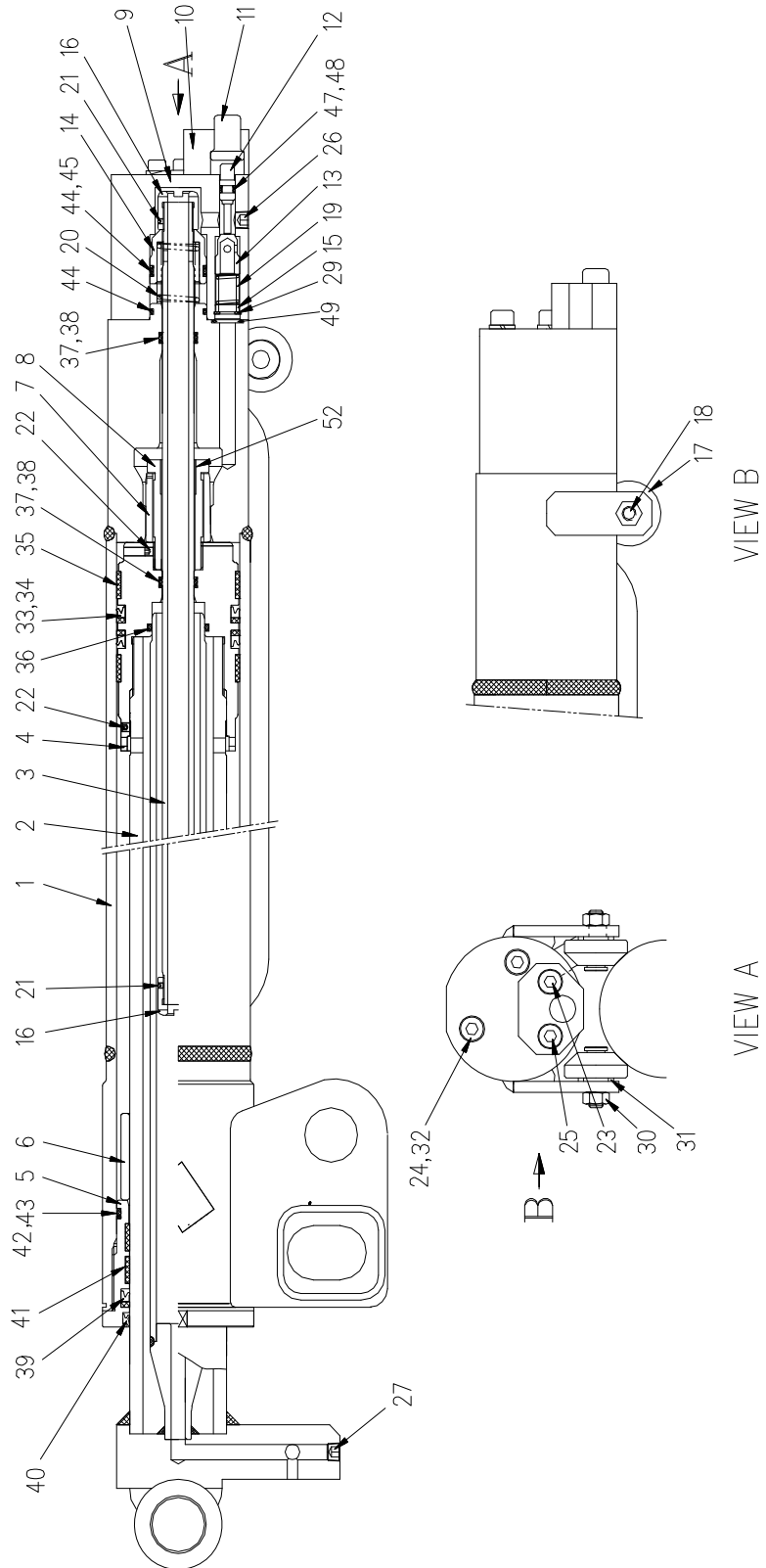
SECTION 27.1 - 1st EXTENSION CYLINDER ASS'Y

1st EXTENSION CYLINER ASS'Y				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	H01-2963-1	1st EXTENSION CYLINDER ASS'Y	HN39T1000
1	1	KL08T1010-01	TUBE ASS'Y	
2	1	KL08T1020-02	ROD ASS'Y	
3	1	KL08T1A30	RETRACTOR	
4	1	SH15T1040	PISTON	
5	1	SH46T1040	ROD COVER	
6	1	KL08T1040	SPACER	
7	1	SS33T1070	CUSHION RING	
8	1	SS33T1080	CUSHION GUIDE	
9	1	SH16T1090	VALVE BODY	
10	1	SS33T10A1	BRACKET-A	
11	1	SS33T10A2	SPOOL GUIDE	
12	1	SS33T10A3	SPOOL	
13	1	SS33T10A4	VALVE-A	
14	1	SS33T10A5	VALVE-B	
15	1	SS33T10A6	WASHER-A	
16	2	HN01T1060	STOP NUT	
17	2	KL08T10P1	ROLLER PAD	Φ36×14W
18	2	KL08T1070	ROLLER SHAFT	Φ13×33.5L
19	1	SS33T10S1	SPRING-A	
20	1	SS33T10S2	SPRING-B	
21	2	2350336	SCREW, SET	BV5×5
22	2	2350446	SCREW, SET	BV6×6
23	1	2610937	BOLT, HEX. SOCKET HEAD	BB8×35
24	2	2611067	BOLT, HEX. SOCKET HEAD	BB8×100
25	1	2611087	BOLT, HEX. SOCKET HEAD	BB8×110
26	2	S01-0493	PLUG	PT1/8×7L
27	3	S01-0398	PLUG	PT1/4×9L
28	2	BOO-0002-1-23	SCREW, SET	#6900ZZ

SECTION 27.1 - 1st EXTENSION CYLINDER ASS'Y

1st EXTENSION CYLINER ASS'Y

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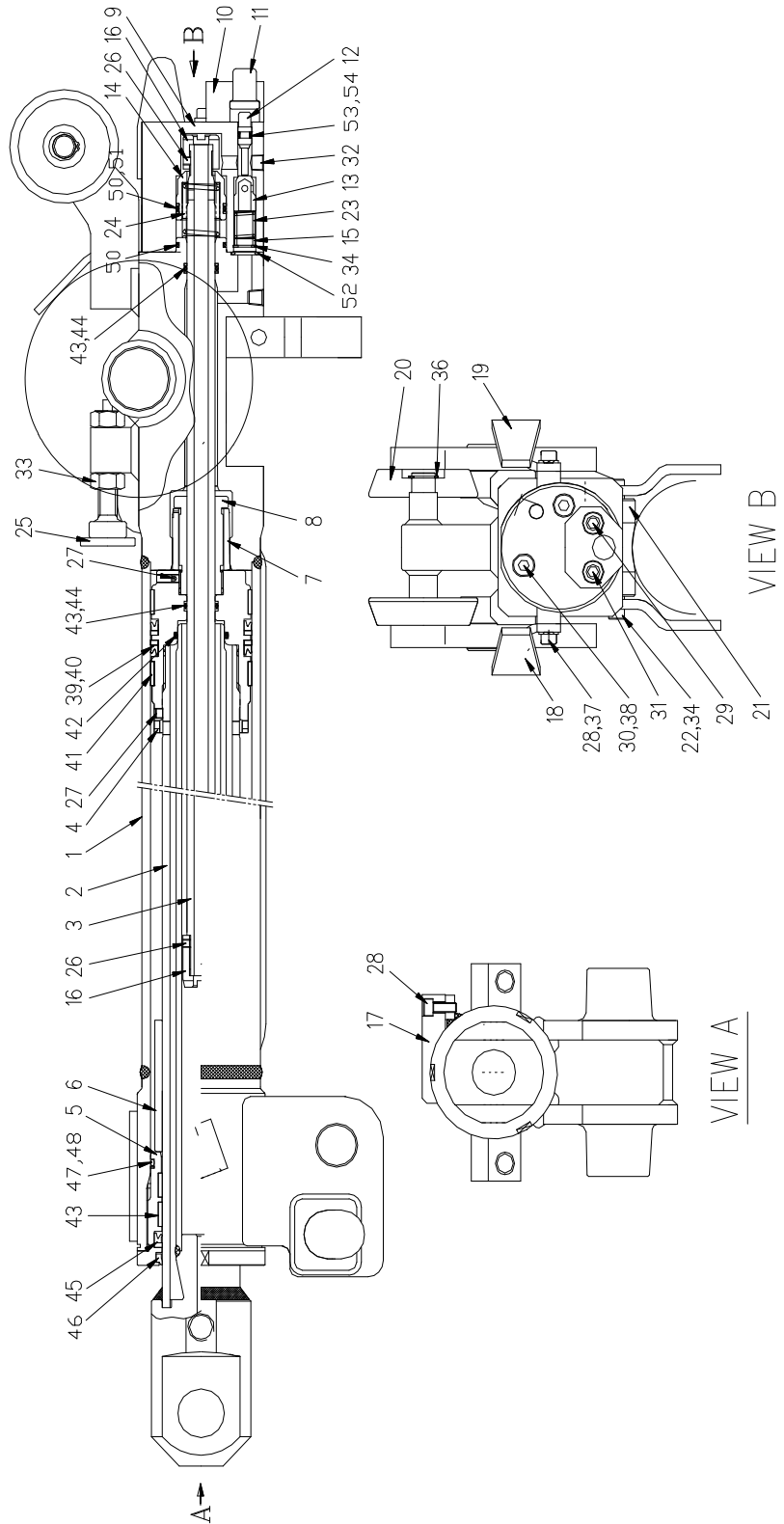
SECTION 27.1 - 1st EXTENSION CYLINDER ASS'Y

1st EXTENSION CYLINER ASS'Y					00
Pos.	Q'ty	Part No.	Description	Remarks	
29	1	4525151	SCREW, SET	RB14	
30	2	3112086	NUT, HEX.	NA8	
31	2	3511101	WASHER, PLAIN	WP10	
32	2	3612081	WASHER, SPRING	WS8	
33	2	6510161	PACKING, PISTON	OSI 70×60×6	
34	2	6037141	RING, BACK UP	60×70×3	
35	2	S01-0641	RING, WEAR	65×70×15	
36	1	6012262	O-RING	1B G30	
37	2	6010212	O-RING	1B P18	
38	4	6034191	RING, BACK UP	T3 P18	
39	1	S01-	PACKING, ROD	NI-300	
40	1	6115651	SEAL, DUST	KDD 55×63×5×6.5	
41	2	S01-0872	RING, WEAR	55×60×3	
42	1	6012332	O-RING	1B G65	
43	1	6035741	RING, BACK UP	T3 G65	
44	2	S01-0907	O-RING	JASO #2028	
45	2	S01-0908	RING, BACK UP	55×33×1.25	
46	2	6010172	O-RING	1B P14	
47	1	S01-0906	O-RING	1A P6	
48	1	6034061	RING, BACK UP	T3 P6	
-	1	S01-2127	SEAL KIT	* MARKED	

SECTION 28.1 - 2nd EXTENSION CYLINDER ASS'Y

2nd EXTENSION CYLINDER ASS'Y

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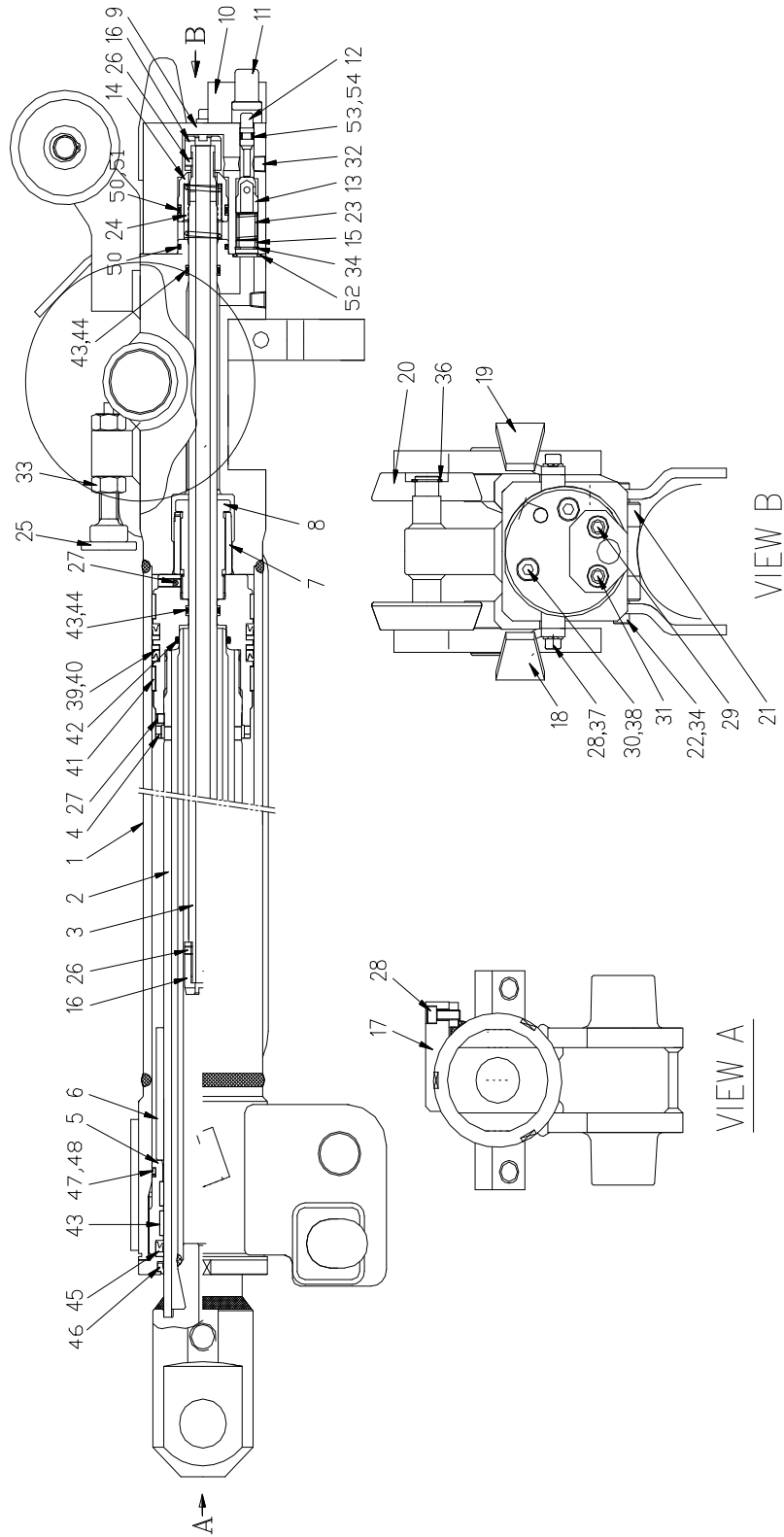
SECTION 28.1 - 2nd EXTENSION CYLINDER ASS'Y

2nd EXTENSION CYLINER ASS'Y				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	H01-2963-2	2nd EXTENSION CYLINDER ASS'Y	HN39T2000-01
1	1	H01-2963-2-1	TUBE ASS'Y	
2	1	KL08T2020	ROD ASS'Y	
3	1	KL08T2A30	RETRACTOR	
4	1	SS33T2040	PISTON	
5	1	SH46T2040	ROD COVER	
6	1	KL08T2040	SPACER	
7	1	SS33T2070	CUSHION RING	
8	1	SS33T1080	CUSHION GUIDE	
9	1	SS33T1090	VALVE BODY	
10	1	SS33T10A1	BRACKET-A	
11	1	SS33T10A2	SPOOL GUIDE	
12	1	SS33T10A3	SPOOL	
13	1	SS33T10A4	VALVE-A	
14	1	SS33T10A5	VALVE-B	
15	1	SS33T10A6	WASHER-A	
16	1	HN01T1060	STOP NUT-A	
17	1	KL08T2070	PAD	
18	1	KL08T2A40-02	BRACKET-R	
19	1	KL08T2A50-02	BRACKET-L	
20	2	SH17T2040	GUIDE, ROLLER	
21	1	KL08ET180	PAD, ROLLER	
22	1	KL05ET190	SHAFT, ROLLER	
23	1	SS33T10S1	SPRING-A	
24	1	SS33T10S2	SRPING-B	
25	1	SS33T20B1-01	BOLT, CHECK	M12×P1.75×85
26	2	2350116	SCREW, SET	BV3×3
27	2	2350446	BOLT, HEX. SOCKET HEAD	BV6×6
28	6	2610467	BOLT, HEX. SOCKET HEAD	BB6×16

SECTION 28.1 - 2nd EXTENSION CYLINDER ASS'Y

2nd EXTENSION CYLINDER ASS'Y

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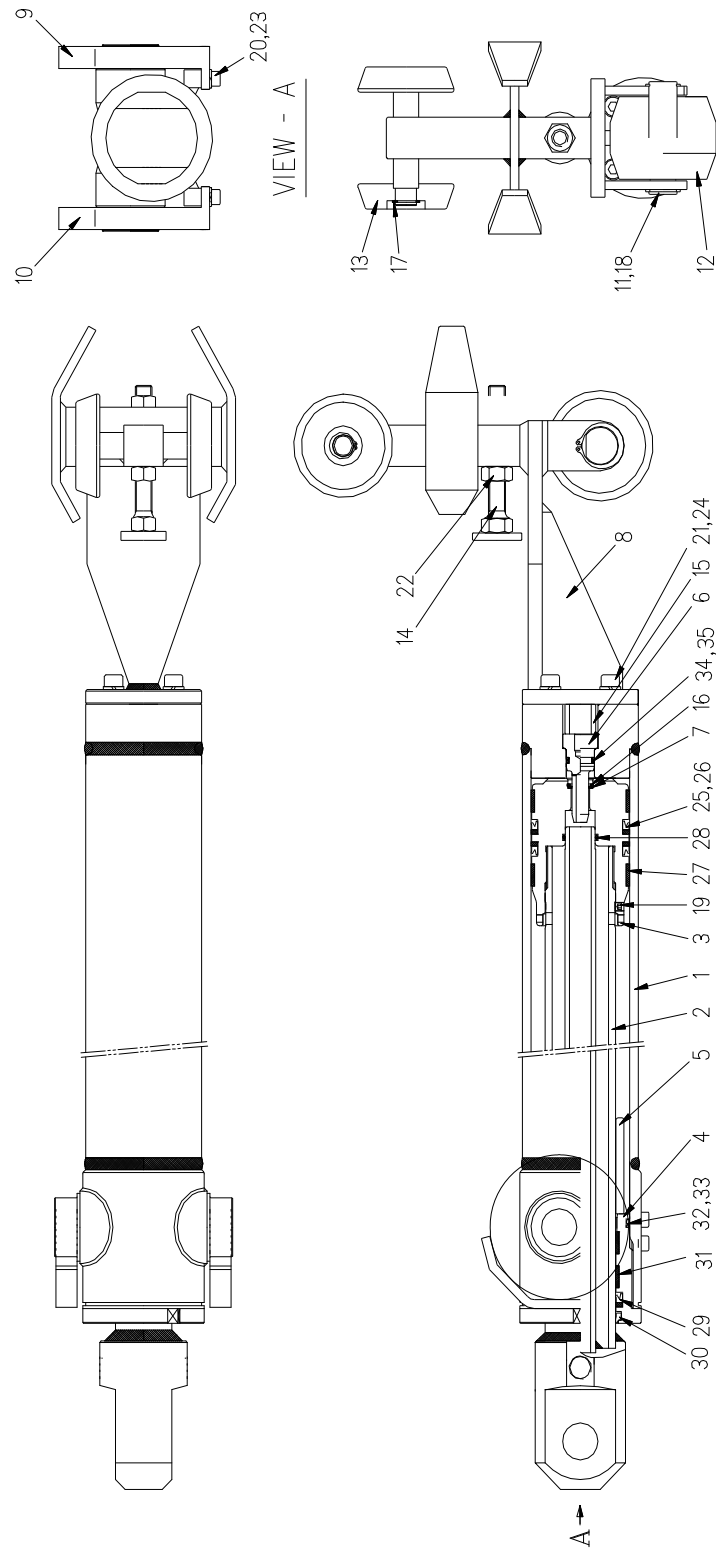
SECTION 28.1 - 2nd EXTENSION CYLINDER ASS'Y

2nd EXTENSION CYLINER ASS'Y				00
Pos.	Q'ty	Part No.	Description	Remarks
29	1	2610937	BOLT, HEX. SOCKET HEAD	BB8×35
30	2	2611067	BOLT, HEX. SOCKET HEAD	BB8×100
31	1	2611087	BOLT, HEX. SOCKET HEAD	BB8×110
32	2	BOO-0002-2-26	PLUG	PT1/8×7L
33	3	3112126	NUT, HEX.	NA12
34	1	1310681	PIN, SPLIT	PW3×25
35	1	4525151	RING, RETAINING	RB14
36	2	4510161	RING, RETAINING	RS15
37	4	3612061	WASHER, SPRING	WS6
38	2	3612081	WASHER, SPRING	WS8
39	2	6510141	PACKING, PISTON	OSI 65×55×6
40	2	6037071	RING, BACK UP	55×65×3
41	2	S01-0878	RING, WEAR	60×65×15
42	1	6012262	O-RING	1B G30
43	2	6010212	O-RING	1B P18
44	4	6034191	RING, BACK UP	T3 P18
45	1	S01-	PACKING, ROD	NI-300
46	1	6115631	SEAL DUST	KDD 50×58×5×6.5
47	2	S01-0877	RING, WEAR	50×55×15
48	1	6012322	O-RING	1B G60
49	1	6035731	RING, BACK UP	T3 G60
50	2	S01-0907	O-RING	JASO #2028
51	2	S01-0908	RING, BACK UP	28×33×1.25
52	2	6010172	O-RING	1B P14
53	1	S01-0906	O-RING	1A P6
54	1	6034061	RING, BACK UP	T3 P6
-	1	S01-2128	SEAL KIT	* MARKED

SECTION 29.1 - 3rd EXTENSION CYLINDER ASS'Y

3rd EXTENSION CYLINER ASS'Y

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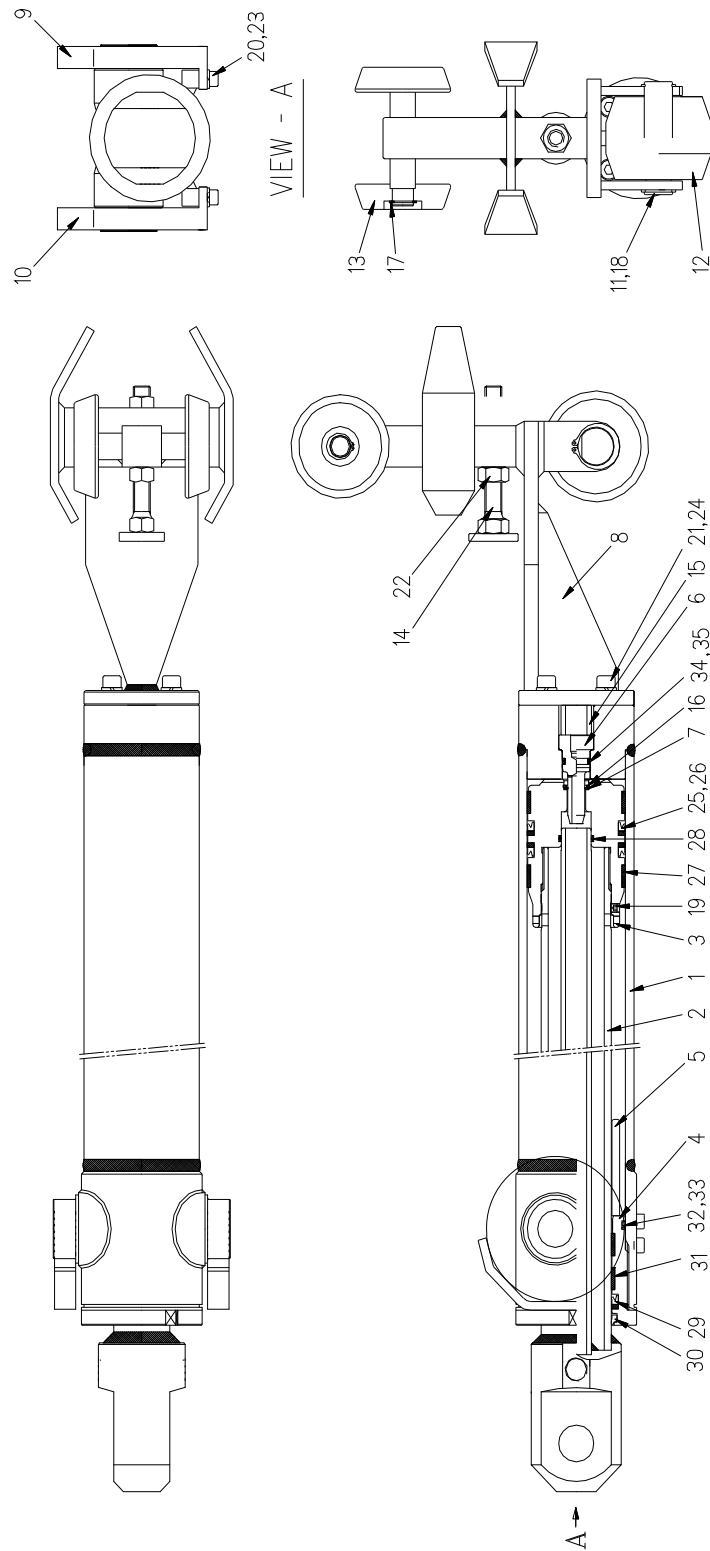
SECTION 29.1 - 3rd EXTENSION CYLINDER ASS'Y

3rd EXTENSION CYLINER ASS'Y				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	H01-2963-3	3rd EXTENSION CYLINDER ASS'Y	HN39T3000-01
1	1	KL08T3010	TUBE ASS'Y	
2	1	KL08T3020	ROD ASS'Y	
3	1	KL07T3030-01	PISTON	
4	1	SH46T3030	ROD COVER	
5	1	SH15T3050	SPACER	
6	1	SS25T2050	CUSHION	
7	1	SS25T2060	RING, CUSHION	
8	1	H01-2963-3-2	BRACKET	
9	1	KL25T3030	BRACKET-R	
10	1	KL25T3040	BRACKET-L	
11	1	SH39T2060	SHAFT, ROLLER	Φ20×79L
12	1	SH09T2040	ROLLER, GUIDE	Φ75×56L
13	2	SH17T2040	ROLLER, GUIDE	Φ70×17L
14	1	SS33T20B1-01	BOLT, CHECK	M12×P1.75×90L
15	1	SH16T3070	SPACER-B	Φ15.3×Φ21.7×20
16	1	BOO-0002-3-13	RING, RETAINING	AR18
17	2	4510161	RING, RETAINING	RS15
18	2	4510211	RING, RETAINING	RS20
19	1	2350446	SCREW, SET	BV6×6
20	4	2610467	BOLT, HEX. SOCKET HEAD	BB6×16
21	4	2610927	BOLT, HEX. SOCKET HEAD	BB8×30
22	3	3112126	NUT, HEX.	NA12
23	4	3612061	WASHER, SPRING	WS6
24	4	3612081	WASHER, SPRING	WS8
25	2	6510161	PACKING, PISTON	ISO 70×60×6
26	2	6037141	RING, BACK UP	60×70×3
27	2	S01-0641	RING, WEAR	65×70×15
28	2	6010222	O-RING	1B P20

SECTION 29.1 - 3rd EXTENSION CYLINDER ASS'Y

3rd EXTENSION CYLINER ASS'Y

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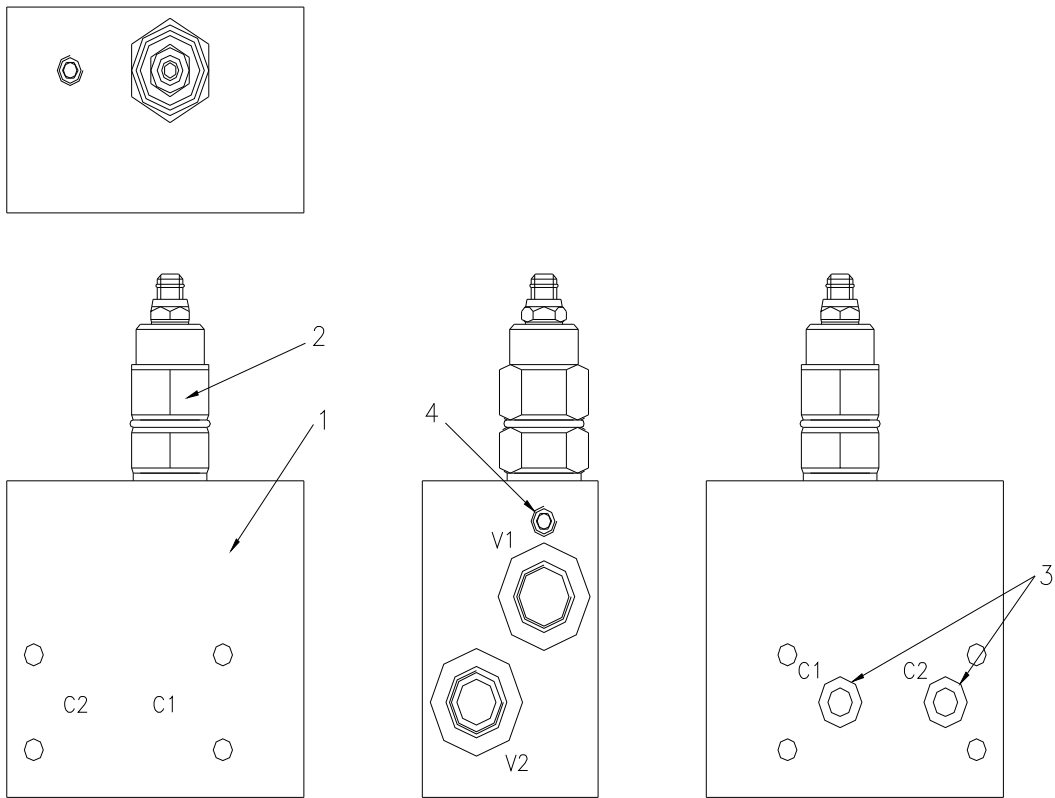
SECTION 29.1 - 3rd EXTENSION CYLINDER ASS'Y

3rd EXTENSION CYLINER ASS'Y					00
Pos.	Q'ty	Part No.	Description	Remarks	
29	1	S01-2857	ROD PACKING	NI-300	
30	1	6115631	DUST SEAL	KDD 50×58×5×6.5	
31	2	S01-0657	RING, WEAR	Φ50×Φ55×15W	
32	1	6012332	O-RING	1B G65	
33	1	6035741	RING, BACK-UP	(T-3) G65	
34	1	6010192	O-RING	1B P16	
35	1	6032181	RING, BACK-UP	(T-2) P16	
-	1	S01-2129	SEAL KIT	* MARKED	

SECTION 30.1 - VALVE, COUNTER BALANCE

VALVE, COUNTER BALANCE

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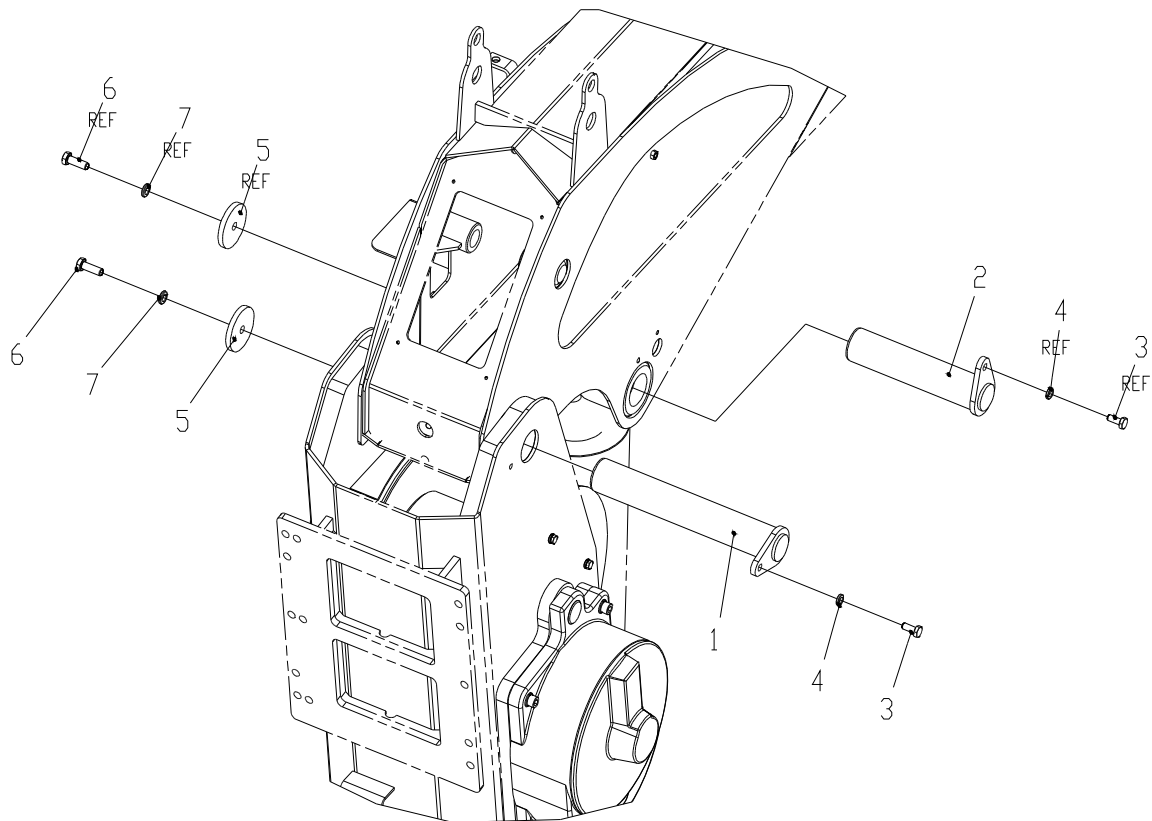
SECTION 30.1 - VALVE, COUNTER BALANCE

VALVE, COUNTER BALANCE					00
Pos.	Q`ty	Part No.	Description	Remarks	
-	1	H02-0201	VALVE, COUNTER BALANCE		
1	1	2-A2141-01-0	BODY		
2	1	H02-0198	COUNTER BALANCE VALVE	9706-04D-P09	
3	2	6010142	O-RING	1B P12	
4	2	S01-0493	PLUG, WERNCH	PT1/8	

SECTION 31.1 - MAIN SHAFT ASS'Y

MAIN SHAFT ASS'Y

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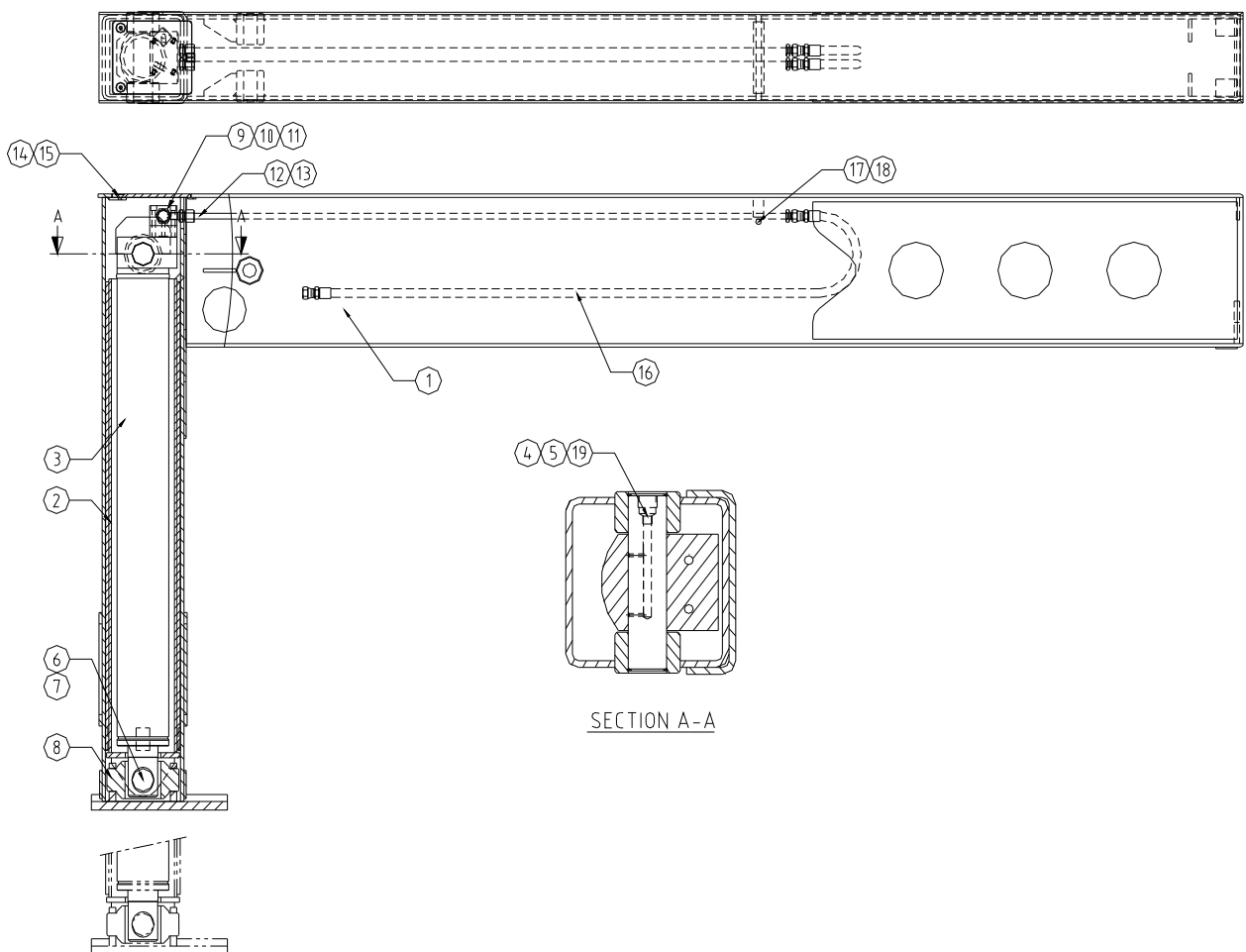
SECTION 31.1 - MAIN SHAFT ASS'Y

MAIN SHAFT ASS'Y				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	A01-4810	MAIN SHAFT ASS'Y	
1	1	TA-2002	SHAFT	
2	1	TA-2003	SHAFT	Φ55×242L
3	2	2111206	BOLT, HEX. HEAD	BA10×25
4	2	3613101	WASHER, SPRING	WS10
5	2	TA-6002	PLATE	
6	2	2111537	BOLT, HEX. HEAD	BA12×35
7	2	3612121	WASHER, SPRING	WS12

SECTION 32.1 - OUTRIGGER ASS'Y

OUTRIGGER ASS'Y

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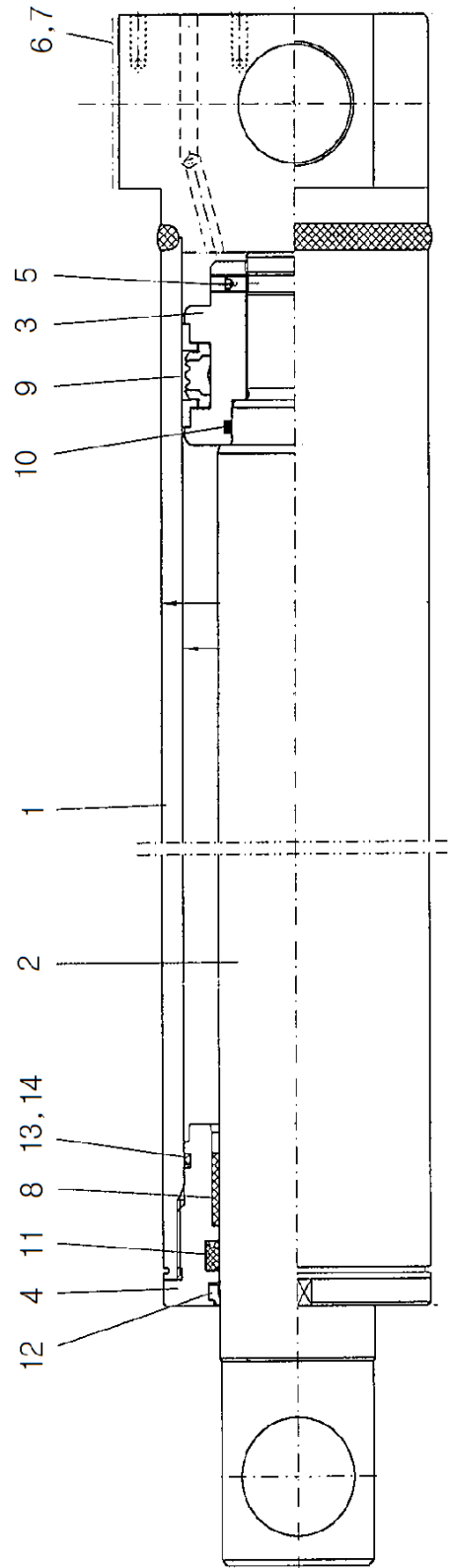
SECTION 32.1 - OUTRIGGER ASS'Y

OUTRIGGER ASS'Y					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	2	OR1-4002	OUTRIGGER ASS'Y		
1	1	OR1-4003	BEAM, OUTRIGGER		
2	1	A05-0970	TUBE, INNER		
3	1	H01-2066	OUTRIGGER CYLINDER ASS'Y		
4	1	S01-5272	PIN		
5	2	4525351	RING, RETAINING	RB40	
6	1	S01-0754	SHAFT		
7	1	4510351	RING, RETAINING	RS40	
8	1	OR1-4011	SHOE ASS'Y		
9	1	H02-1058	VALVE, PILOT CHECK		
10	2	6010141	O-RING	P12	
11	4	2615586	BOLT, HEX. SOCKET HEAD	BB6×60	
12	2	9002512	NIPPLE, PF PIPE	NTP10-PT1/4	
13	2	H01-1355	PIPE, HYDRAULIC	Φ10×t1.5	
14	1	A05-0755	COVER		
15	2	2241606	SCREW, MACHINE	BC6×8	
16	2	7113892	HOSE, H.P (HUCF)	1/4×1220-PF1/4	
17	1	M01-0745	PIN		
18	2	4551121	RING, RETAINING	RE9	
19	1	4610702	NIPPLE GREASE	A-PT1/8	

SECTION 33.1 - OUTRIGGER JACK CYLINDER ASS'Y

OUTRIGGER JACK CYLINDER ASS'Y

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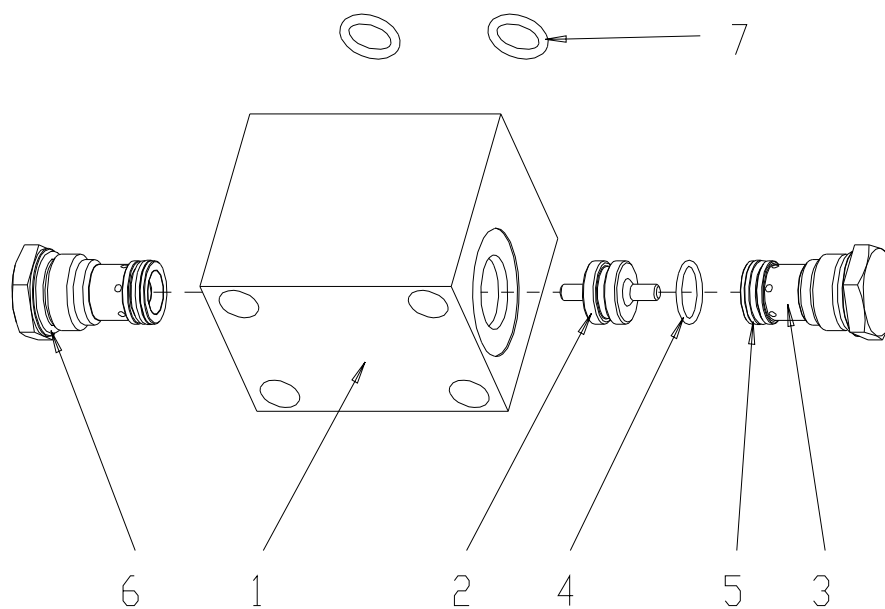
SECTION 33.1 - OUTRIGGER JACK CYLINDER ASS'Y

OUTRIGGER JACK CYLINDER ASS'Y					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	2	H01-2066	OUTRIGGER CYLINDER ASS'Y	HN25J0A00	
1	1	HN25J0010	TUBE ASS'Y		
2	1	HN24J0A10	ROD		
3	1	SH10J8030	PISTON		
4	1	HN04J0030	COVER, ROD		
5	1	-	SCREW, SET	M8×P1.25×6L	
6	1	-	DUST COVER	50×70×3.2t	
7	2	-	HEX. BOLT	M6×P1.0×10L	
8	1	-	DU BUSH	MB 5525 DU	
9	1	-	SEAL, COMPACT	PCB 0A0800-NCRO	
10	1	-	O-RING	1B G45	
11	1	-	U-PACKING	NI-300 (Φ55×Φ65×10W)	
12	1	-	DUST WIPER	KDD 55	
13	1	-	O-RING	1B G75	
14	1	-	RING, BACK-UP	(T-3) G75	
-	1	S01-2125	KIT, SEAL		

SECTION 34.1 - PILOT CHECK VALVE (D)

PILOT CHECK VALVE (D)

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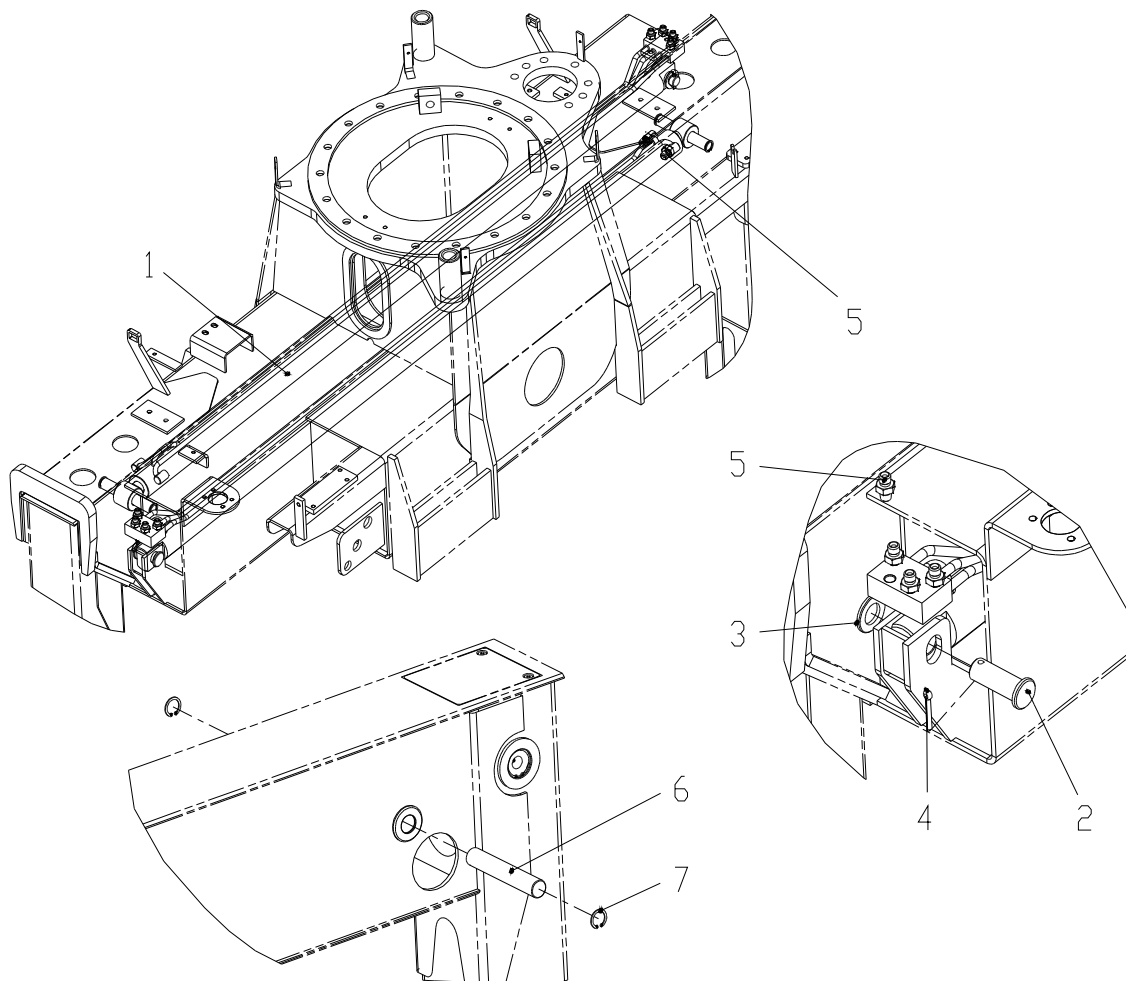
SECTION 34.1 - PILOT CHECK VALVE (D)

PILOT CHECK VALVE (D)				00
Pos.	Q`ty	Part No.	Description	Remarks
-	2	H02-1058	PILOT CHECK VALVE (D)	
1	1	D25-0005-1	BODY, VALVE	50×55×73
2	1	D25-0005-2	SPOOL	
3	2	D25-0005-3	VALVE, CHECK	
4	3	D25-0005-4	O-RING	Φ13×t1.5
5	2	D25-0005-5	RING, BACK-UP	Φ13×t1.2
6	2	D25-0005-6	O-RING	Φ16.5×t2.0
7	2	6010142	O-RING	P12

SECTION 35.1 - OUTRIGGER EXTENSION CYLINDER KIT

OUTRIGGER EXTENSION CYLINDER KIT

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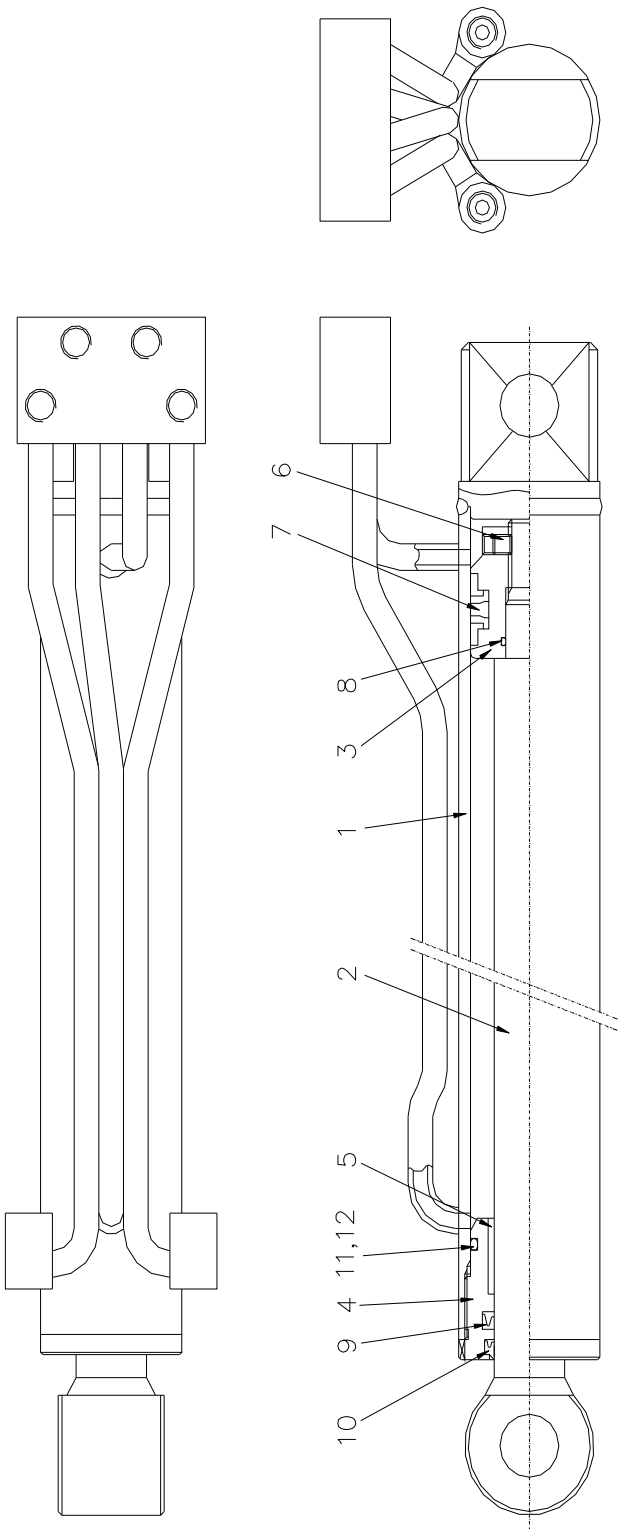
SECTION 35.1 - OUTRIGGER EXTENSION CYLINDER KIT

OUTRIGGER EXTENSION CYLINDER KIT				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	A01-5286	OUTRIGGER EXTENSION CYLINDER KIT	
1	2	H01-2063	SIDE SHIFT CYLINDER ASS'Y	
2	2	OR1-3004	SHAFT	
3	2	3511241	WASHER, PLAIN	WP24
4	2	1311321	PIN, SPLIT	PW6.3×40
5	12	9211062	NIPPLE, PT HOSE	PF1/4×PT1/4
6	2	N01-0932	PIN	
7	4	4525251	RING, RETAINING	RB25

SECTION 36.1 - SIDE SHIFT CYLINDER ASS'Y

SIDE SHIFT CYLINDER ASS'Y

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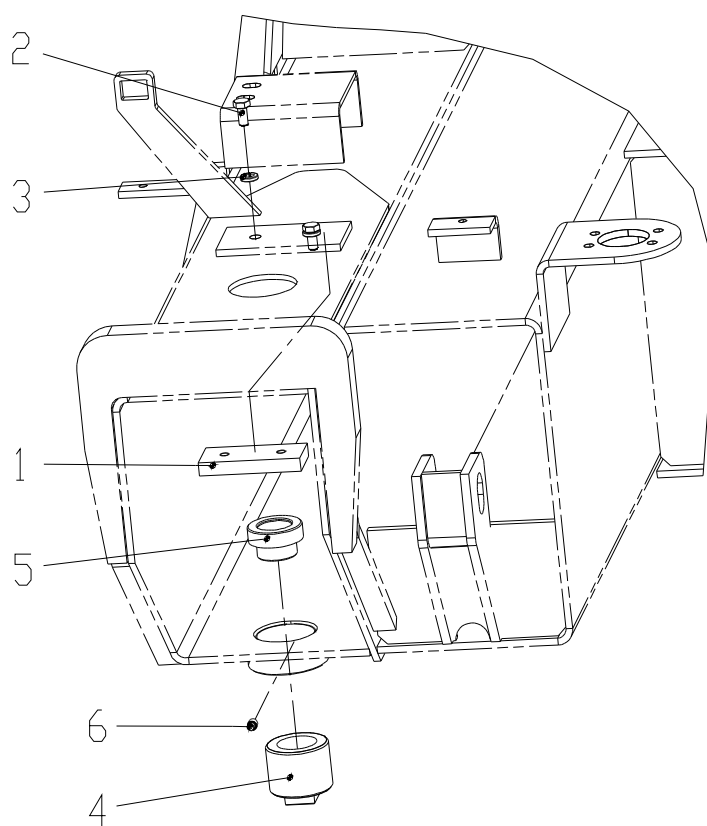
SECTION 36.1 - SIDE SHIFT CYLINDER ASS'Y

SIDE SHIFT CYLINDER ASS'Y					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	2	H01-2063	SIDE SHIFT CYLINDER ASS'Y		
1	1	HN21SSA10	TUBE ASS'Y		
2	1	HN03SS020	ROD ASS'Y		
3	1	SS35C0030	PISTON		
4	1	SS32FE030-01	COVER, ROD		
5	1	S01-2209	RING, WEAR	Φ30×Φ34×20W	
6	1	2350446	SCREW, SET	BV6×6	
7	1	6211211	SEAL, COMPACT	PCB 1A0500-NCRO	
8	1	6010212	O-RING	1B P18	
9	1	S01-2684	PACKING, ROD	NI-300	
10	1	6115571	SEAL, DUST	KDD 30	
11	1	6012292	O-RING	1B G45	
12	1	6035701	RING, BACK-UP	T3 G45	
-	1	S01-2123	KIT, SEAL	PART REMARKS *	

SECTION 37.1 - OUTRIGGER PAD KIT

OUTRIGGER PAD KIT

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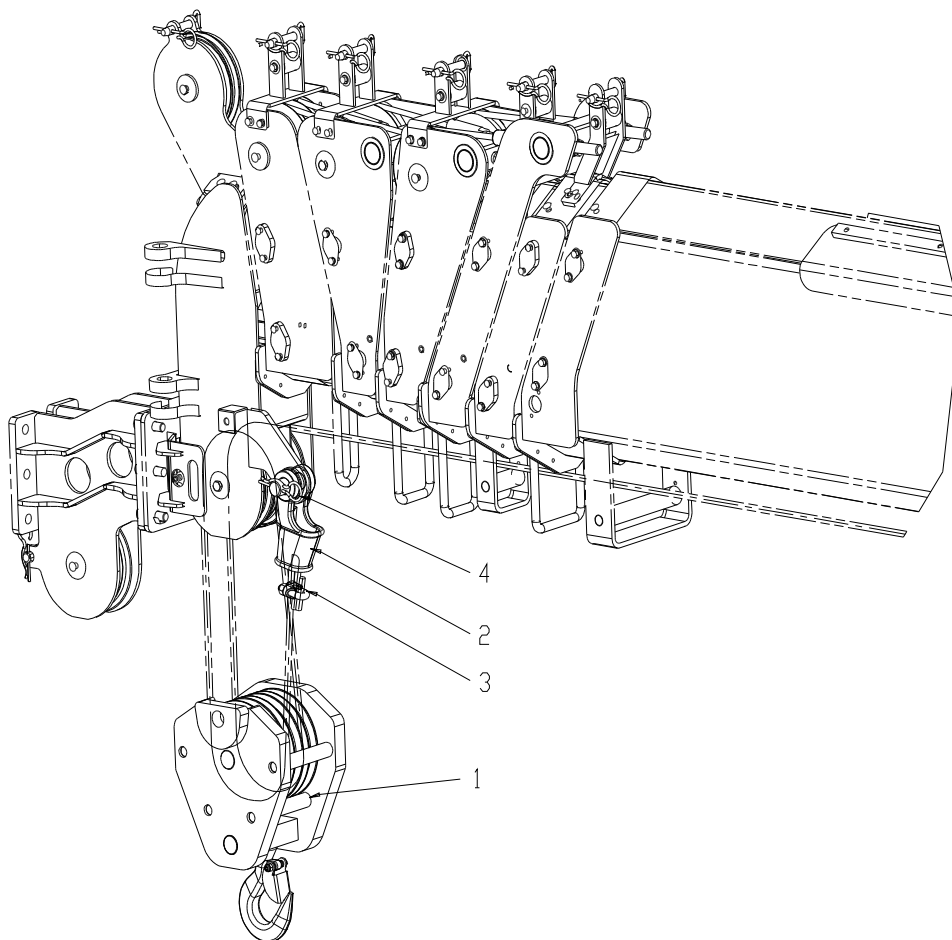
SECTION 37.1 - OUTRIGGER PAD KIT

OUTRIGGER PAD KIT					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	A01-5287	OUTRIGGER PAD KIT		
1	2	OR1-0009	PAD, SLIDE		
2	4	2110867	BOLT, HEX. HEAD	BA8×16	
3	4	3612081	WASHER, SPRING	WS8	
4	2	OR-0062	BOLT, ADJUST		
5	2	OR-0063	PAD, ADJUST	MC	
6	2	2340576	SCREW, SET (STICK END)	BM8×12	

SECTION 38.1 - HOOK ASS'Y

HOOK ASS'Y

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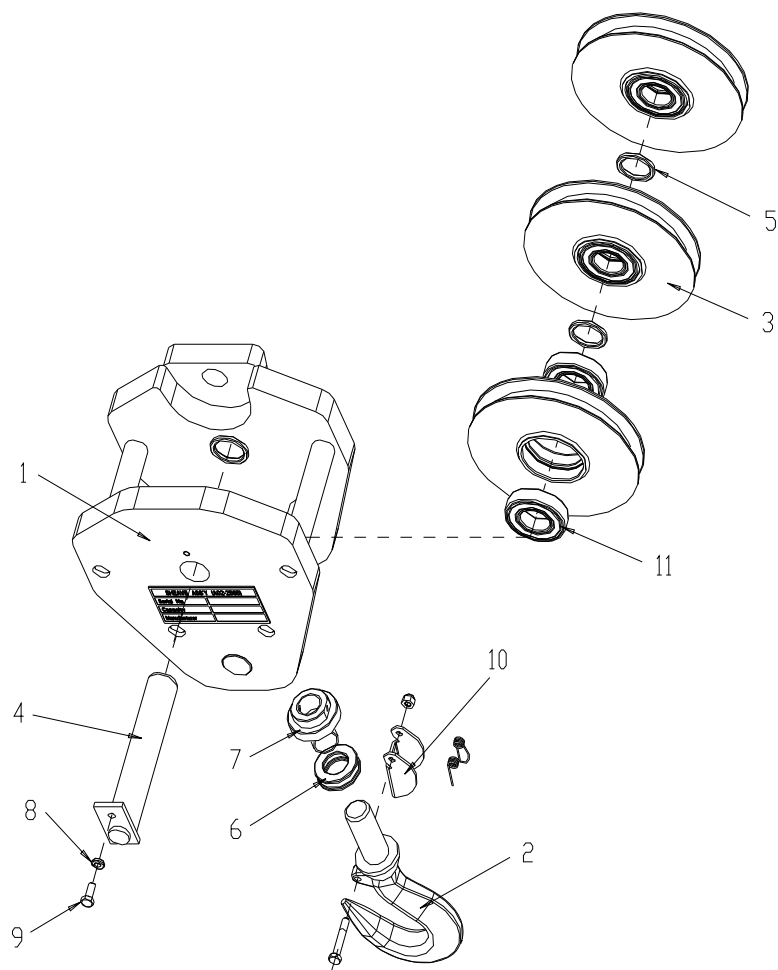
SECTION 38.1 - HOOK ASS'Y

HOOK ASS'Y				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	123 1637	HOOK ASS'Y	
1	1	A02-2565	SHEAVE ASS'Y	11 ton
2	1	A02-1264	SOCKET ASS'Y	7/16 inch
3	1	N01-1722	CLIP, WIRE ROPE	
4	1	S01-0002	PIN, LOCK	

SECTION 39.1 - SHEAVE ASS'Y

SHEAVE ASS'Y

00



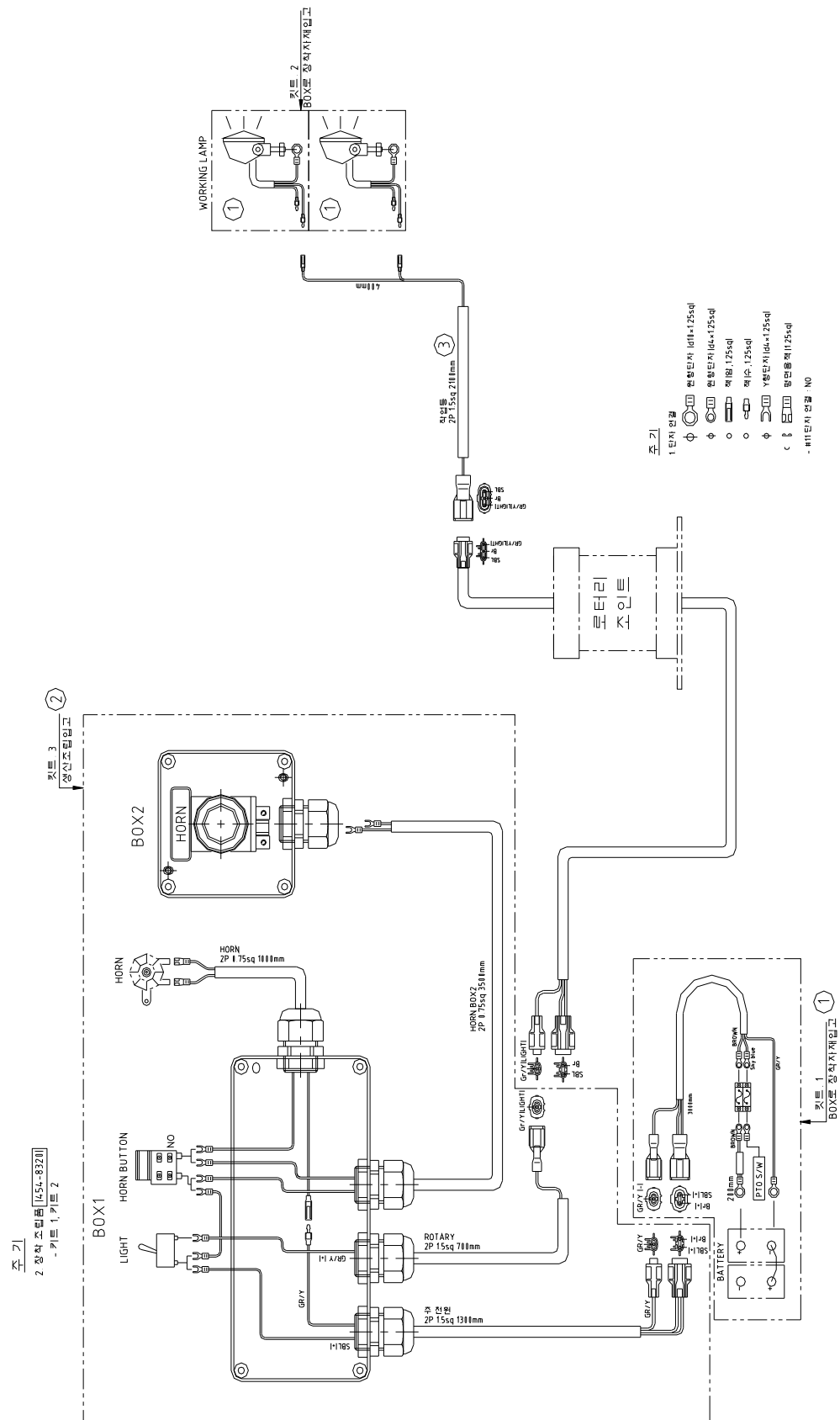
SECTION 39.1 - SHEAVE ASS'Y

SHEAVE ASS'Y				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	A02-2565	SHEAVE ASS'Y	
1	1	HK-1002	BODY	
2	1	M01-1780	HOOK	
3	3	HK-1006	SHEAVE	
4	1	HK-0008	SHAFT	
5	4	HK-0010	SPACER	
6	1	4211060	BEARING, THRUST BALL	#51206
7	1	HK-0015	NUT, LOCK	M30×P3.0
8	1	3612081	WASHER, SPRING	WS8
9	1	2110887	BOLT, HEX. HEAD	BA8×20
10	1	S01-1596	CLIP ASS'Y, SAFETY	
11	2	4112074	RADIAL BALL BEARING	B 6207ZZ

SECTION 40.1 - ELECTRIC WIRING DIAGRAM

ELECTRIC WIRING DIAGRAM

01





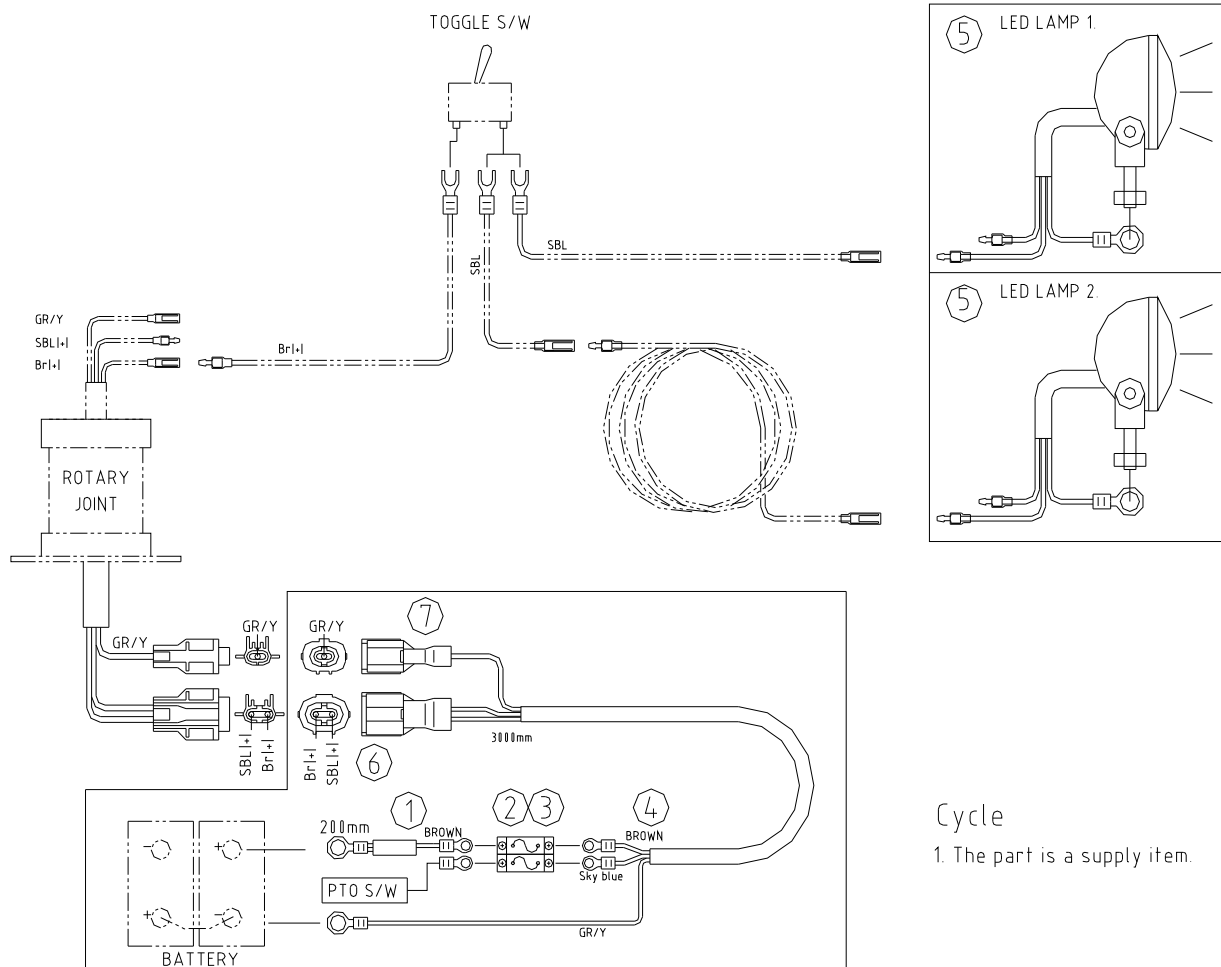
SECTION 40.1 - ELECTRIC WIRING DIAGRAM

ELECTRIC WIRING DIAGRAM				01
Pos.	Q'ty	Part No.	Description	Remarks
-	1	E00-0127	ELECTRIC WIRING DIAGRAM	MOUNTING PART PRODUCTION
1	1	454 8320	ELECTRIC ASS'Y	
2	1	E01-1196	ELECTRIC ASS'Y	
3	1	E01-1143	ELECTRIC WIRE ASS'Y	

SECTION 41.1 - ELECTRIC ASS'Y (MOUNTING PART)

ELECTRIC ASS'Y (MOUNTING PART)

01



Cycle

1. The part is a supply item.

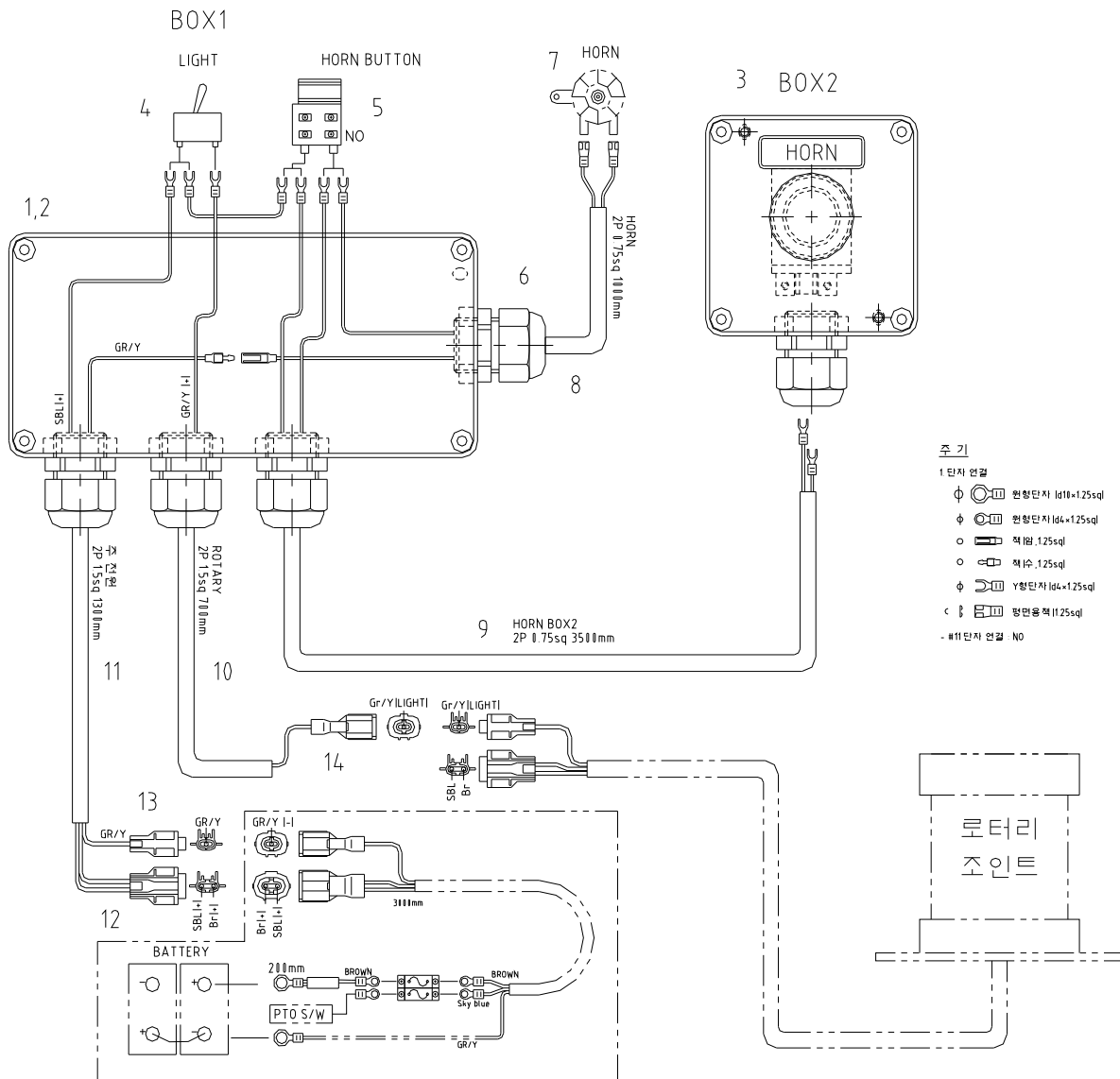
SECTION 41.1 - ELECTRIC ASS'Y (MOUNTING PART)

ELECTRIC ASS'Y (MOUNTING PART)					01
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	454 8320	ELECTRIC ASS'Y		
1	1	E01-0909	ELCE. WIRE ASS'Y	1.5SQ	
2	1	E01-0908	HOLDER, FUSE	D6, 2P	
3	2	E01-0010	FUSE	D6, 15A	
4	1	E01-0910	ELEC. WIRE ASS'Y	1.5SQ	
5	2	454 8500	LED LIGHT		
6	1	S01-5661	2P MALE ASS'Y		
7	1	S01-5659	1P MALE ASS'Y		

SECTION 42.1 - ELECTRIC ASS'Y (PRODUCTION)

ELECTRIC ASS'Y (PRODUCTION)

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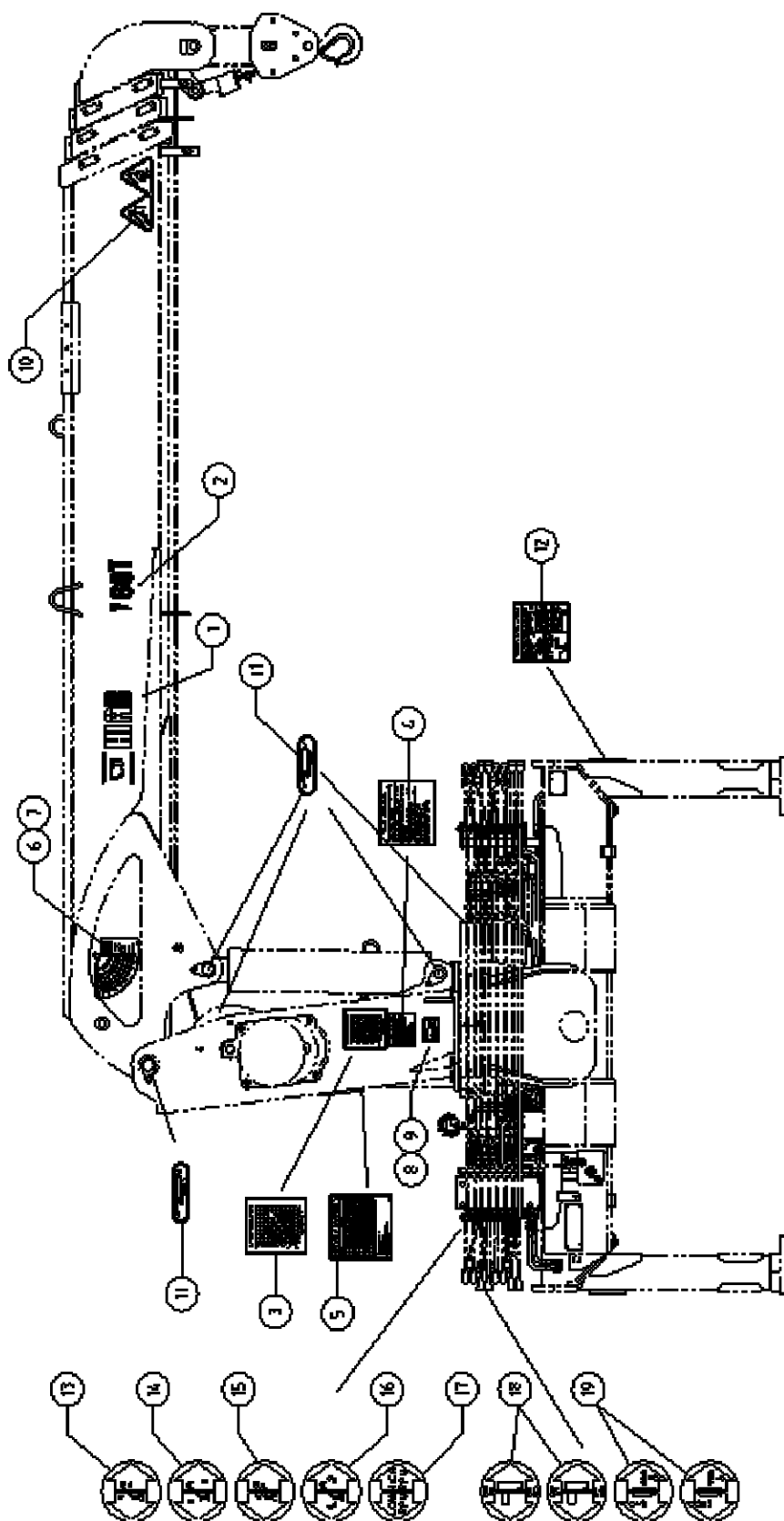
SECTION 42.1 - ELECTRIC ASS'Y (PRODUCTION)

ELECTRIC ASS'Y (PRODUCTION)					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	E01-1196	ELECTRIC ASS'Y		
1	1	E01-1197	BOX, FI	AL 081806	
2	6	S01-4582	SCREW, ROUND HEAD	M4×12	
3	1	A05-2924	BOX, SWITCH	AL 081806	
4	1	E01-0053	SWITCH, TOGGLE ON/OFF	WJT-2210	
5	1	E01-0095	SWITCH, PUSH BUTTON	JK-4025	
6	5	CS-9026	CONNECTOR	PG13.5	
7	1	E01-0082	HORN		
8	1	E01-1105	ELECTRIC WIRE ASS'Y	2P 0.75sq 1000mm	
9	1	E01-1107	ELECTRIC WIRE ASS'Y	2P 0.75sq 3500mm	
10	1	E01-1108	ELECTRIC WIRE ASS'Y	2P 1.5sq 700mm	
11	1	E01-1109	ELECTRIC WIRE ASS'Y	2P 1.5sq 1300mm	
12	1	S01-5660	2P FEMALE ASS'Y		
13	1	S01-5658	1P FEMALE ASS'Y		
14	1	S01-5659	1P MALE ASS'Y		

SECTION 43.1 - ARRANGEMENT, STICKER

ARRANGEMENT, STICKER

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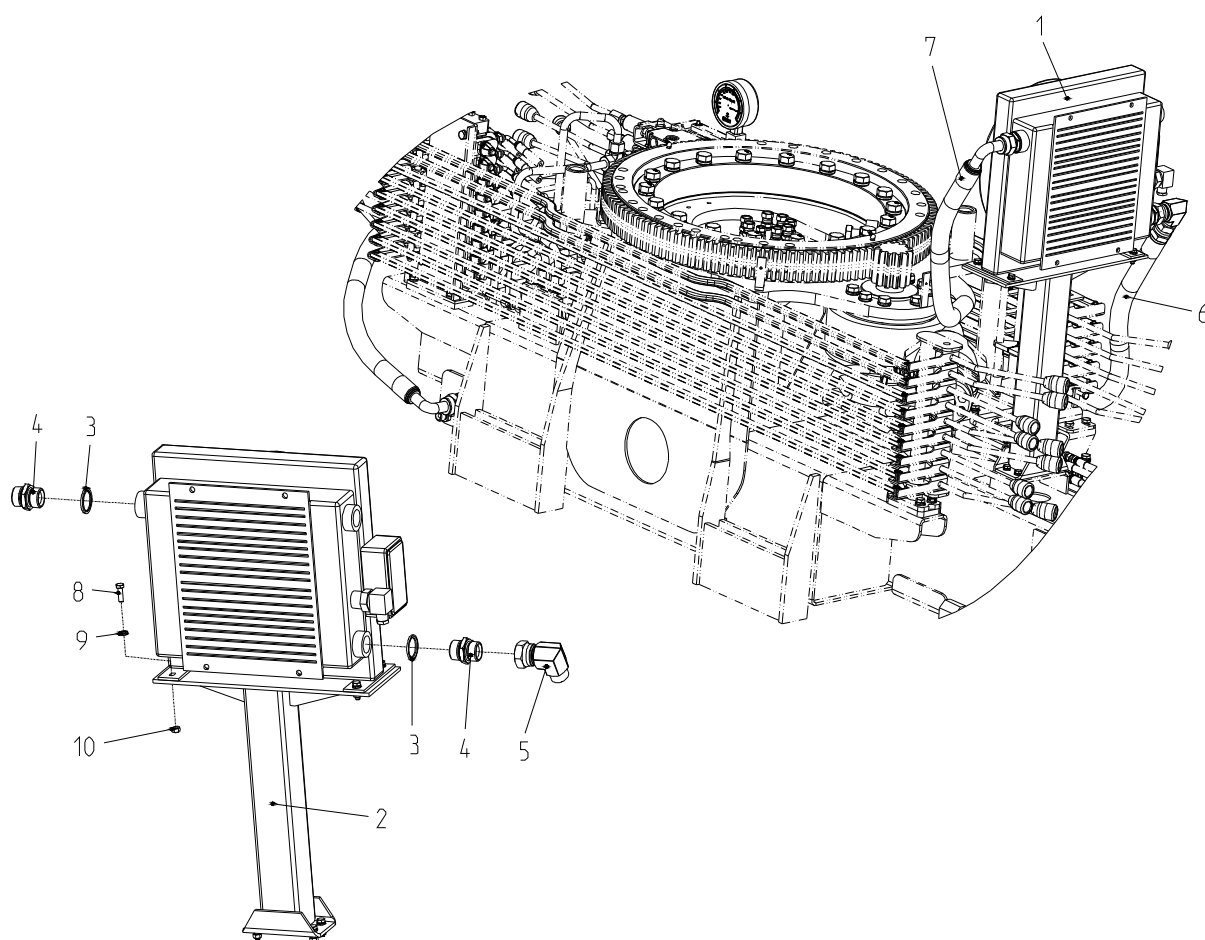
SECTION 43.1 - ARRANGEMENT, STICKER

ARRANGEMENT, STICKER				00
Pos.	Q`ty	Part No.	Description	Remarks
-	1	A01-4750	ARRANGEMENT, STICKER	
1	2	P01-3029	LABEL, LOGO	
2	2	P01-3050	LABEL, MODEL NAME	
3	1	P01-2776	WORKING RADIUS	
4	1	P01-0142	LABEL, WARNING	
5	1	P01-2777	LOAD CHART	
6	1	P01-2778	PLATE, BOOM ANGLE [RIGHT]	
7	1	P01-2779	PLATE, BOOM ANGLE [LEFT]	
8	1	SK-5007	PLATE, NAME	
9	4	S01-0182	RIVET, BLIND	
10	2	P01-0124	PLATE, WARNING	
11	1	P01-0159	STICKER, GREASE	
12	7	P01-0358	PLATE, O/R WARNING	
13	2	P01-0130	PLATE, CONTROL	DERRICK
14	2	P01-0131	PLATE, CONTROL	WINCH
15	2	P01-0132	PLATE, CONTROL	EXTENSION
16	2	P01-0133	PLATE, CONTROL	SLEWING
17	2	P01-0134	PLATE, CONTROL	RPM
18	4	P01-0264	STICKER, SIDE SHIFT	
19	4	P01-0265	STICKER, JACK	

SECTION 44.1 - OIL COOLER MOUNTING

OIL COOLER MOUNTING

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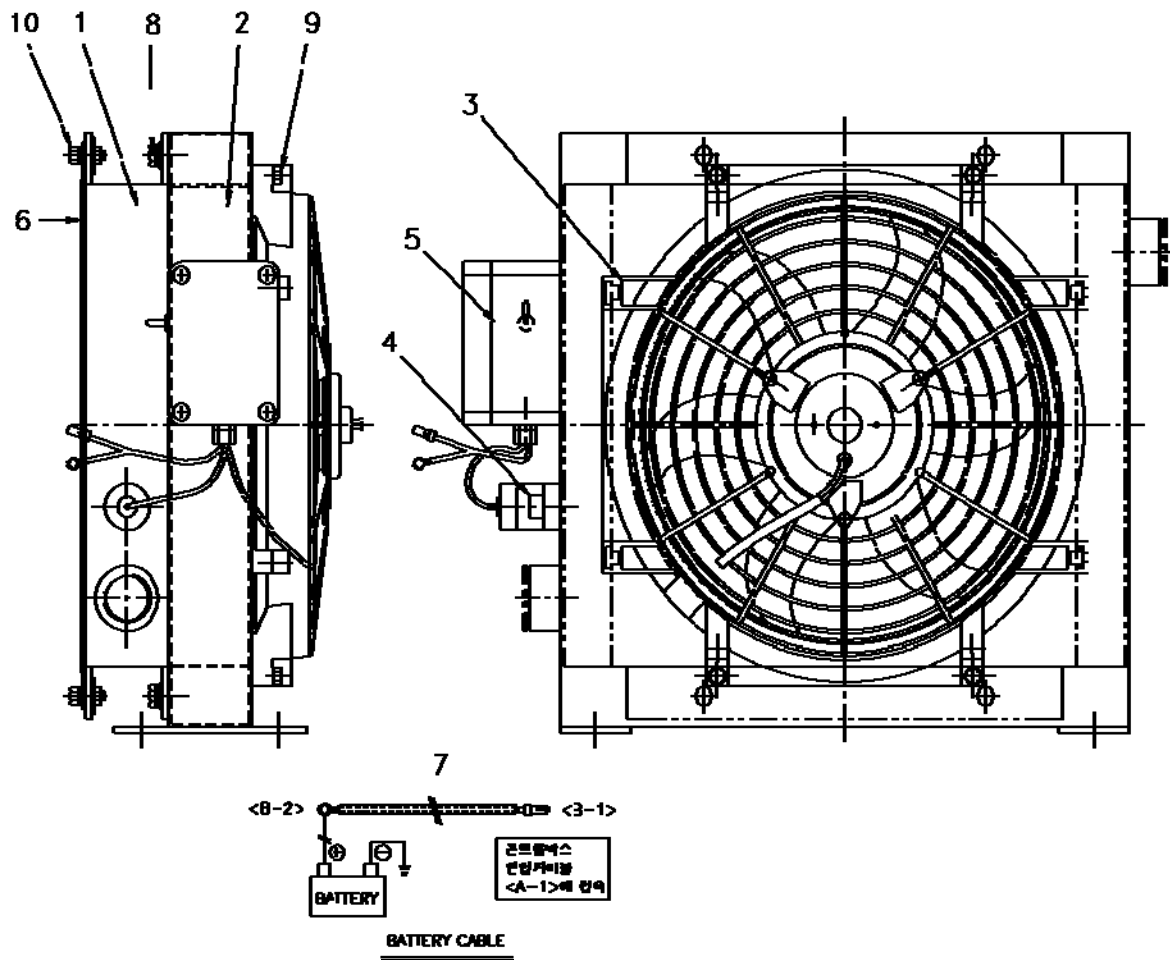
SECTION 44.1 - OIL COOLER MOUNTING

OIL COOLER MOUNTING					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	A01-5277	OIL COOLER MOUNTING		
1	1	H01-1268	OIL COOLER	DC24V	
2	1	A05-4402	COOLER BRACKET ASS'Y		
3	2	6800161	WASHER, SEALING	NF1	
4	2	9270102	UNION, CONICAL	1/1	
5	1	S01-1495	ELBOW, SWIVEL	2-PF 1"	
6	1	7180321	HOSE, H.P (HECF)	1×2600-PF1	
7	1	7180381	HOSE, H.P (HECF)	1×2900-PF1	
8	4	2110906	BOLT, HEX. HEAD	BA8×25	
9	4	3612081	WASHER, SPRING	WS8	
10	4	3111081	NUT, HEX.	NA8	

SECTION 45.1 - OIL COOLER

OIL COOLER

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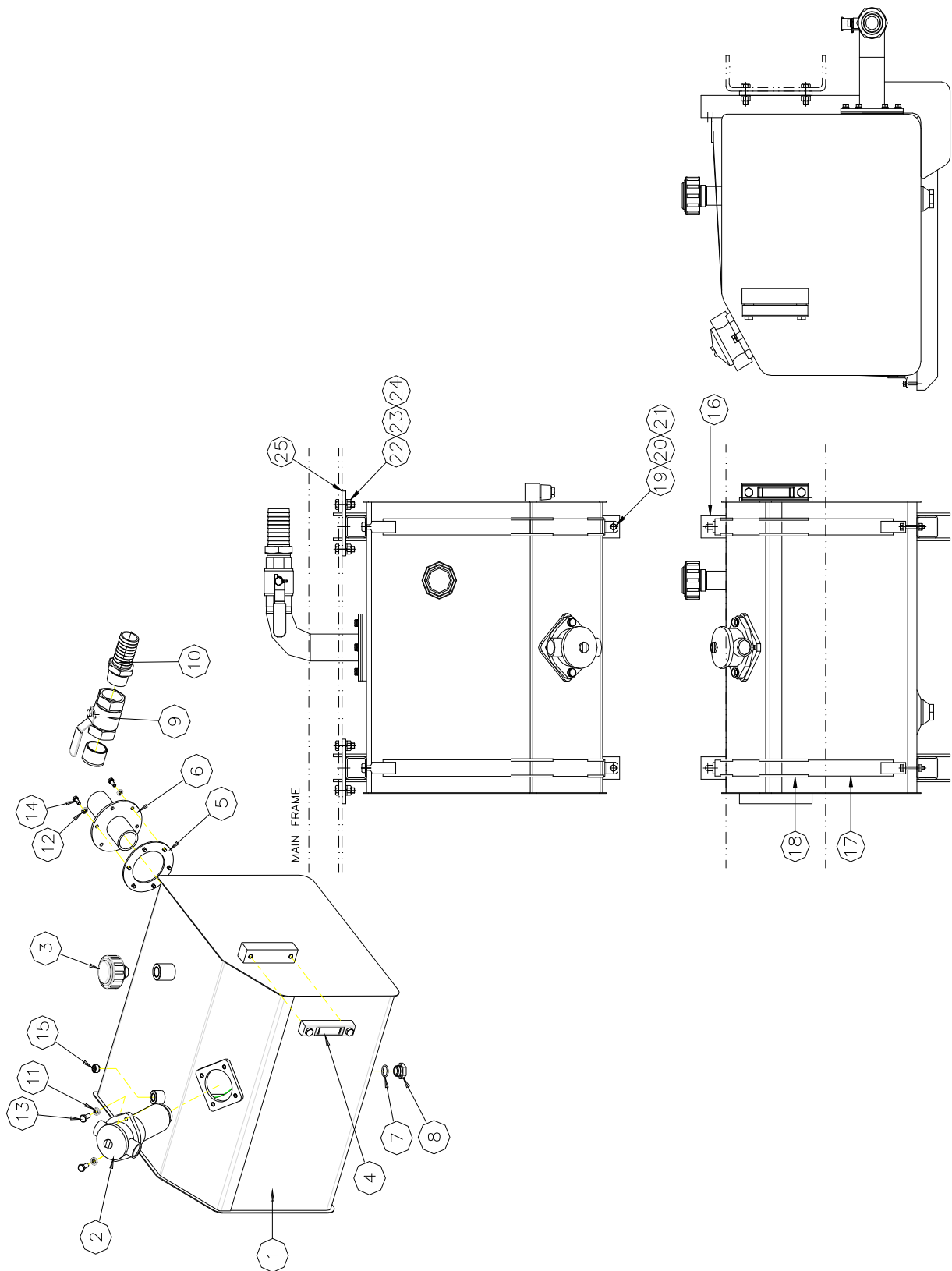
SECTION 45.1 - OIL COOLER

OIL COOLER					00
Pos.	Q'ty	Part No.	Description	Remarks	
-	1	H01-1268	OIL COOLER ASS'Y		
1	1	-	OIL COOLER		
2	1	-	SHROUD		
3	1	-	FAN & MOTOR ASS'Y		
4	1	-	THERMO SWITCH	55±5°C	
5	1	-	CONTROL BOX ASS'Y		
6	1	-	SCREEN		
7	1	-	BATTERY CABLE		
8	4	-	BOLT	M8×1.25P×20L	
9	4	-	BOLT	M6×1.0P×30L	
10	4	-	BOLT	M6×1.0P×20L	

SECTION 46.1 - OIL TANK ASS'Y (OPTION)

OIL TANK ASS'Y (OPTION)

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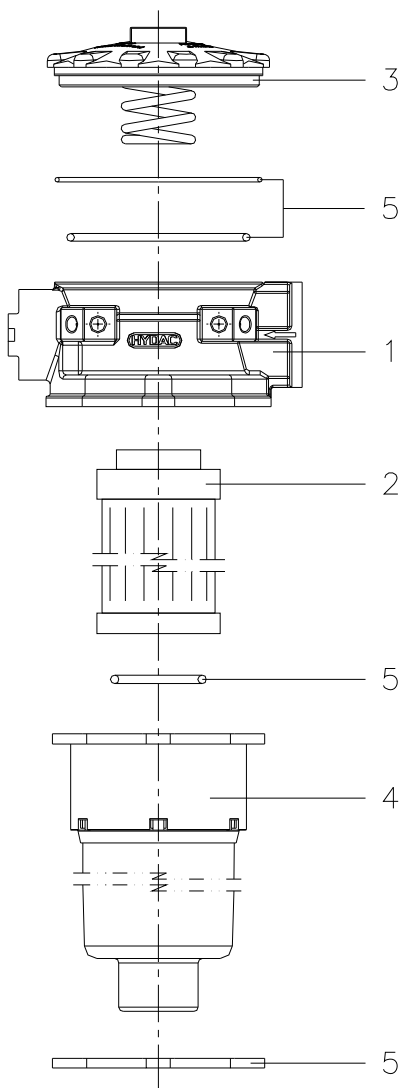
SECTION 46.1 - OIL TANK ASS'Y (OPTION)

OIL TANK ASS'Y (OPTION)				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	A02-2652	OIL TANK ASS'Y	170 ℓ
1	1	A05-3260	BODY, OIL TANK	
2	1	H01-1287	FILTER, RETURN	RFM165
3	1	S01-2475	BRETHHER, AIR	BF30
4	1	S01-2474	GAUGE, OIL LEVEL	FSA
5	1	OT-0013	GASKET	t4×150
6	1	A05-1570	FLANGE, SUCTION	PT2"×50A
7	1	6010282	O-RING	P24
8	1	S01-0076	PLUG	PF3/4
9	1	S01-0740	VALVE, HAND STOP	2-PT2"
10	1	S01-0741	NIPPLE, SUCTION	PT2,O.D52
11	2	3612101	WASHER, SPRING	WS8
12	6	3612081	WASHER, SPRING	WS10
13	2	2111206	BOLT, HEX. HEAD	BA10×25
14	6	2110886	BOLT, HEX. HEAD	BA8×20
15	1	BA-0079	PLUG, PT	PT1/2
16	2	A02-3261	BRACKET, TANK	
17	2	A02-3262	OIL TANK BAND	
18	4	S01-0046	BAND	
19	2	2115976	BOLT, HEX. HEAD	BA8×55
20	4	3511081	WASHER, PLAIN	WP8
21	2	3111081	NUT, HEX.	NA8
22	8	2111546	BOLT, HEX. HEAD	BA12×40
23	8	3612121	WASHER, SPRING	WS12
24	8	3111121	NUT, HEX.	NA12
25	2	W01-5641	PLATE	t9

SECTION 47.1 - FILTER, RETURN (OPTION)

FILTER, RETURN (OPTION)

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SECTION 47.1 - FILTER, RETURN (OPTION)

FILTER, RETURN (OPTION)				00
Pos.	Q'ty	Part No.	Description	Remarks
-	1	H01-1287	FILTER, RETURN	
1	1	-	HOUSING	
2	1	H01-1288	FILTER, ELEMENT	
3	1	S01-2480	COVER	
4	1	S01-3414	BOWL	
5	3	S01-4977	KIT, SEAL	

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